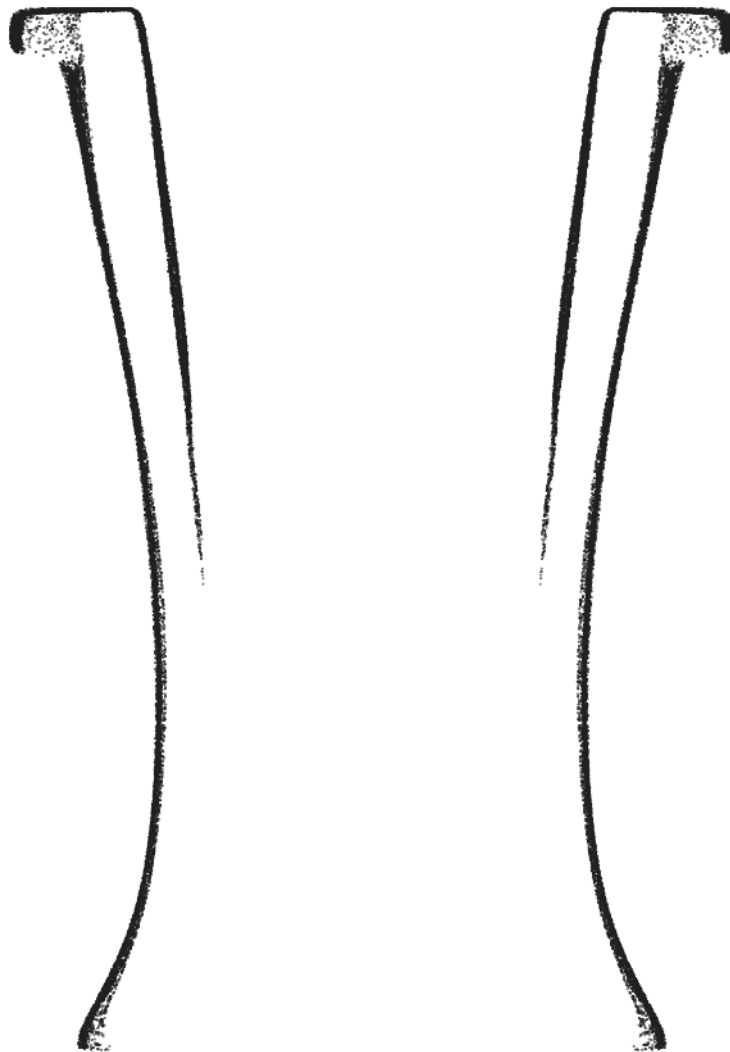


MV012 - Concave Beaker

An Exploration of Precision



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Date: 2025-07-21
Version: 01.20



Petrie Museum, CC BY-NC-SA

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Artifact Information

Artifact Data

Collection	Petrie Museum of Egyptian Archaeology
Provenance ¹	Petrie Museum of Egyptian Archaeology
Provenience ²	Unknown
Attribution	6th Dynasty

Museum information

Ref.	LDUCE-UC41050
Description	Basalt cylinder jar with flared base, one large chip on rim, several smaller on base. Otherwise intact.
URL	https://collections.ucl.ac.uk/Details/collect/46311

Maijers vessel classification³

Short classification	Concave Beaker
Long classification	The vessel is created in an open form classified as a beaker with a concave shape, the vessels Concave curves are ending in a squared flat rim.

Physical properties

Precision score ⁴	6
Height (approximate)	195 mm 7.68 in
Width (approximate)	134 mm 5.28 in
Material	Basalt
Mohs Hardness ⁵	6 - 7 (Basalt)
Weight	

Scan information

Source	Scanned by Artifact Foundation
Source file name	UC41050_base_0.09_decimated
Scan method	Laser
Scanner	FreeScan Combo+
Rated scan accuracy	Not specified
Scan date	2024-10-07
Scanned by	Adam Young and Károly Póka
Mesh decimation	Topology-Preserving Quadratic Edge Collapse Decimation ⁶
Number of vertices	15 594 650
Mesh density ⁷	52 µm 2.04 thou
Max vertex distance	173 µm 6.798 thou
Min vertex distance	0 µm 0.000 thou
Vertices per cm ²	16 540 (approximated)
Vertices per in ²	106 709 (approximated)

¹The verifiable chain of custody of an artifact

²The location or site where an artifact was recovered

³Vessel artifact classification developed by W. Arnold Maijer and described in his publication Masters of Stone, ISBN 978-90-829212-0-5

⁴The precision score metric is described in Precision Score Of The Artifact, p. 49

⁵The Mohs scale is an ordinal scale, from 1 to 10, describing the materials resistance to abrasion (the ability of harder material to scratch softer material)

⁶<https://arxiv.org/pdf/2411.16874>, <https://www.cs.cmu.edu/~garland/Papers/quadrics.pdf>

⁷Median distance between vertices

Alignment In The Cartesian Coordinate System

For precise and valid measurements of the vessel's geometry to be possible, the points of the scanned dataset must first and foremost be placed optimally in a Cartesian coordinate system. Several alignment methods and algorithms have been tested on a number of different vessels to determine the best way to achieve optimal alignment.

Any misalignment of the artifact will increase the error of the precision measurements, due to the distortion/wobble effect caused by the misaligned object. To visualize this distortion, we can consider a representation of the three-dimensional point cloud data, folded to a two-dimensional plane. This folded representation is obtained by rotating all scanned points around an assumed center axis to $y = 0, x > 0$, thus resulting in a two-dimensional profile representation of all scanned vertices in the object.

Figure 1 illustrates this effect on a ideal ellipsoid. In the first image, the ellipsoid is perfectly aligned, resulting in a narrow and precise two-dimensional folded profile. As misalignments are introduced, the two-dimensional profile increases in width, visually showing the distortion, causing the error in the precision measurements to increase. While easy to understand visually, this distortion can also be objectively quantified, and as such used to compare the fitness of different assumed center axes against each other, and further to create an automated and solid process for optimal Cartesian alignment of the scan data.

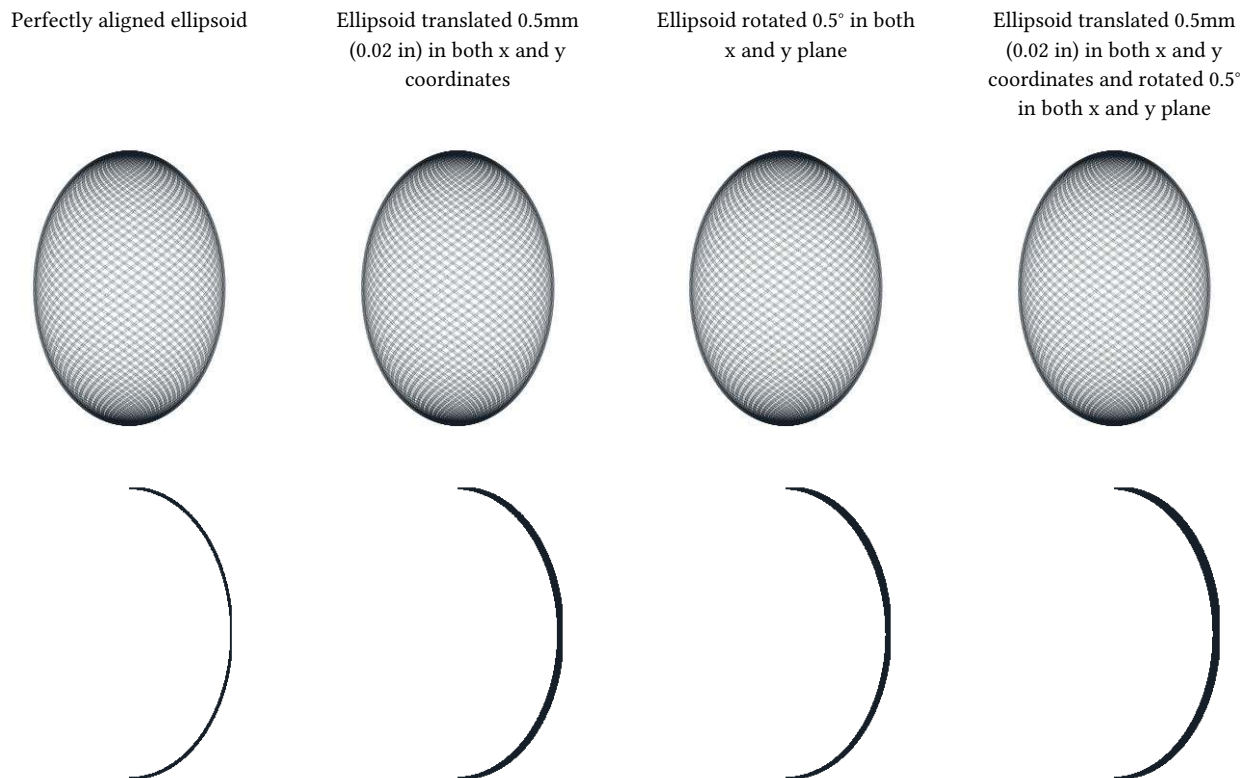


Figure 1: Distortion caused by a misalignment of the artifact

In contemporary metrology analysis of modern production objects, it is common to align the object in a Cartesian coordinate system by fitting a flat surface of the object to a reference plane in the coordinate system, cylindrical features to an ideal cylinder etc., or by using specific markers placed on the object in the design process. This methodology, however, is inadequate for the ancient objects in question. Most scanned artifacts, do not have a valid flat surface which could be aligned to a plane in the Cartesian coordinate system; most surfaces seem to be curved. Some artifacts do have a flat base, however this is often a worn area of the artifact and practical tests have shown that alignment to such surfaces will not produce optimal alignment of the scan data.

As conventional methods of alignment do not always yield good results with these types of artifacts, a more adequate method of alignment has been developed to enable precise measurements and statistical analysis of the scan data.

To find the optimal position of the vessel in the coordinate system, a range of rotation and translation tests are carried out to find the best fit of the central axis.

Based on the assumption that the analyzed object was created using a rotational process, and thus have symmetry around a central axis, the alignment of the artifact is carried out in a two-step process. An overview of this process is given below.

The artifact is placed in a Cartesian coordinate system, in an initially unaligned state. The first step in the alignment process estimates the central rotational axis of the vessel, by analyzing the coaxiality of thin cross-section slices of the vessel. The slices will be as thin as possible based on the mesh density of the scan, while still ensuring enough data points in each slice to be statistically valid.

For each slice, circular regression⁸ (estimate of best fit circle) is used to estimate the center point of this slice. Combined over the total Z-axis range of the vessel, these center points provide us with an indicator of the incline and position of the vessel's central axis.

The next step will optimize the center axis alignment by progressively minimizing the deviation (perpendicular to the surface curvature) of the two-dimensional profile, see Figure 1. By ascertaining and comparing the resulting fit of many thousands of different potential rotations, the best fit alignment of the scan data can be estimated, and an optimal center axis (in relation to the data points) can be reconstructed. The actual three-dimensional point-cloud is then aligned to this axis, by rotating and translating the scanned data points to match the Z-axis of the Cartesian coordinate system.

To enable extensive analysis of the full surface of the artifact, the mesh is split into exterior and interior surfaces. The exterior surface is aligned independently of interior data points, providing a baseline for exterior quality assessment. The interior surface is represented by two alignments:

- Aligned with the exterior mesh to analyze concentricity, and
- Aligned separately to assess its precision and compare the true tilt/displacement between interior and exterior surfaces.

⁸Circle regression algorithm used: Kenichi Kanatani, Prasanna Rangarajan, "Hyper least squares fitting of circles and ellipses" Computational Statistics & Data Analysis, Vol. 55, pages 2197-2208, (2011)

Statistics used throughout the report

This section provides an overview of the key statistical and model-evaluation metrics employed throughout the report to analyze dataset variability, model fit, and predictive accuracy.

Each measure is introduced with its mathematical formulation, practical interpretation, and explicit reference to how it is calculated in the context of the evaluated models and residuals. Together, these metrics quantify:

- Data variability (e.g., MAD, Standard Deviation, Range).
- Model accuracy (e.g., MSD, RMSD).
- Robustness vs. sensitivity to extreme values and central tendencies.

Mean Squared Deviation (MSD), also known as Mean Squared Error (MSE).

$$\text{MSD} = \frac{\sum_{i=1}^n (y_i - \hat{y})^2}{n}$$

The Mean Squared Deviation (MSD) measures the average magnitude of squared differences between observed (y_i) and predicted ($\hat{y}$) values, calculated as the mean of squared residuals, and is used as a measure of discrepancy in regression and model-fitting contexts.

This measure amplifies the influence of larger deviations through squaring, emphasizes imperfections in the observed data, but retains sensitivity to outliers.

Root Mean Squared Deviation (RMSD), also known as Root Mean Squared Error (RMSE).

$$\text{RMSD} = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y})^2}{n}}$$

The Root Mean Square Deviation (RMSD) measures the magnitude of differences between observed (y_i) and predicted ($\hat{y}$) values by calculating the square root of the average of squared residuals.

RMSD is a commonly used measure of discrepancy in regression and model-fitting contexts. It quantifies the average magnitude of residuals while retaining sensitivity to larger deviations (via squaring), making it particularly useful for evaluating model accuracy.

Standard Deviation (SD)

$$s = \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n - 1}}$$

The Standard Deviation measures the spread of data (y_i) around the mean ($\bar{y}$) by calculating the square root of the average of squared differences between each value and the mean.

It is sensitive to outliers as it amplifies their influence through squaring, in contrast to MAD.

Throughout this report, the Standard Deviation is calculated using the absolute residuals from regression models.

Median Absolute Deviation (MedianAD)

$$\text{MedianAD} = \text{median}(|y_i - \text{median}(y)|)$$

The Median Absolute Deviation (MAD) measures the spread of data around the median by calculating the median of absolute differences between each value and the median.

MAD is a robust measure of spread, analogous to the interquartile range (a robust measure centered on the middle 50% of data), and differs from the standard deviation in that it minimizes the impact of outliers.

Throughout this report, the MAD is calculated using the absolute values of residuals from regression models.

Range

$$\max(y_i) - \min(y_i)$$

The Range measures the spread of a dataset by calculating the difference between the maximum and minimum values.

The Range is a simple measure of spread, capturing the full extent of variability. Range is very sensitive to extreme values, as it is entirely determined by the two most extreme data points.

Throughout this report, the Range is calculated using the full range of residuals from regression models.

Precision

To explore the manufacturing precision of the artifact in depth, the following analysis have been carried out:

- Circularity around the axis of symmetry is examined in detail at selected cross-sections.
- Overall circularity around the axis of symmetry is measured for the full height of the vessel (areas of the vessel with extensive damage are not taken into account for this metric).
- Concentricity of the vessel between selected cross-sections are examined in detail to determine if the existence of an axis of rotation in the manufacture of the object can be established.
- The coaxiality of the vessel is analyzed to explore the precision of the central axis of the object.
- The surface variability is analyzed and visualized on through a heatmap.

Circularity

Circularity is the measurement of how round the surface of an object is, optionally in reference to a datum axis. The *circularity tolerance* is the radial distance of two circles, each with their centers in the datum axis, and each of them conforming, respectively, to the minimum and maximum deviations of the data-set to a true circle, see Figure 2.

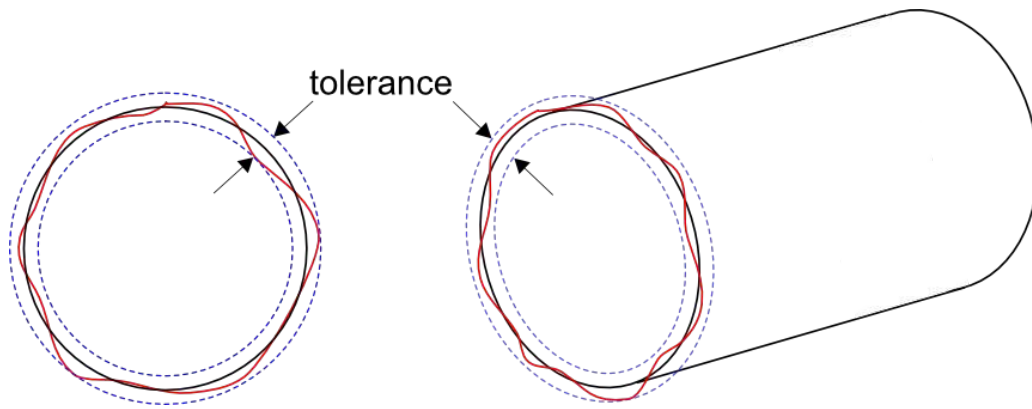


Figure 2: Circularity tolerance.

Circularity is examined at different cross-sections of the vessel, using the established Z-axis as the datum axis (axis of symmetry). The distance between the scanned points in the local datum plane is measured to determine the range between the two concentric circles encompassing the measured points, see Figure 3.

Referencing all of the individual circularity measurements to the global (reconstructed) axis of symmetry of the object, allows us to ascertain not only circularity of local features of the object, but how well circularity was *maintained* over the entire manufacturing process. This is an important distinction, which may be able to provide valuable insights into requirements of the construction methods. For reference, and seeing that the variance in local circularity also holds interest, measurements of circularity of the vessel without reference to the axis of symmetry can additionally be found in the Concentricity, p. 27.

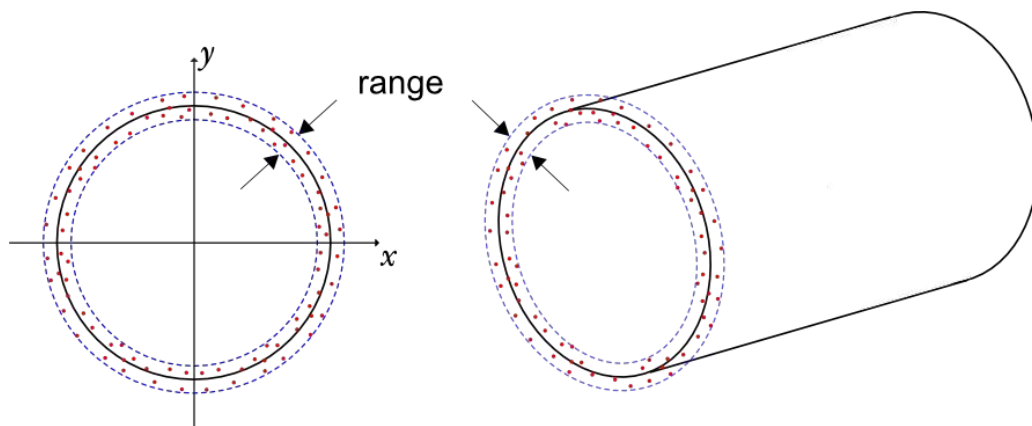


Figure 3: Circularity measurements.

If the circularity is determined from slices of the vessel exclusively in the *Z-plane* (actually measuring the cylindricity of a very thin slices of the vessel, in an attempt to approximate circularity), this would - in some areas - introduce significant distortion (increasing measurement errors) in the samples, due to the curvature of the vessel's surface.

Each sample slice of the vessel is therefore obtained perpendicular to the surface curvature, see Figure 6 to Figure 12. The measurements are taken conservatively without filtration of potential outliers.

To explore the potential distortion caused by obtaining samples in the Z-plane only, please refer to Appendix A, where measurements in the Z-plane and measurements perpendicular to surface curvature are compared side by side.

Detailed circularity measurements of selected points

Circularity measurements across a range of selected slices of the vessel (see Table 1) have been analyzed in-depth, and detailed plots of each measurement is provided. Furthermore, full circularity measurements are shown for each available scanned surface including a detailed plot to visualize the circularity of all areas of the vessel.

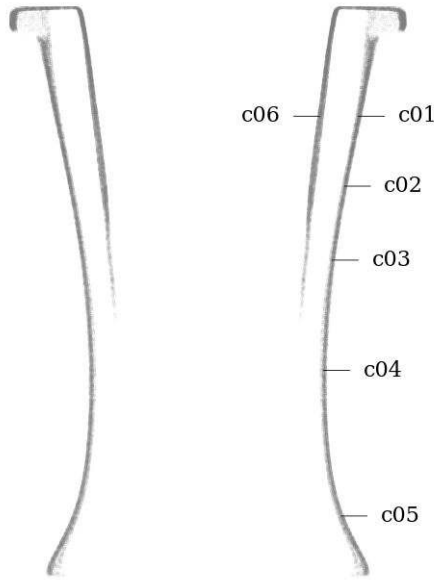


Figure 4: Circularity measurement sample locations, full mesh aligned with exterior surface

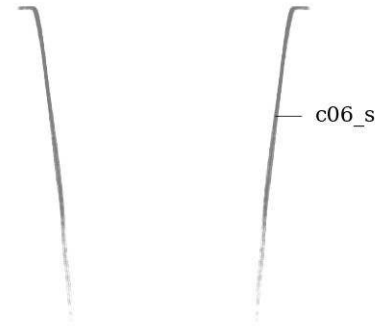


Figure 5: Circularity measurement sample location, separately aligned interior mesh

Metric

Tag	Area	Measured deviation ⁹	Residuals				Sample size	Slice		
			Range	RMSD ¹⁰	MAD ¹¹	SD		Height	Z coord.	Radius ¹²
		mm	mm	mm	mm	mm		mm	mm	mm
c01	exterior	Ø102.587±0.982	1.856	0.404	0.157	0.221	2969	0.050	157.004	51.293
c02	exterior	Ø93.084±0.762	1.324	0.333	0.123	0.182	2256	0.050	133.214	46.542
c03	exterior	Ø84.726±1.012	1.717	0.366	0.148	0.234	2056	0.050	108.050	42.363
c04	exterior	Ø78.956±1.124	1.879	0.411	0.162	0.276	1841	0.050	70.549	39.478
c05	exterior	Ø90.715±1.045	1.695	0.476	0.157	0.252	2032	0.050	21.022	45.357
c06	interior	Ø76.449±1.039	1.999	0.463	0.159	0.211	2987	0.050	157.004	38.224
c06_s	interior sep.	Ø76.508±0.550	1.092	0.208	0.081	0.107	2969	0.050	157.004	38.254

Imperial

Tag	Area	Measured deviation ⁹	Residuals				Sample size	Slice		
			Range	RMSD ¹⁰	MAD ¹¹	SD		Height	Z coord.	Radius ¹²
		in	in	in	in	in		in	in	in
c01	exterior	Ø4.0389±0.0387	0.0731	0.0159	0.0062	0.0087	2969	0.0020	6.1813	2.0194
c02	exterior	Ø3.6647±0.0300	0.0521	0.0131	0.0048	0.0072	2256	0.0020	5.2447	1.8324
c03	exterior	Ø3.3357±0.0398	0.0676	0.0144	0.0058	0.0092	2056	0.0020	4.2539	1.6678
c04	exterior	Ø3.1085±0.0442	0.0740	0.0162	0.0064	0.0109	1841	0.0020	2.7775	1.5542
c05	exterior	Ø3.5714±0.0411	0.0667	0.0187	0.0062	0.0099	2032	0.0020	0.8276	1.7857
c06	interior	Ø3.0098±0.0409	0.0787	0.0182	0.0062	0.0083	2987	0.0020	6.1813	1.5049
c06_s	interior sep.	Ø3.0121±0.0217	0.0430	0.0082	0.0032	0.0042	2969	0.0020	6.1813	1.5061

Table 1: Detailed circularity measurements at selected samples of MV012.

Figure 6 to Figure 12 shows a detailed plots of each circularity measurement.

⁹Sample diameter Ø± maximum measured deviation from measured radius

¹⁰Root mean square deviation (RMSD) also called Root mean square error (RMSE)

¹¹Median absolute deviation

¹²Median sample radius from z-axis

Graphical overview of circularity measurement c01

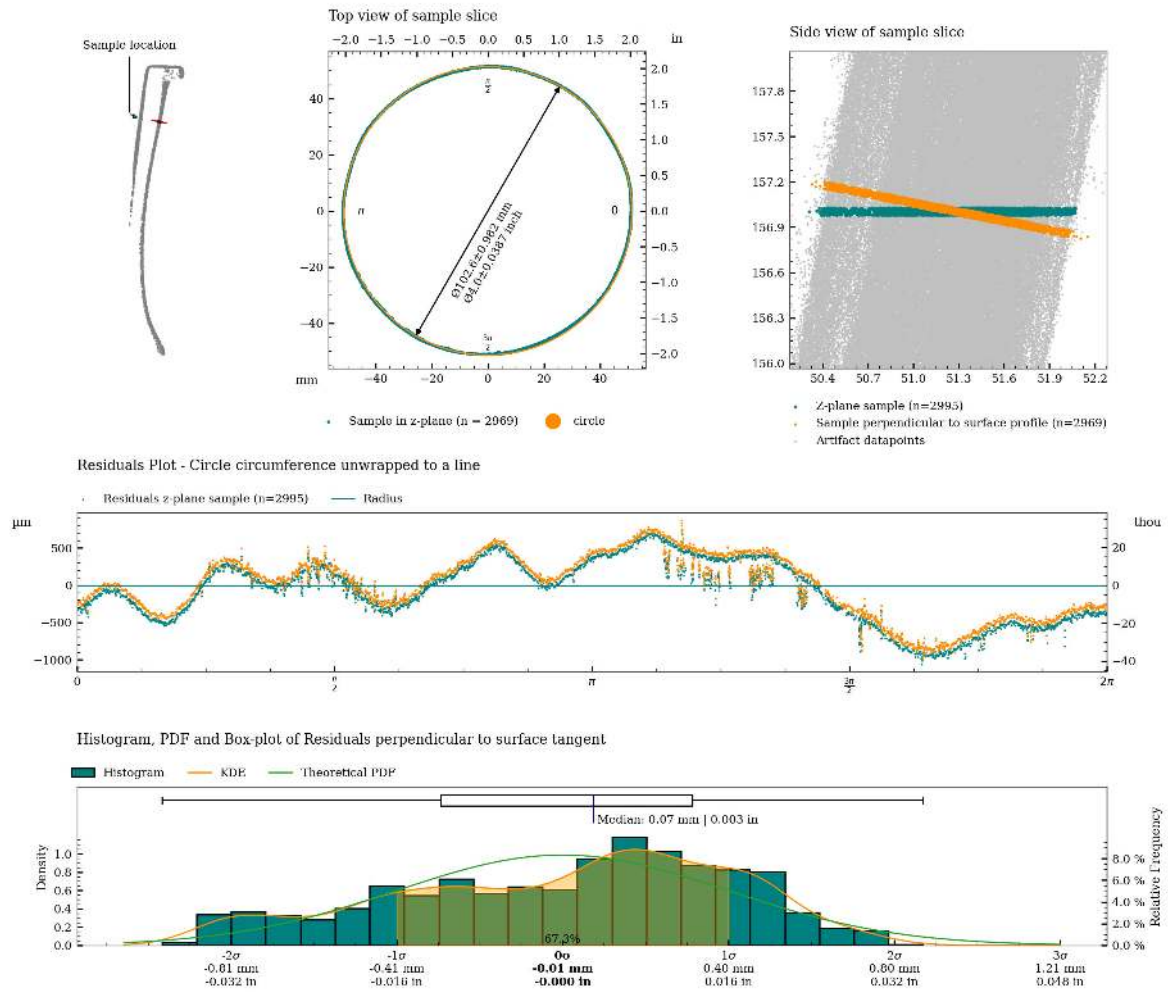


Figure 6: Charts with statistics for the measurement of c01.

Graphical overview of circularity measurement c02

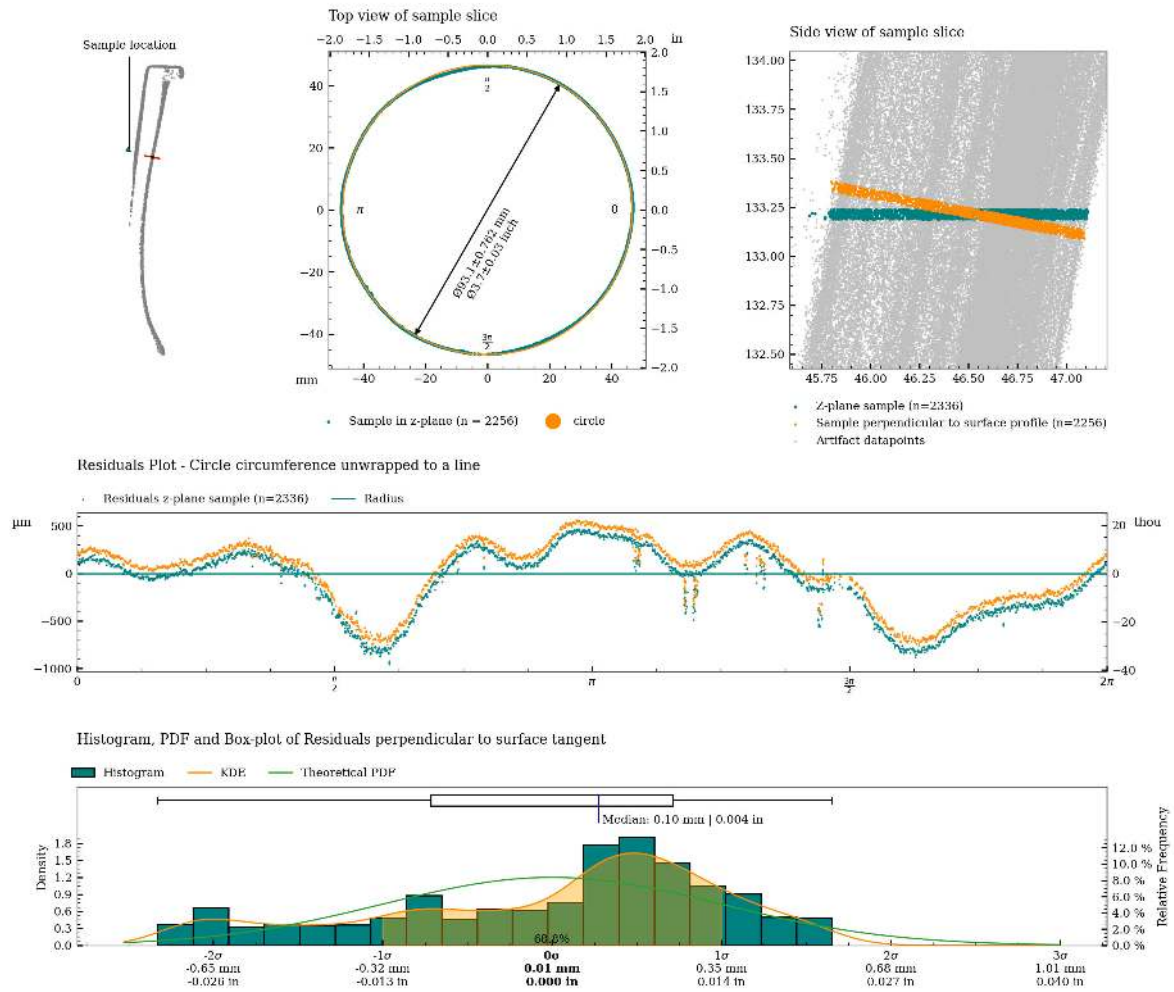


Figure 7: Charts with statistics for the measurement of c02.

Graphical overview of circularity measurement c03

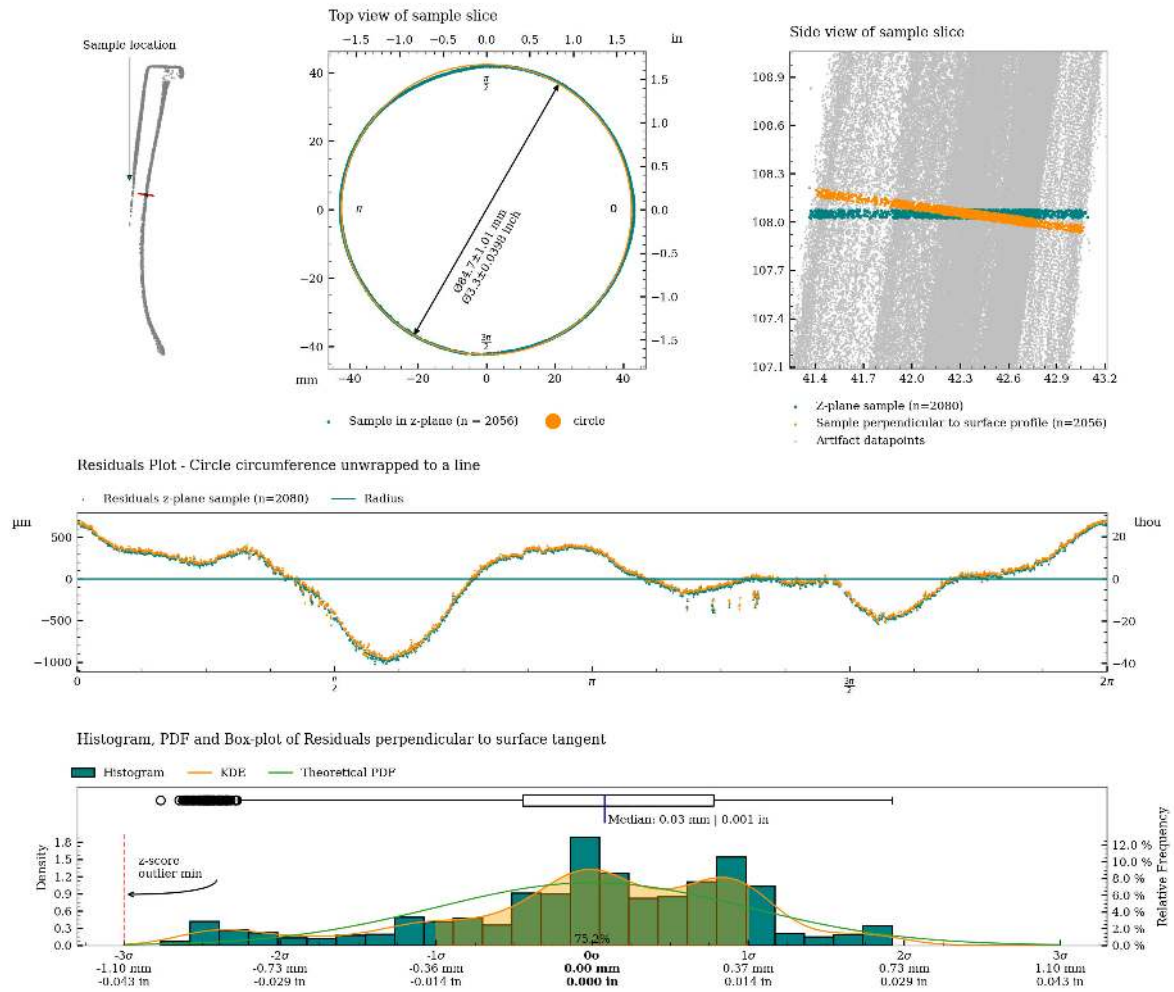


Figure 8: Charts with statistics for the measurement of c03.

Graphical overview of circularity measurement c04

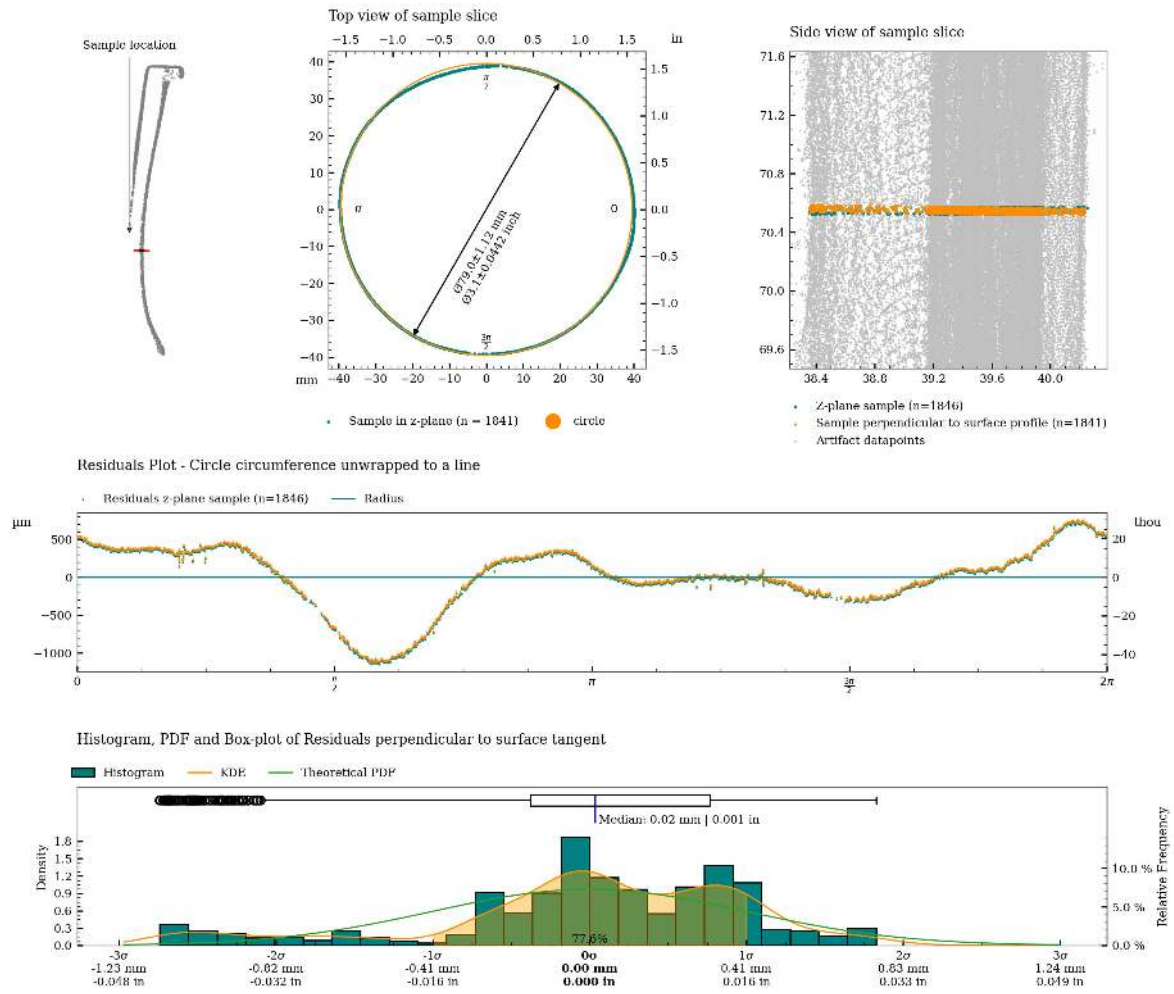


Figure 9: Charts with statistics for the measurement of c04.

Graphical overview of circularity measurement c05

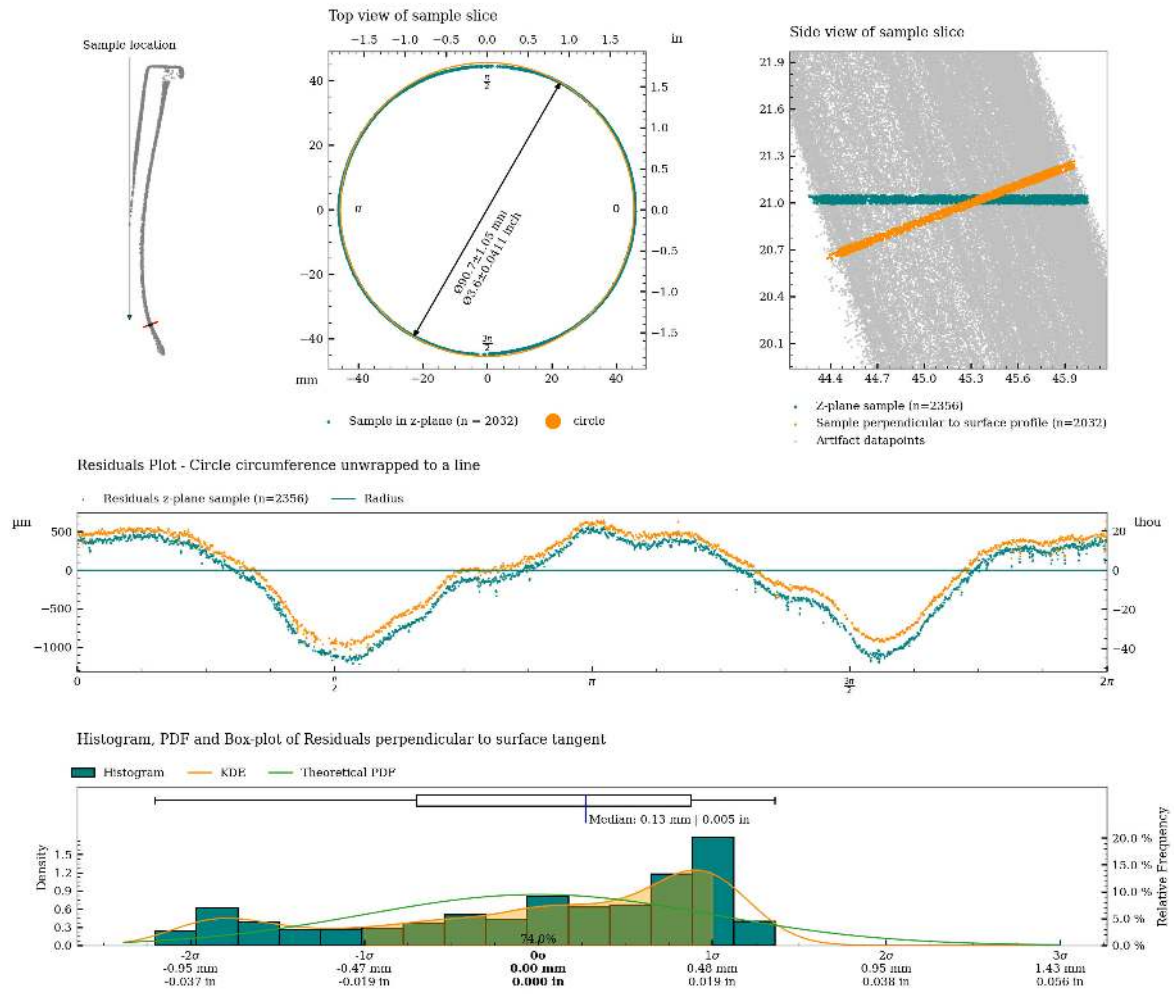


Figure 10: Charts with statistics for the measurement of c05.

Graphical overview of circularity measurement c06

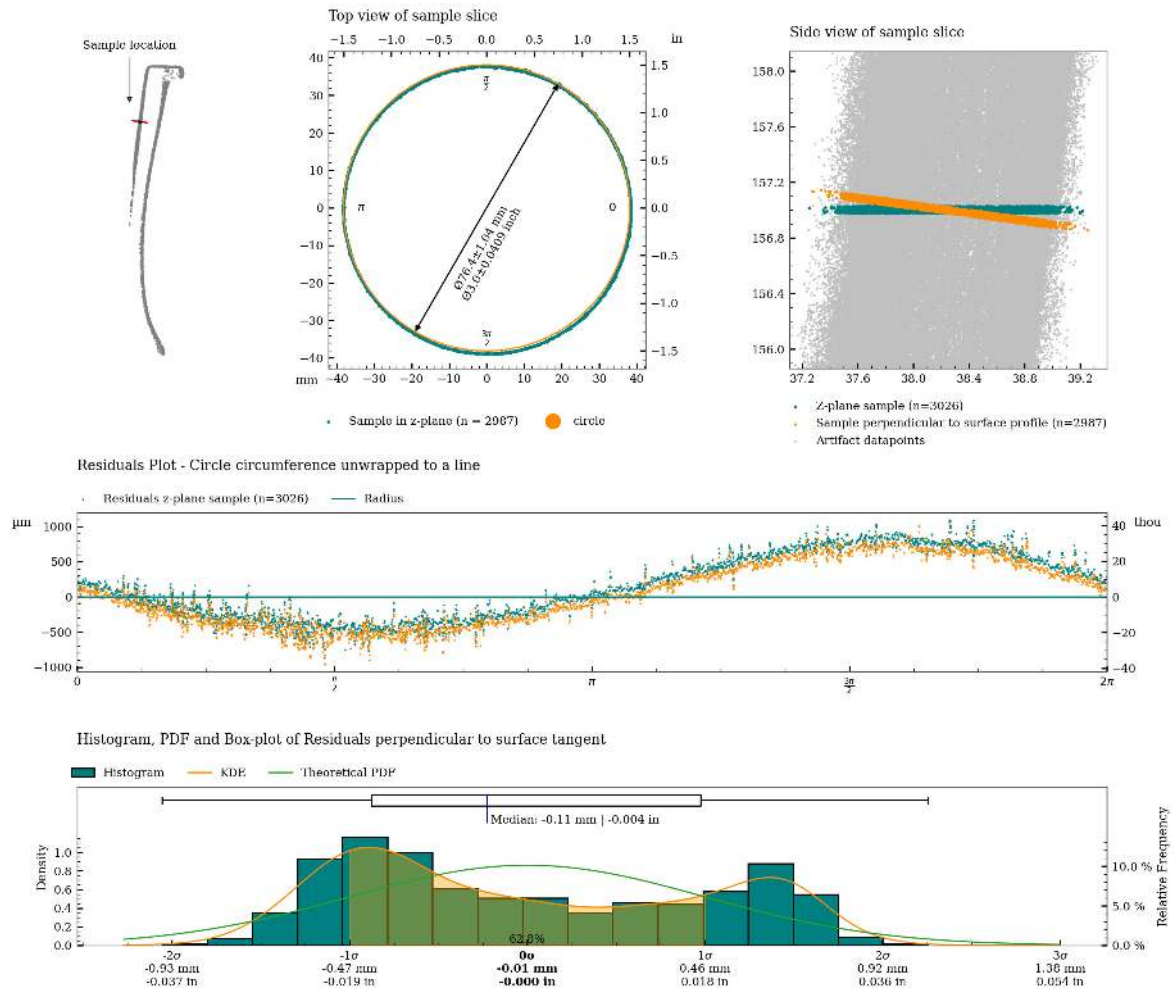


Figure 11: Charts with statistics for the measurement of c06.

Graphical overview of circularity measurement c06_s

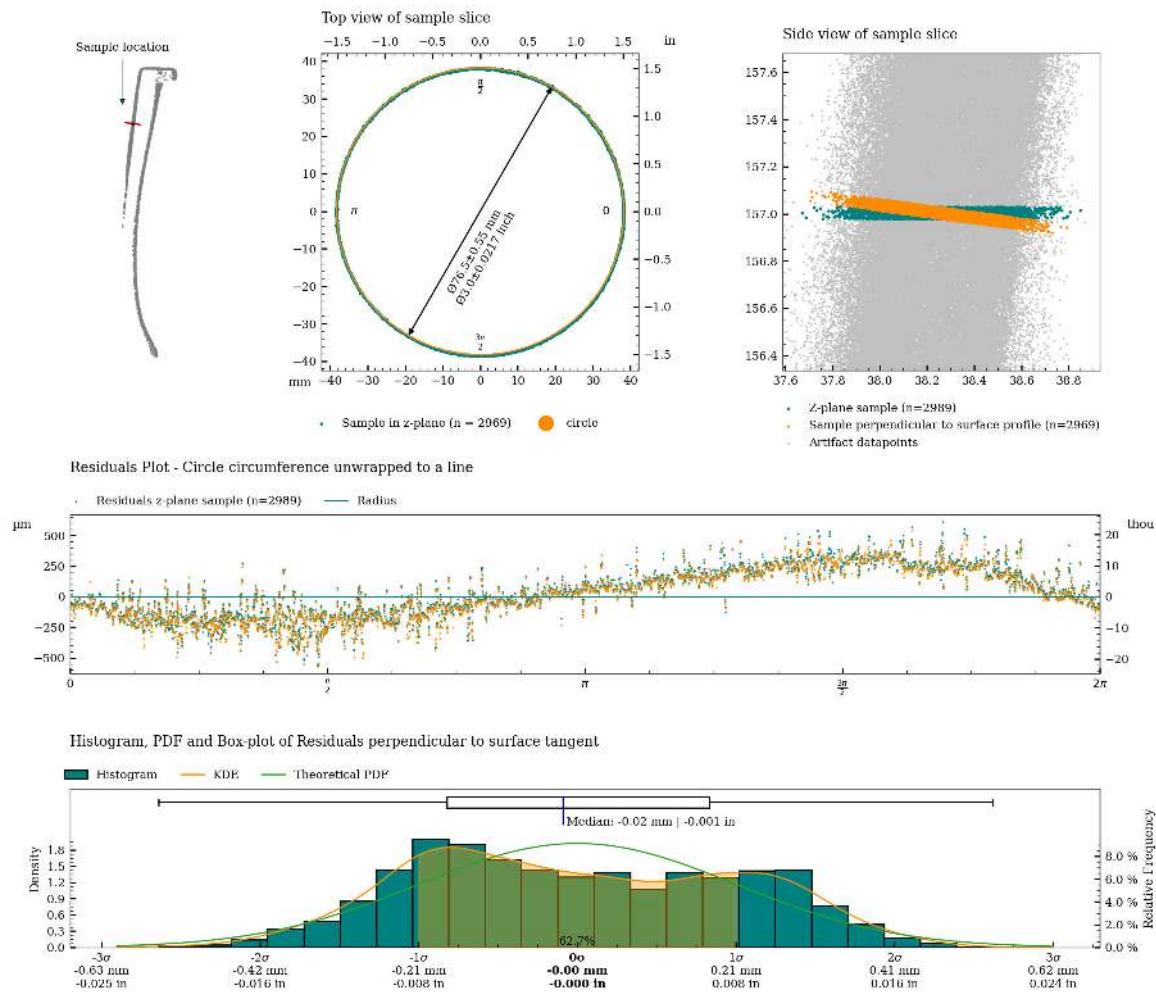


Figure 12: Charts with statistics for the measurement of c06_s.

Table 2 shows statistical measures of the circularity of the vessel, measured along the full height (areas on the artifact scan containing damaged parts have been removed to the best extent possible to reduce the influence of the measurement).

Metric											
Area	Range			Standard Deviation			RMSD			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		
Exterior	1.700	1.306	2.344	0.226	0.178	0.317	0.397	0.316	0.523	3151	0.050
Interior	1.680	1.387	2.152	0.187	0.139	0.228	0.388	0.260	0.518	760	0.050
Interior separate	1.037	0.799	1.656	0.093	0.071	0.145	0.158	0.109	0.268	760	0.050

Imperial											
Area	Range			Standard Deviation			RMSD			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	in	in	in	in	in	in	in	in	in		
Exterior	1.700	1.306	2.344	0.226	0.178	0.317	0.397	0.316	0.523	3151	0.050
Interior	1.680	1.387	2.152	0.187	0.139	0.228	0.388	0.260	0.518	760	0.050
Interior separate	1.037	0.799	1.656	0.093	0.071	0.145	0.158	0.109	0.268	760	0.050

Table 2: Perpendicular Circularity analysis of MV012.

Circularity analysis of exterior surface

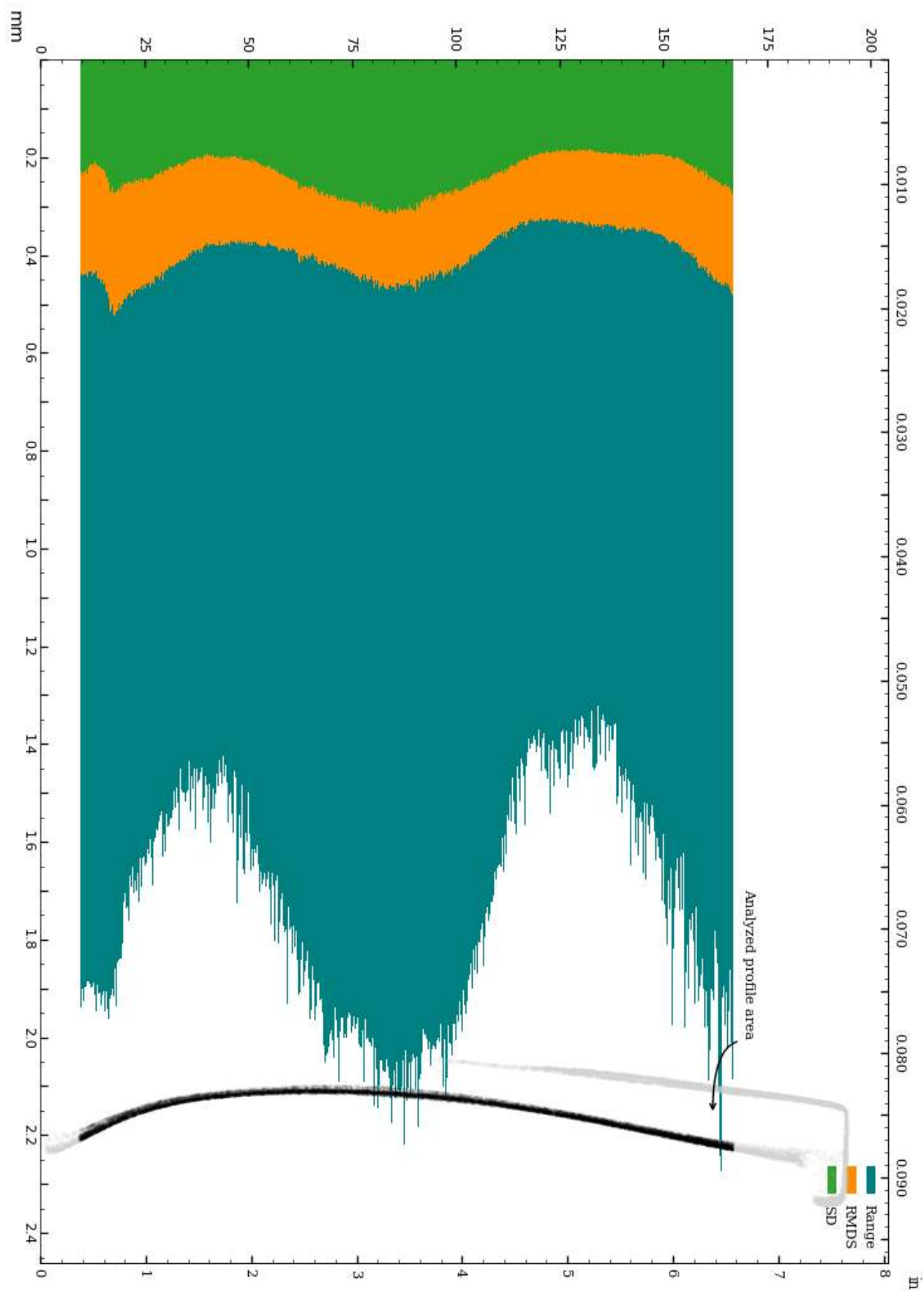


Figure 13: Circularity of exterior surface.

Circularity analysis of exterior surface, Standard Deviation and Root Mean Squared Deviation

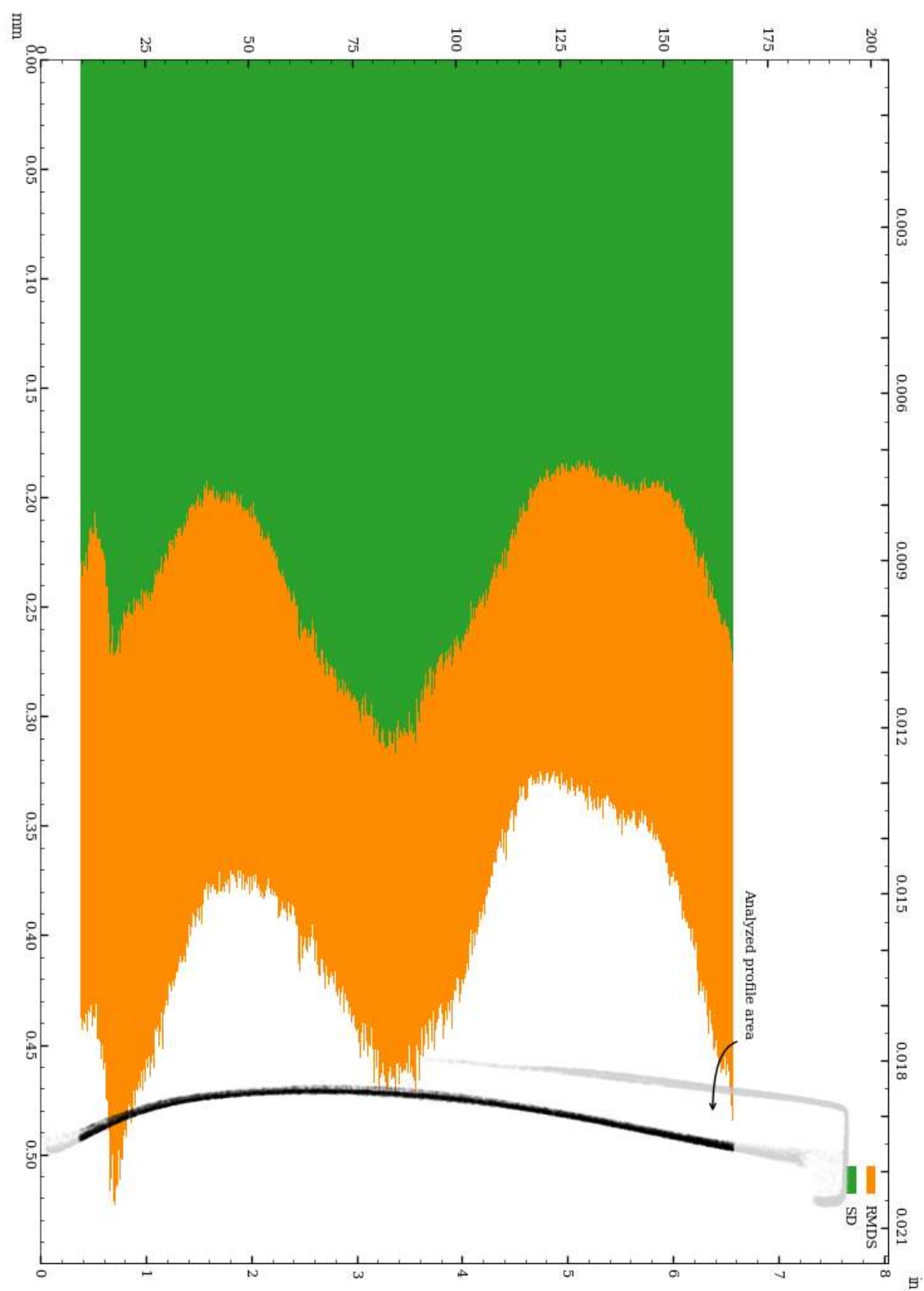


Figure 14: Vessel circularity of exterior surface, standard deviation and median absolute deviation.

The distributions of the circularity measurements across 3151 slices of the exterior surface are shown below.

Range measurement distribution across 3151 slices of exterior surface

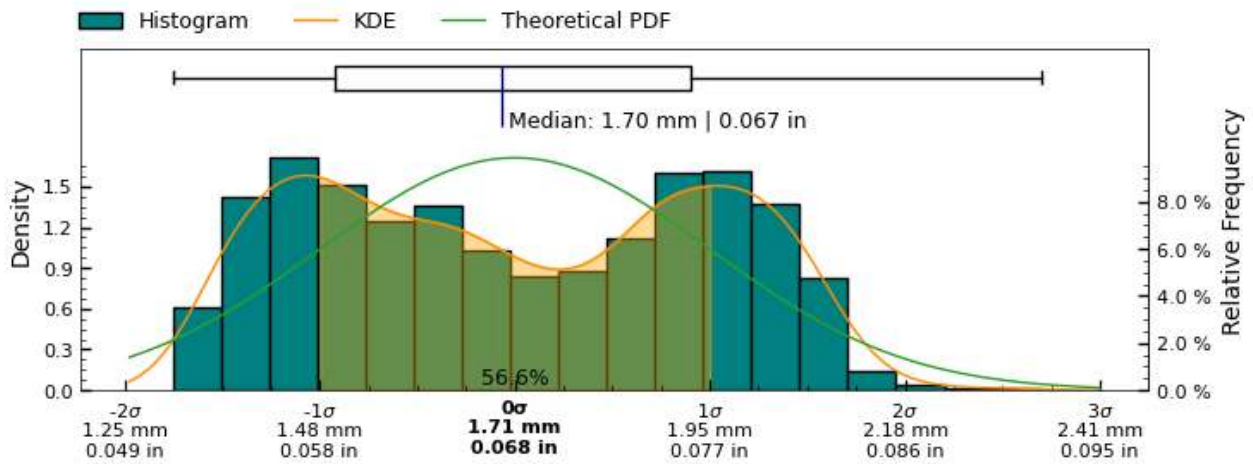


Figure 15: Range measurement distribution across measured slices of exterior surface

Standard Deviation measurement distribution across 3151 slices of exterior surface

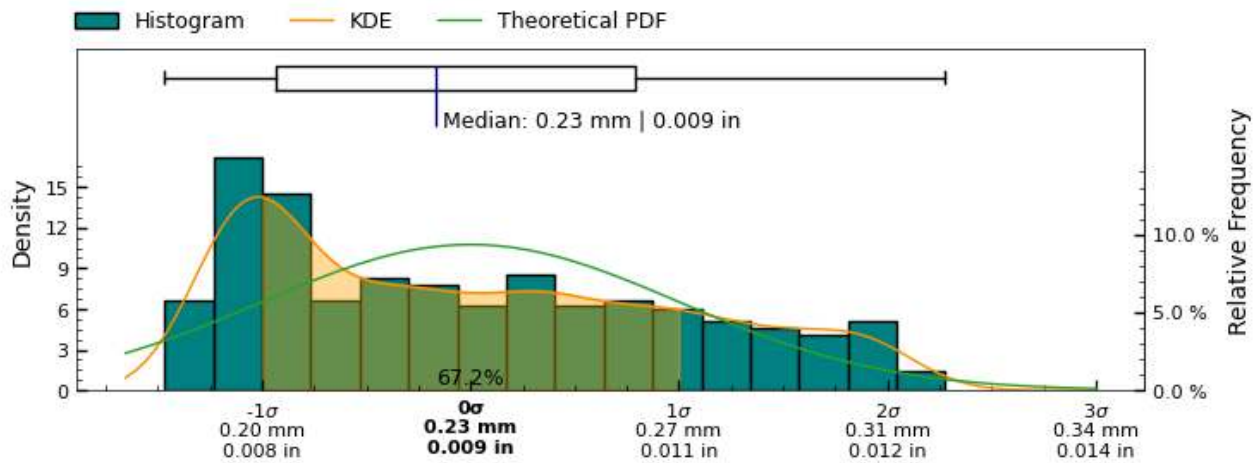


Figure 16: Standard Deviation measurement distribution across measured slices of " + exterior + " surface

Root Mean Squared Deviation measurement distribution across 3151 slices of exterior surface

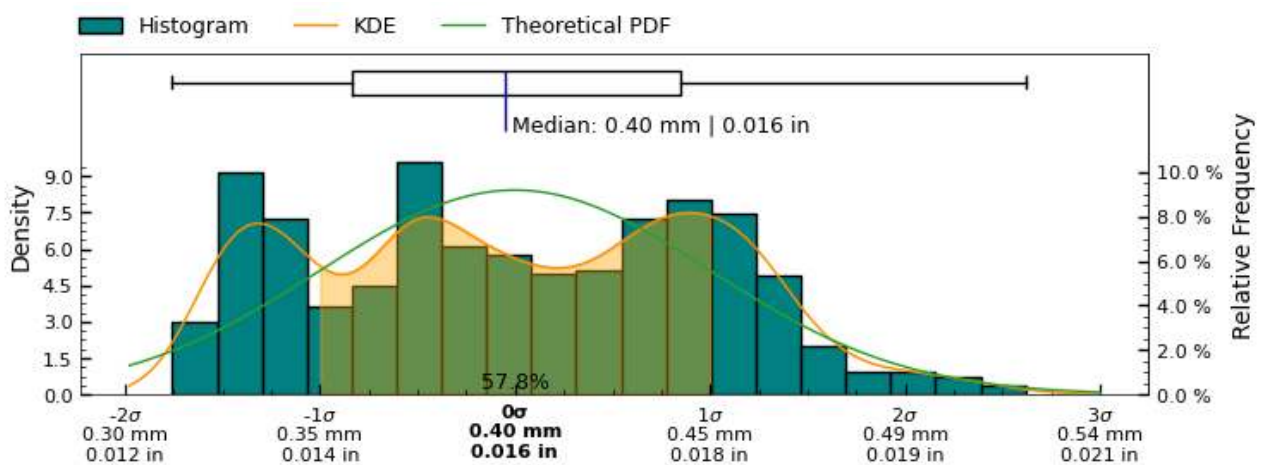


Figure 17: Root Mean Squared Deviation measurement distribution across measured slices of exterior surface

Circularity analysis of interior surface

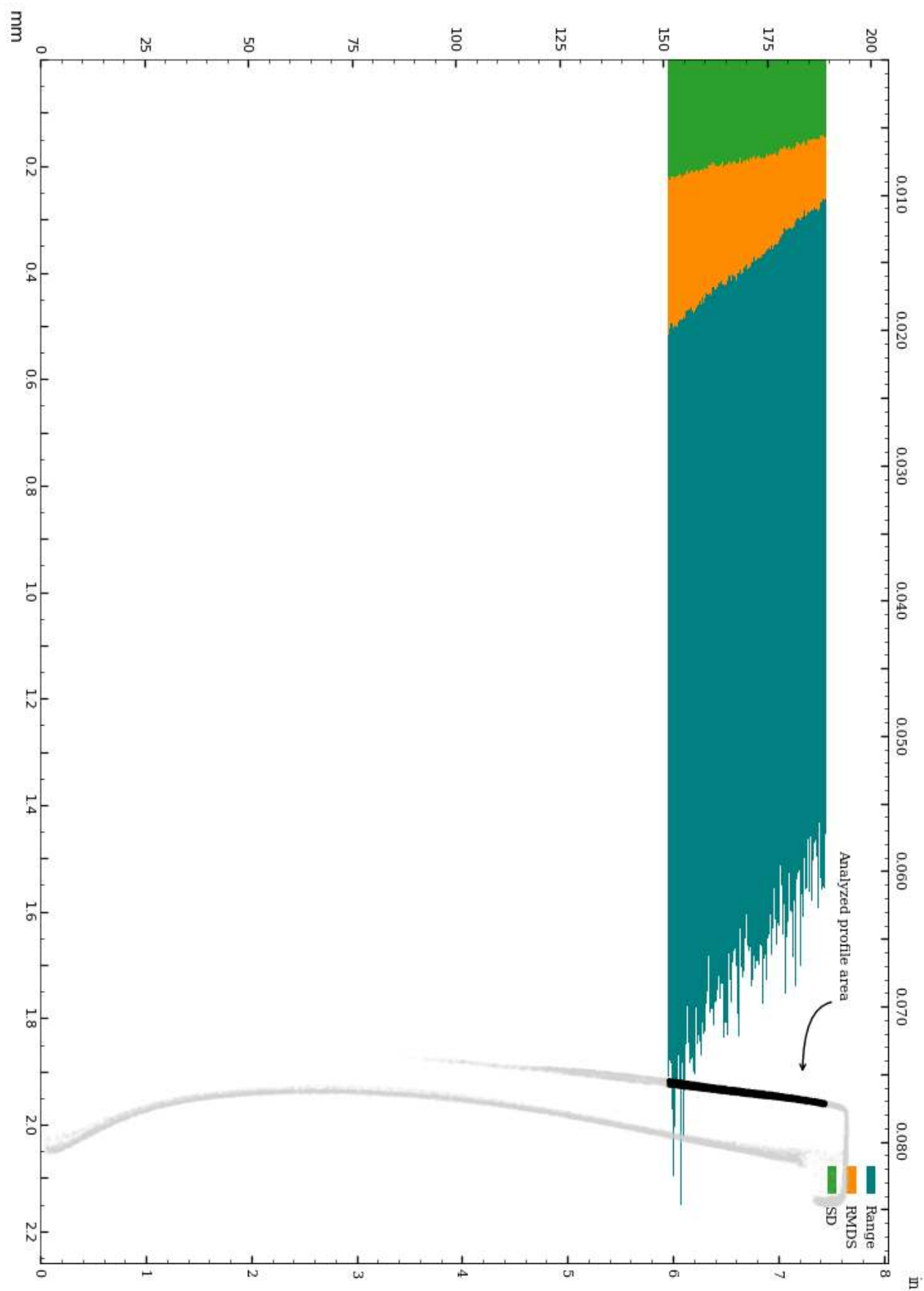


Figure 18: Circularity of interior surface.

Circularity analysis of interior surface, Standard Deviation and Root Mean Squared Deviation

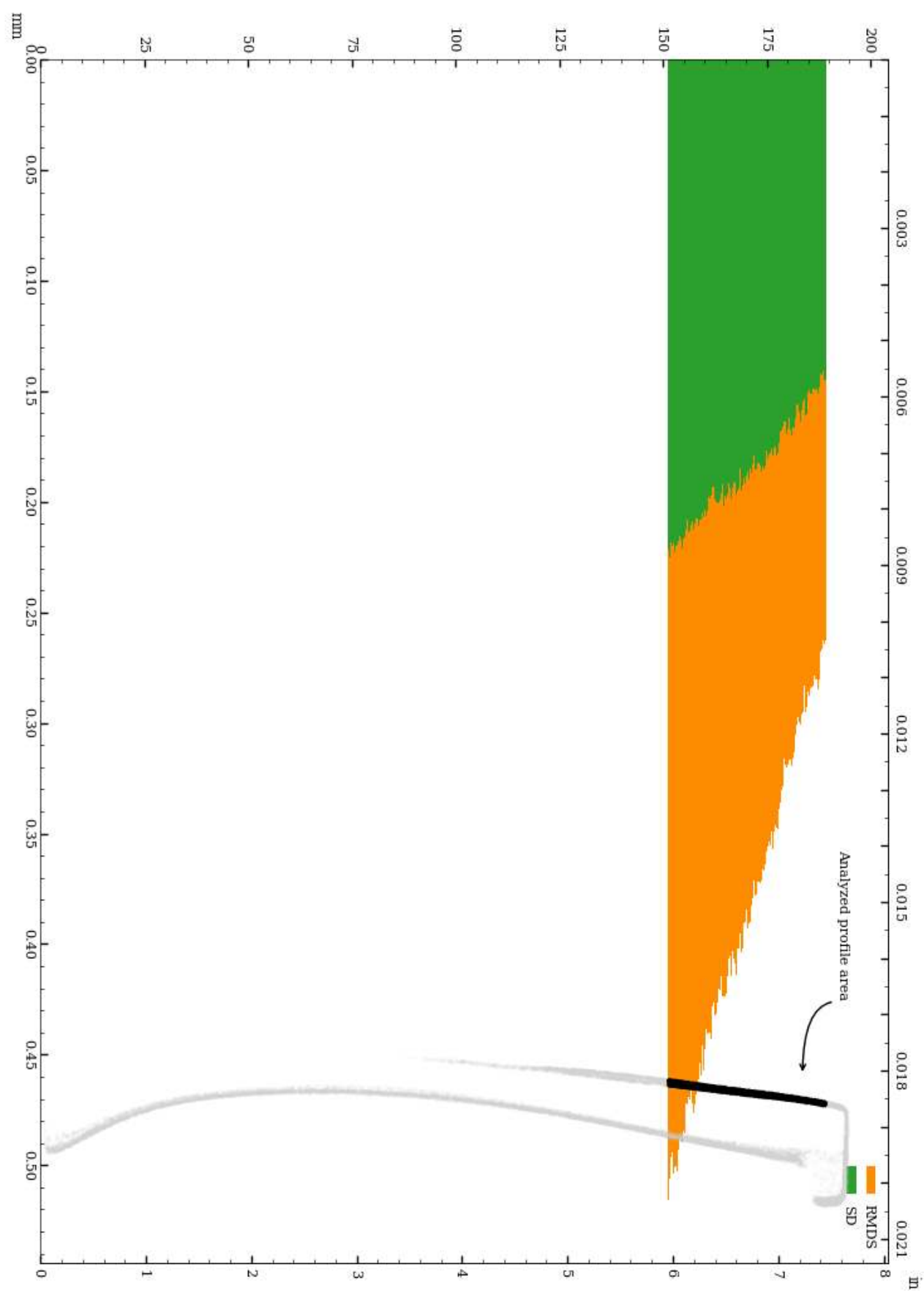


Figure 19: Vessel circularity of interior surface, standard deviation and median absolute deviation.

The distributions of the circularity measurements across 760 slices of the interior surface are shown below.

Range measurement distribution across 760 slices of interior surface

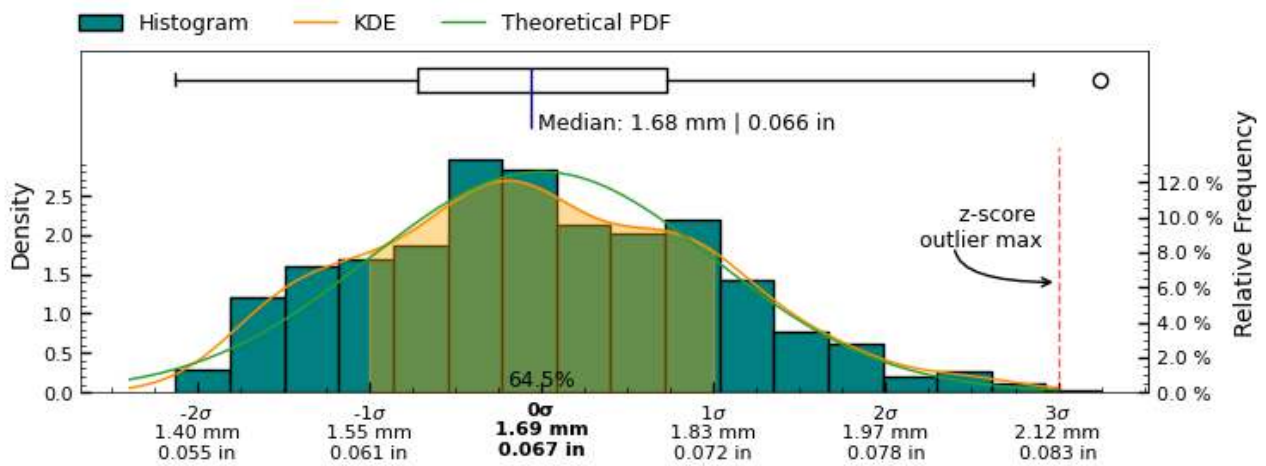


Figure 20: Range measurement distribution across measured slices of interior surface

Standard Deviation measurement distribution across 760 slices of interior surface

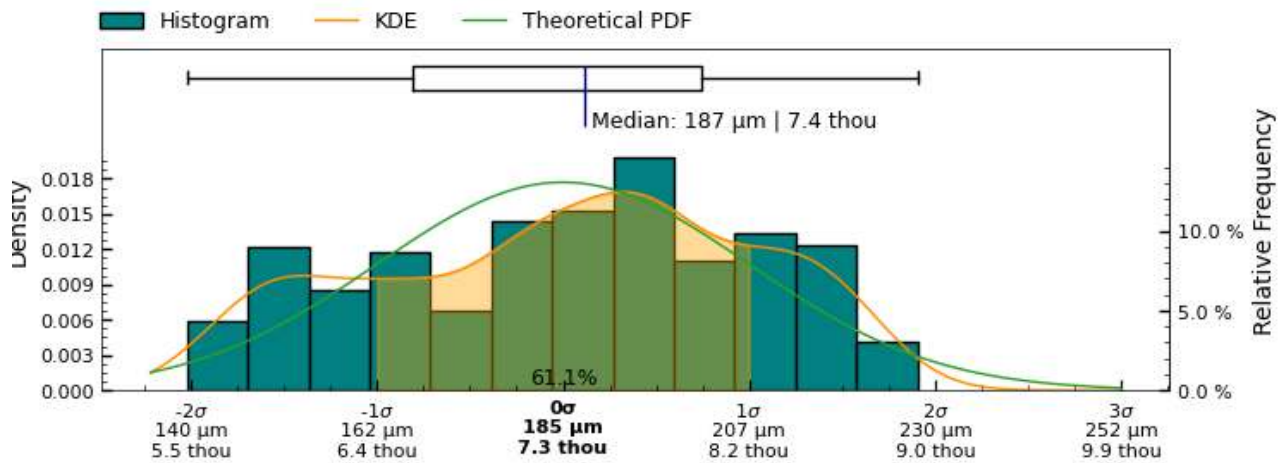


Figure 21: Standard Deviation measurement distribution across measured slices of " + interior + " surface

Root Mean Squared Deviation measurement distribution across 760 slices of interior surface

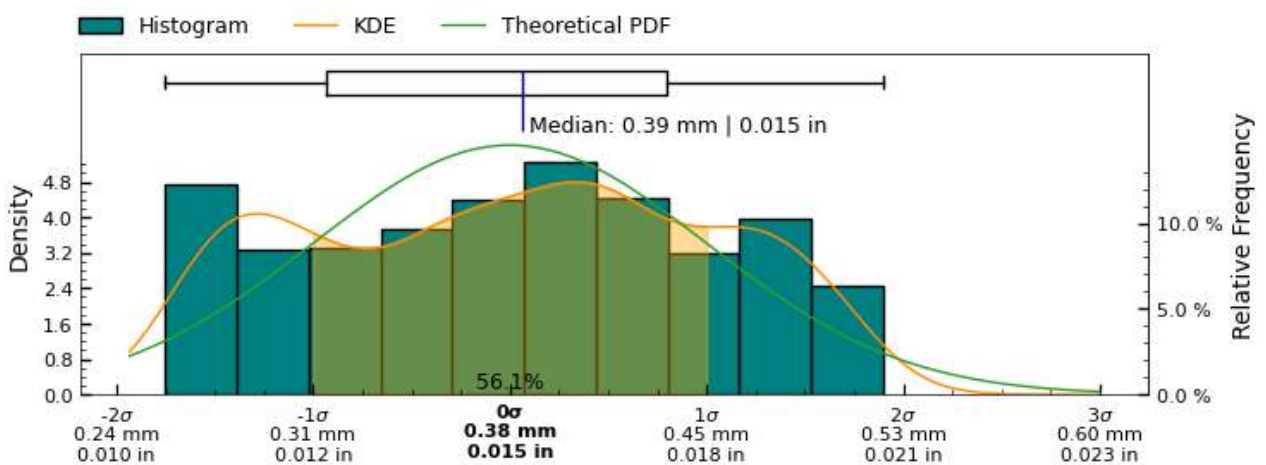


Figure 22: Root Mean Squared Deviation measurement distribution across measured slices of interior surface

Circularity analysis of interior separately aligned surface

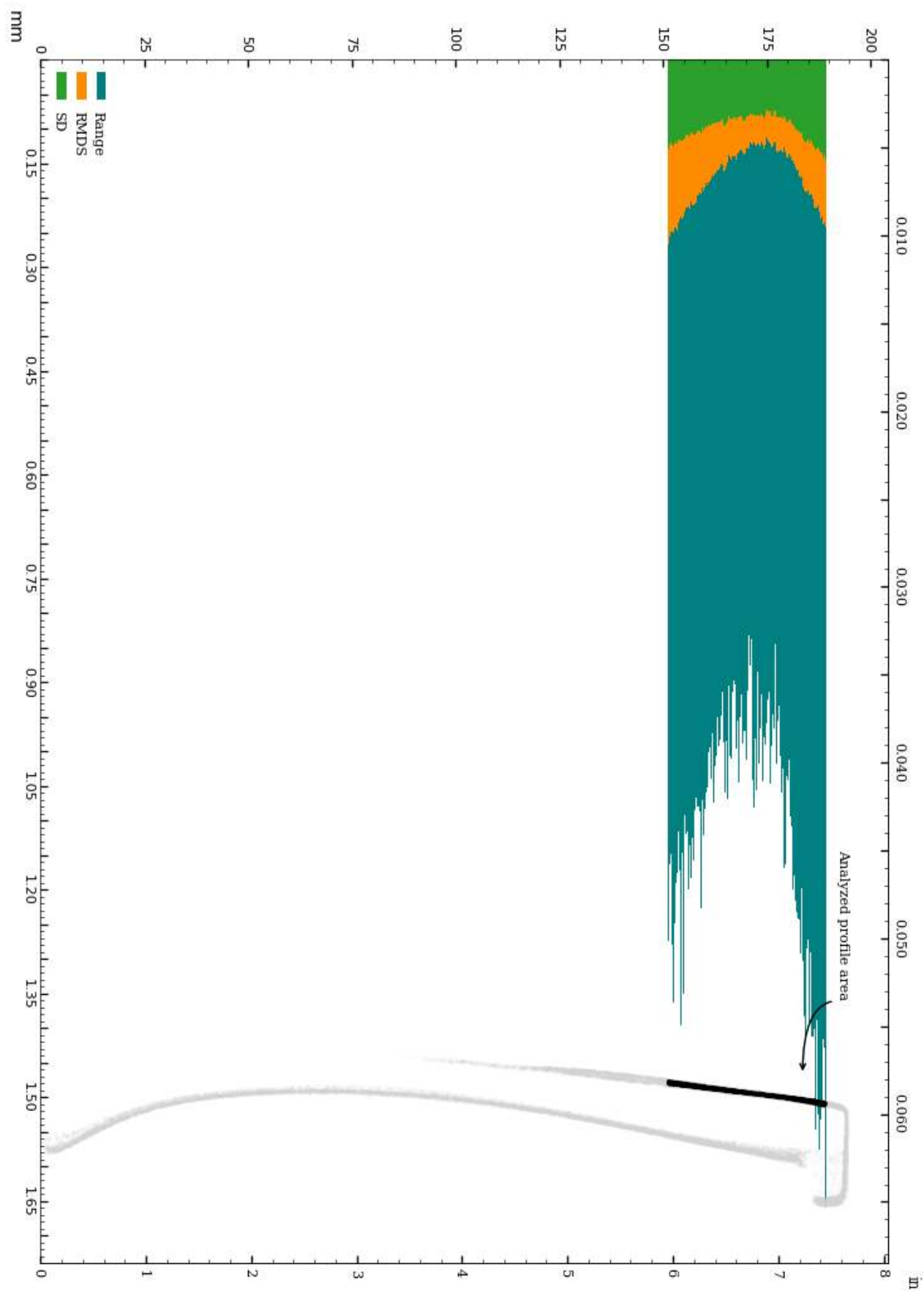


Figure 23: Circularity of interior_separate surface.

Circularity analysis of interior separately aligned surface, Standard Deviation and Root Mean Squared Deviation

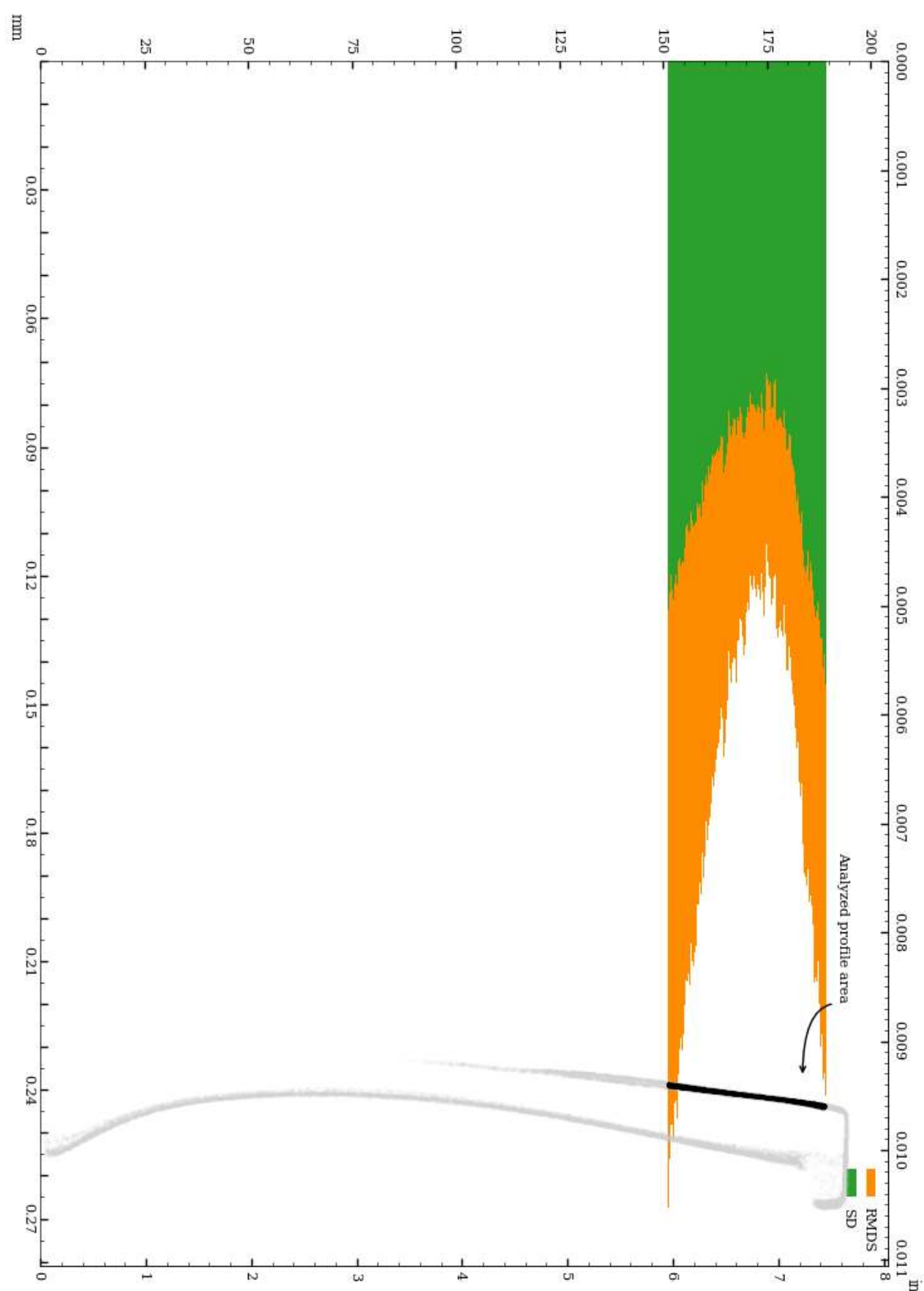


Figure 24: Vessel circularity of interior_separate surface, standard deviation and median absolute deviation.

The distributions of the circularity measurements across 760 slices of the interior_separate surface are shown below.

Range measurement distribution across 760 slices of interior separately aligned surface

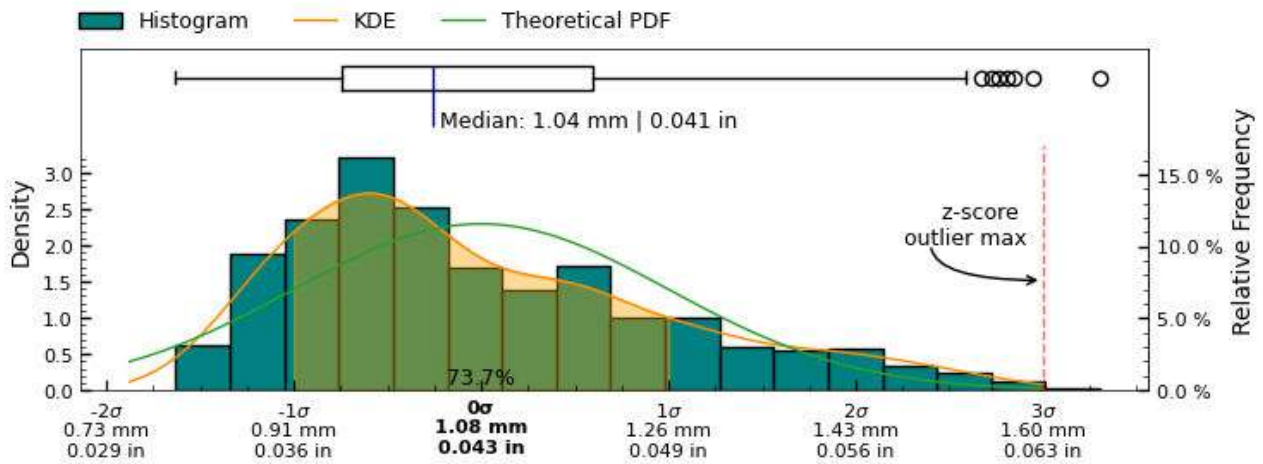


Figure 25: Range measurement distribution across measured slices of interior_separate surface

Standard Deviation measurement distribution across 760 slices of interior separately aligned surface

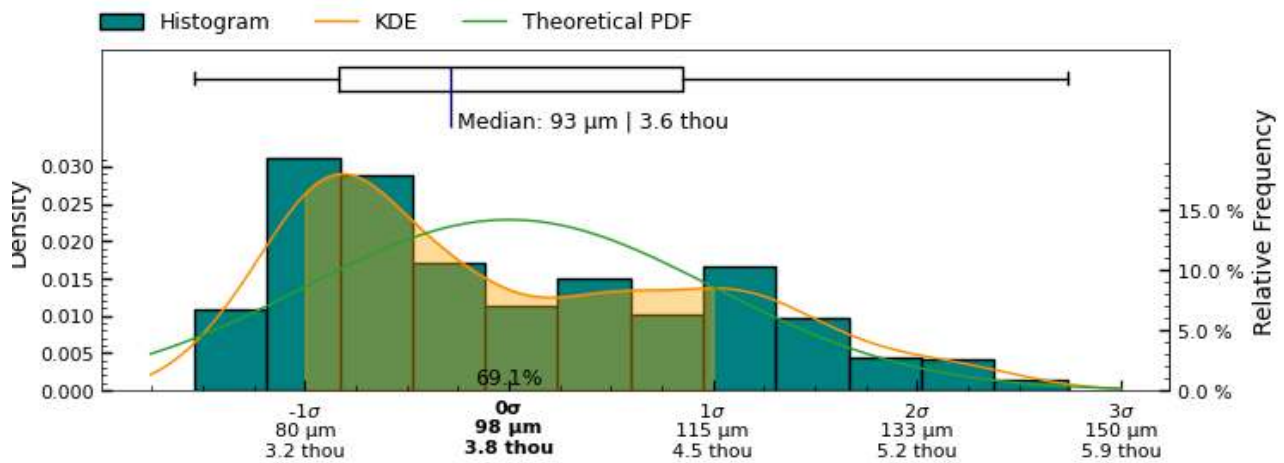


Figure 26: Standard Deviation measurement distribution across measured slices of " + interior_separate + " surface

Root Mean Squared Deviation measurement distribution across 760 slices of interior separately aligned surface

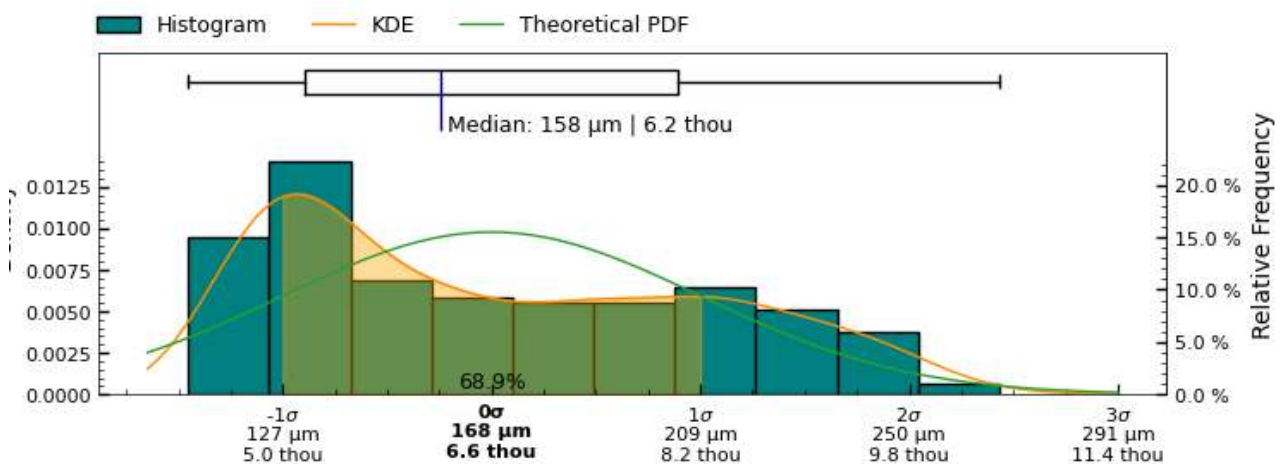


Figure 27: Root Mean Squared Deviation measurement distribution across measured slices of interior separately aligned surface

Concentricity

The concentricity metric describes the deviation in the center-point of the referenced features. As such, it is a measure to determine if several features of the object share the same center point/axis, and how closely. See Figure 28 for a visual representation of this metric.

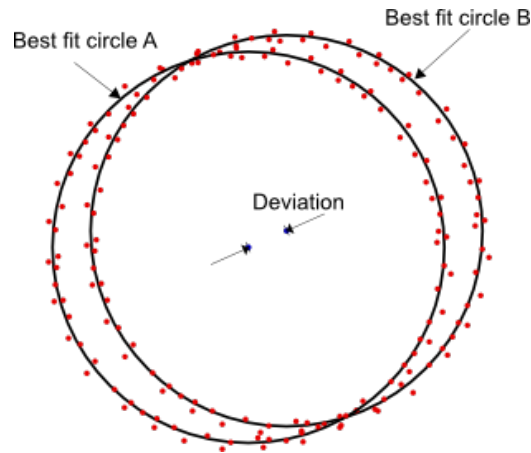


Figure 28: Concentricity measures the deviation (distance) between the center of two circles.

Determination of concentricity has been carried out by establishing the best fit circles of sample slices, using RANSAC (Random sample consensus) algorithm for outlier detection of a least squares circle regression on the scanned data-points at each cross-section, to estimate centers of each cross-section.

The concentricity between both the interior and exterior circular cross-sections is explored for cross-section measurements with the same Z-coordinates.

Additionally, the concentricity between each cross-section measurement defined in Figure 4 and the datum axis $(x, y) = (0, 0)$ has been calculated to establish the deviation of the feature center from the datum axis.

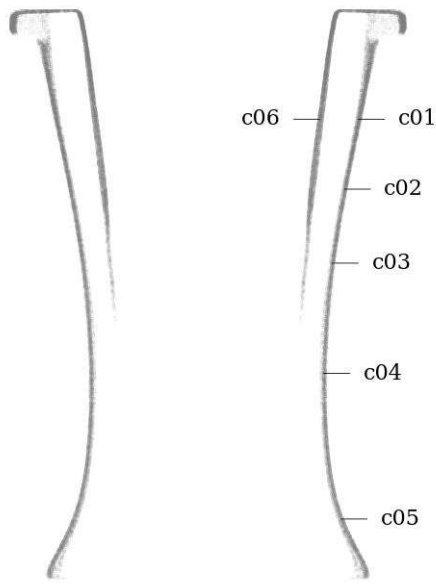


Figure 29: Circularity measurement sample locations, full mesh aligned to exterior surface

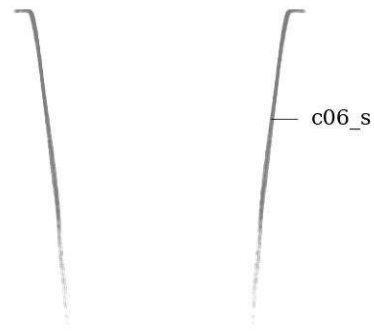


Figure 30: Circularity measurement sample location, separately aligned interior mesh

Metric

Tag	Reference	Deviation	Sample size	Circle fit residuals analysis for sample listed in Tag column						
				Range full	Range inliers	RMSD full	RMDS inliers	SD full	SD inliers	Center (x,y)
		mm		mm	mm	mm	mm	mm	mm	μm
c01	z-axis	0.446	2969	1.773	1.773	0.395	0.395	0.233	0.233	−414, 164
c02	z-axis	0.233	2256	1.385	1.385	0.343	0.331	0.189	0.182	−189, 136
c03	z-axis	0.139	2056	1.723	1.589	0.371	0.331	0.234	0.200	137, 19
c04	z-axis	0.254	1841	1.873	1.873	0.409	0.409	0.284	0.284	233, −103
c05	z-axis	0.073	2032	1.802	1.802	0.481	0.481	0.263	0.263	37, −62
c06	z-axis	0.619	2987	3.171	3.001	0.966	0.964	0.389	0.390	121, −607
c06_s	z-axis	0.250	2969	1.635	1.519	0.429	0.428	0.182	0.179	−31, −248
c01	c06_s	0.563								−384, 412

Imperial

Tag	Reference	Deviation	Sample size	Circle fit residuals analysis for sample listed in Tag column						
				Range full	Range inliers	RMSD full	RMDS inliers	SD full	SD inliers	Center (x,y)
		in		in	in	in	in	in	in	thou
c01	z-axis	0.0175	2969	0.0698	0.0698	0.0156	0.0156	0.0092	0.0092	−16.3, 6.4
c02	z-axis	0.0092	2256	0.0545	0.0545	0.0135	0.0130	0.0074	0.0072	−7.4, 5.3
c03	z-axis	0.0055	2056	0.0678	0.0626	0.0146	0.0130	0.0092	0.0079	5.4, 0.7
c04	z-axis	0.0100	1841	0.0738	0.0738	0.0161	0.0161	0.0112	0.0112	9.2, −4.1
c05	z-axis	0.0029	2032	0.0710	0.0710	0.0189	0.0189	0.0103	0.0103	1.5, −2.5
c06	z-axis	0.0244	2987	0.1249	0.1182	0.0380	0.0380	0.0153	0.0153	4.8, −23.9
c06_s	z-axis	0.0098	2969	0.0644	0.0598	0.0169	0.0168	0.0072	0.0071	−1.2, −9.8
c01	c06_s	0.0222								−15.1, 16.2

Table 3: Concentricity analysis of MV012.

Concentricity analysis of c01

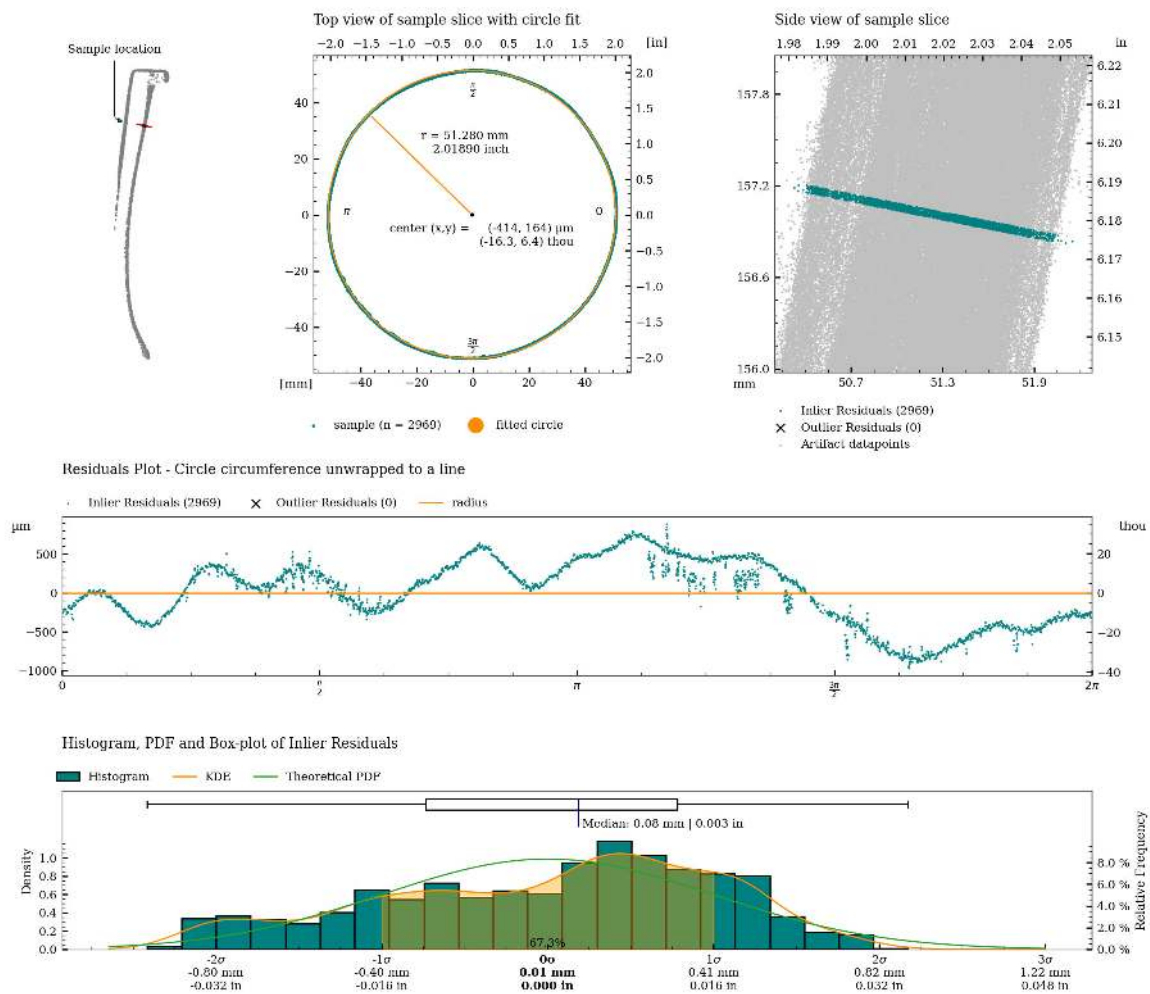


Figure 31: Detailed plot of concentricity measurement for c01.

Concentricity analysis of c02

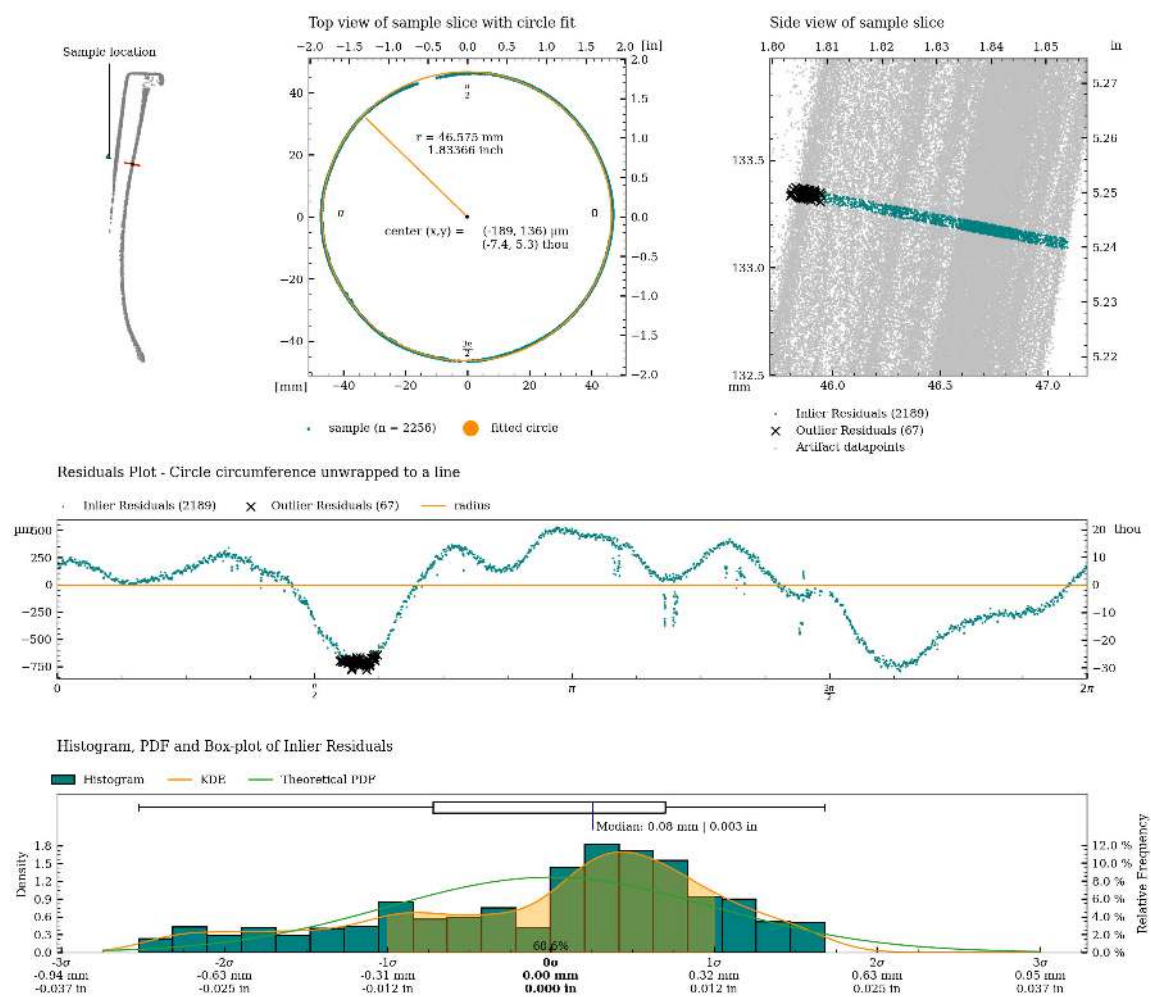


Figure 32: Detailed plot of concentricity measurement for c02.

Concentricity analysis of c03

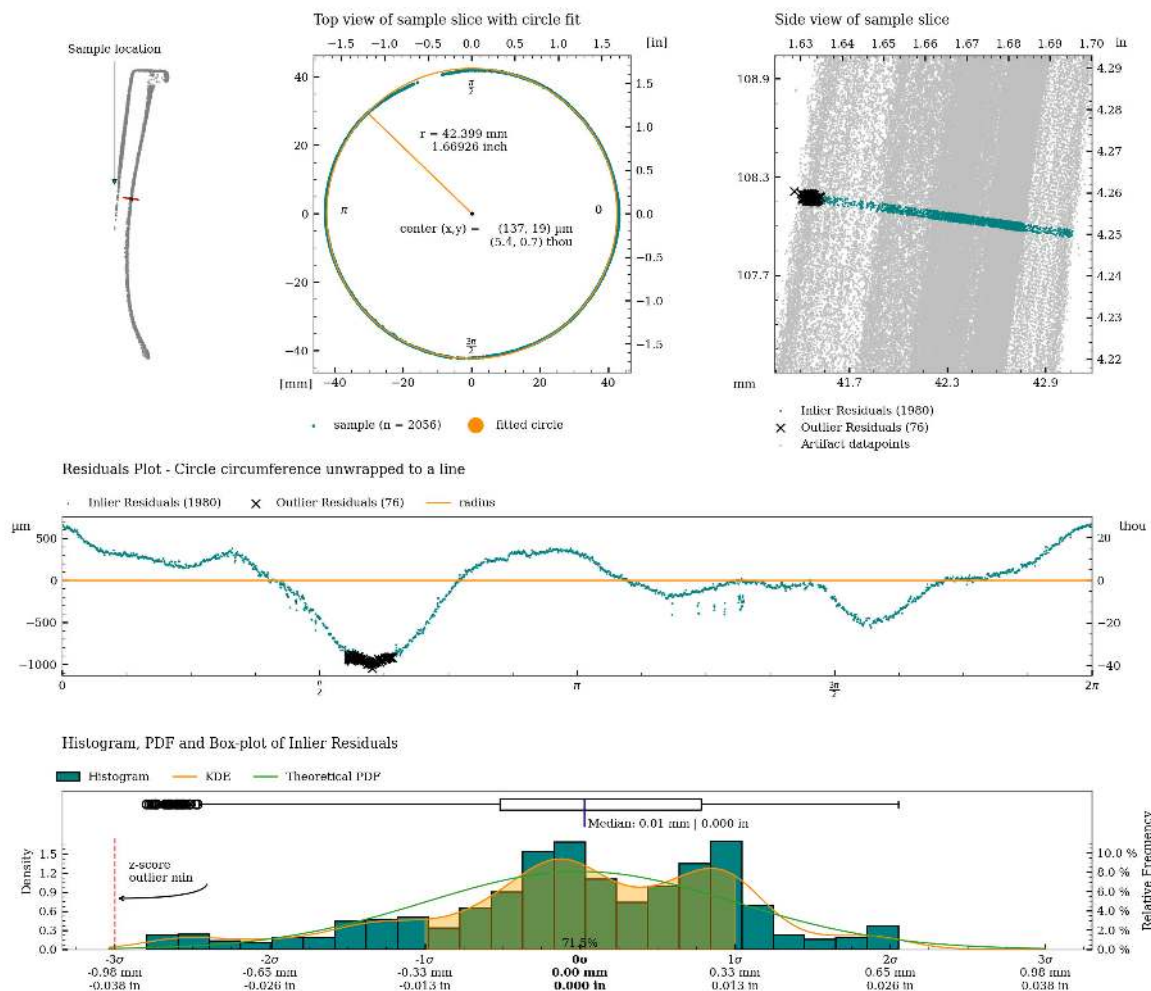


Figure 33: Detailed plot of concentricity measurement for c03.

Concentricity analysis of c04

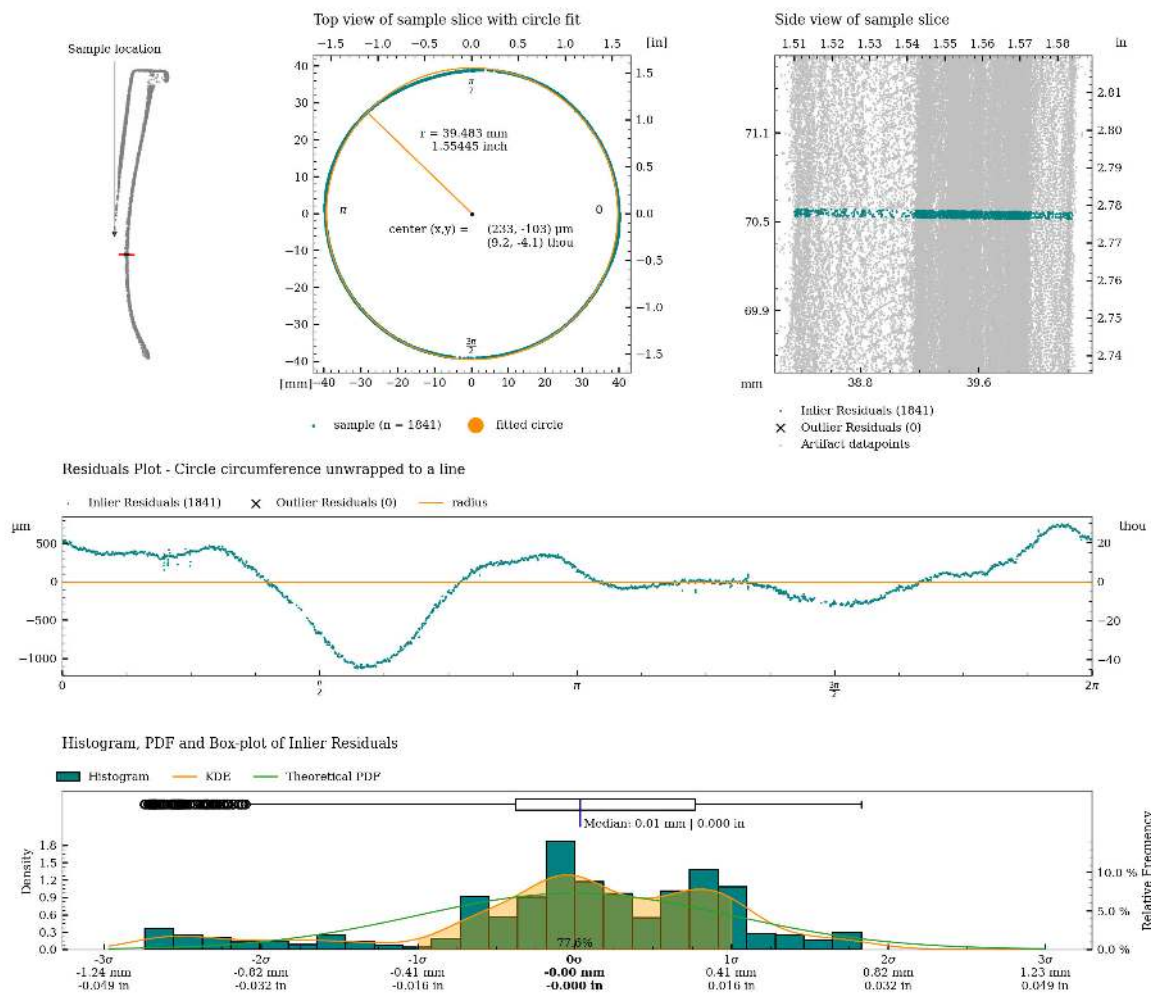


Figure 34: Detailed plot of concentricity measurement for c04.

Concentricity analysis of c05

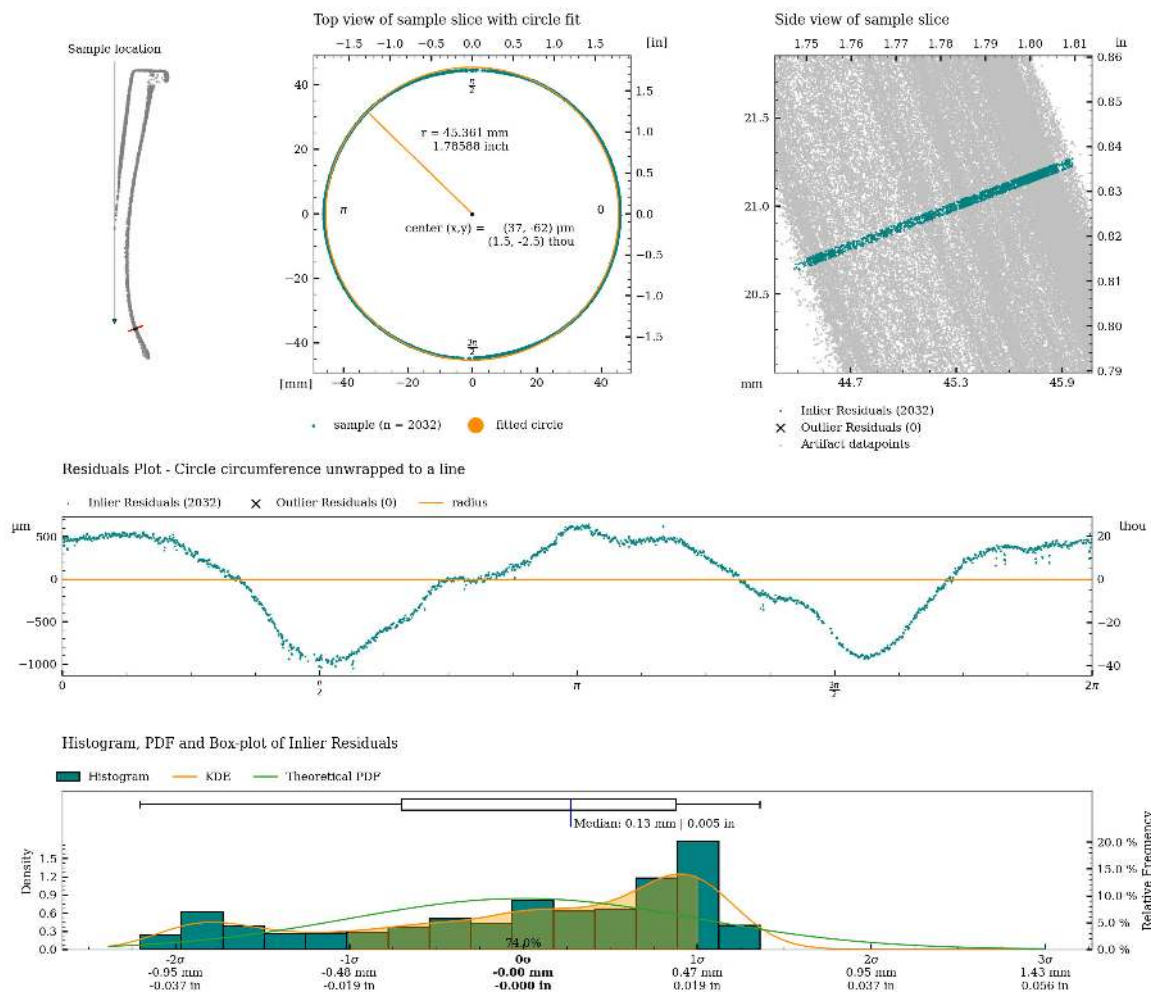


Figure 35: Detailed plot of concentricity measurement for c05.

Concentricity analysis of c06

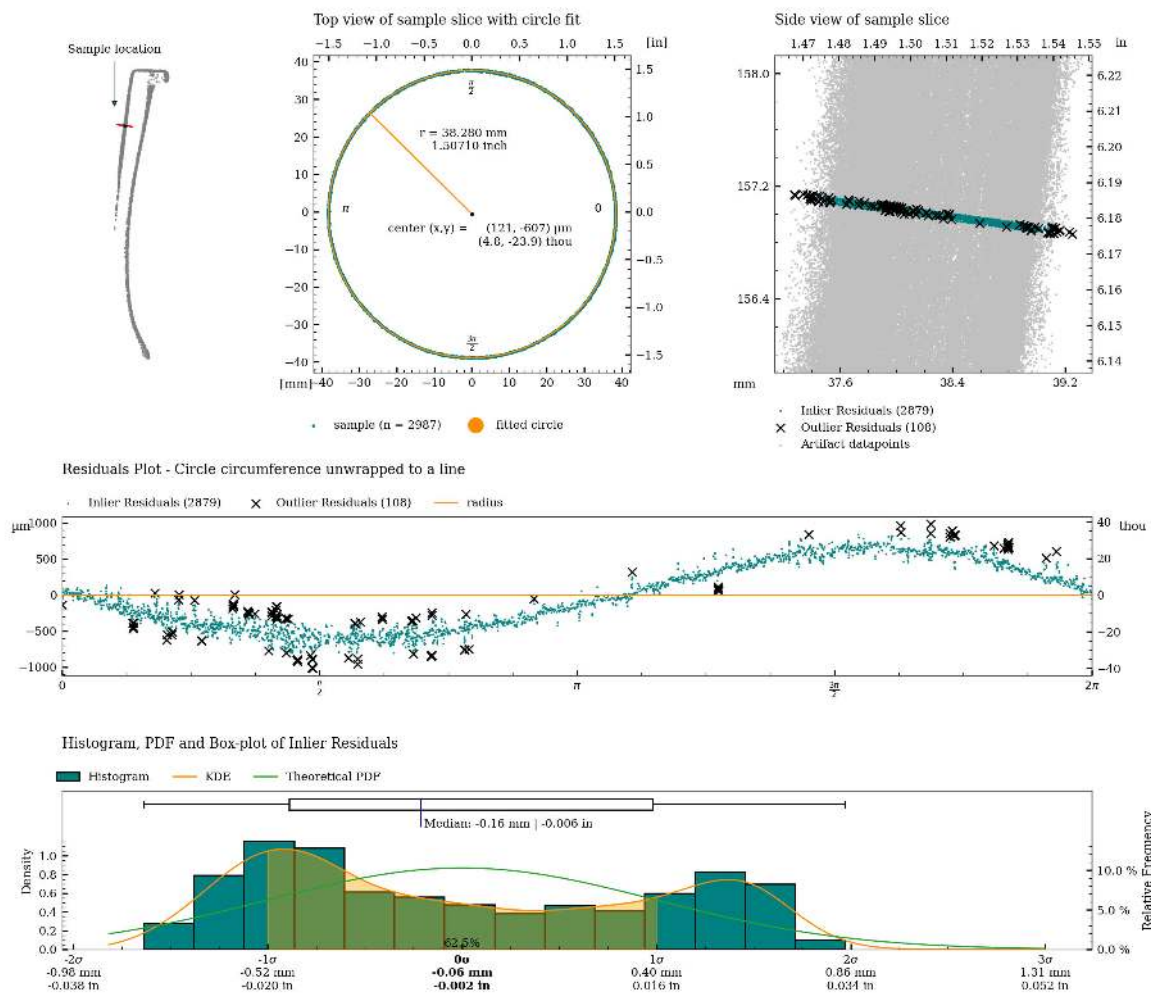


Figure 36: Detailed plot of concentricity measurement for c06.

Concentricity analysis of c06_s

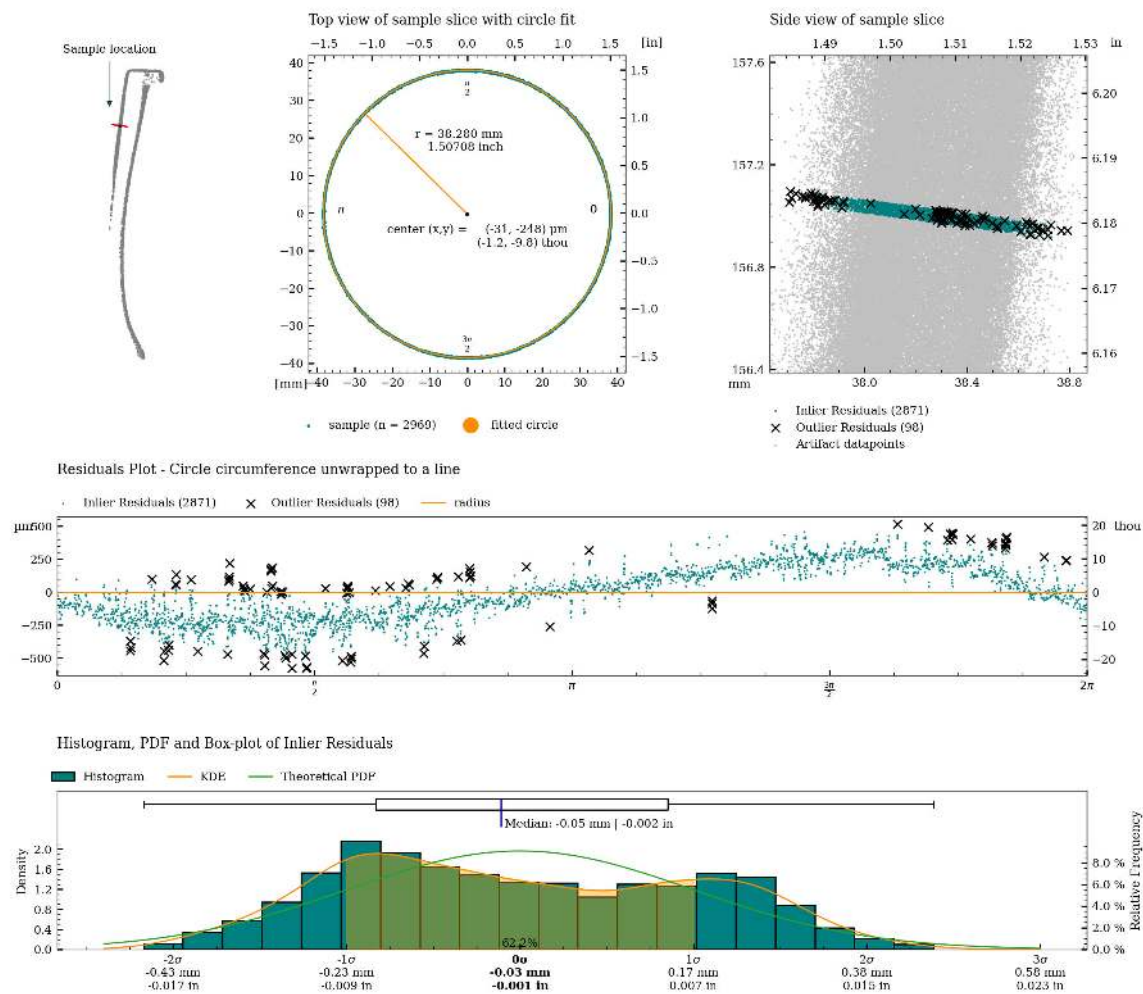


Figure 37: Detailed plot of concentricity measurement for c06_s.

Coaxiality

Coaxiality refers to the straightness and consistency of a central line running through the center of the vase. It measures how aligned the core of the vase remains along its vertical axis.

The coaxiality measurements are calculated using RANSAC (Random sample consensus) algorithm for outlier detection on least squares circle regression on cross-sections of the vessel (excluding potential handles), to estimate the best fit circle centers for each slice of the vessel. A best-fit line connects these centers, showing whether the vessels's shape twists or remains straight. This concept helps describe the symmetry and structural uniformity in a visual and analytical way.

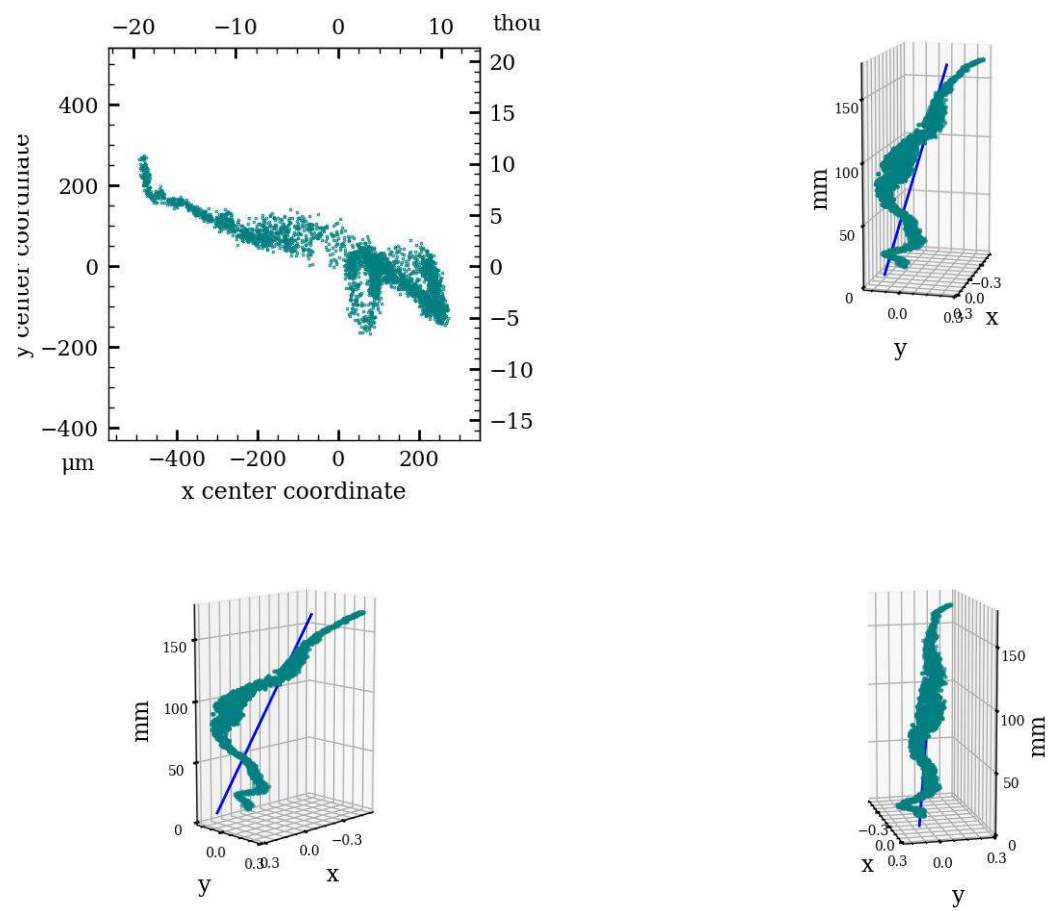
Coaxiality is measured for:

- The exterior surface (excluding handles)
- The interior surface

	Exterior		Interior		Interior separate	
Analyzed Slices	3151		760		760	
Median sample size	2001		3325		3324	
Slice Height	50 μm	2.0 thou	50 μm	2.0 thou	50 μm	2.0 thou
Statistics with Z-axis as Reference						
Median Absolute Deviation (MAD)	198 μm	7.8 thou	517 μm	20.4 thou	151 μm	5.9 thou
Standard Deviation (SD)	116 μm	4.6 thou	105 μm	4.1 thou	86 μm	3.4 thou
Root Mean Square Deviation (RMSD)	234 μm	9.2 thou	516 μm	20.3 thou	173 μm	6.8 thou
Statistics with Best Fit Central Axis as Reference						
Best fit Central Axis Equation (in metric coordinate system with unit [mm])	x = 0.304 + t-0.00323		x = -0.447 + t0.00361		x = -0.402 + t0.00234	
	y = -0.109 + t0.00138		y = -2.488 + t0.01188		y = -2.493 + t0.01426	
	z = 0.000 + t0.99999		z = 0.000 + t0.99992		z = 0.000 + t0.99990	
Axis tilt	-0.185°		0.205°		0.132°	
Median Absolute Deviation (MAD)	163 μm	6.4 thou	15 μm	0.6 thou	15 μm	0.6 thou
Standard Deviation (SD)	77 μm	3.0 thou	9 μm	0.3 thou	9 μm	0.3 thou
Root Mean Square Deviation (RMSD)	169 μm	6.7 thou	19 μm	0.7 thou	18 μm	0.7 thou

Table 4: Coaxiality analysis of vessel MV012.

Coaxiality plots, exterior surface



Coaxiality residuals from fitted axis, exterior surface

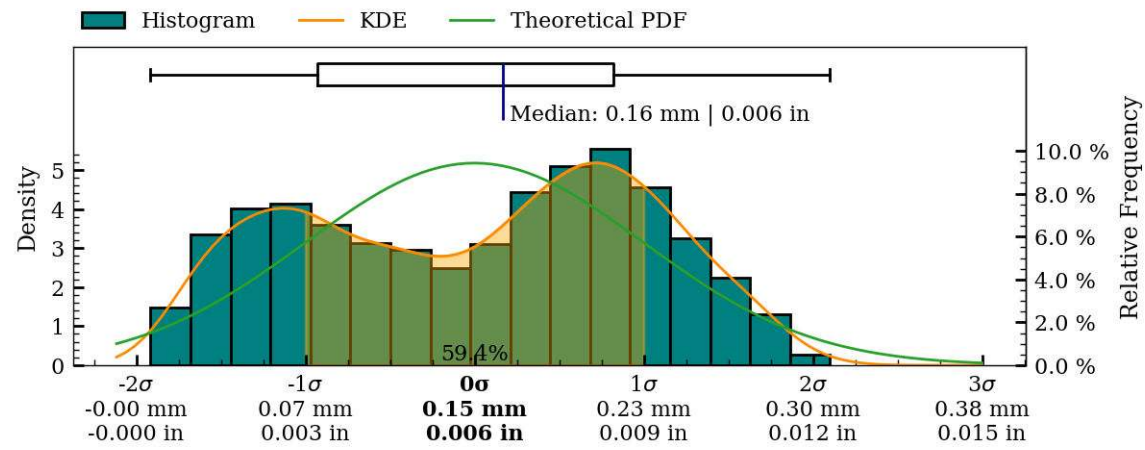
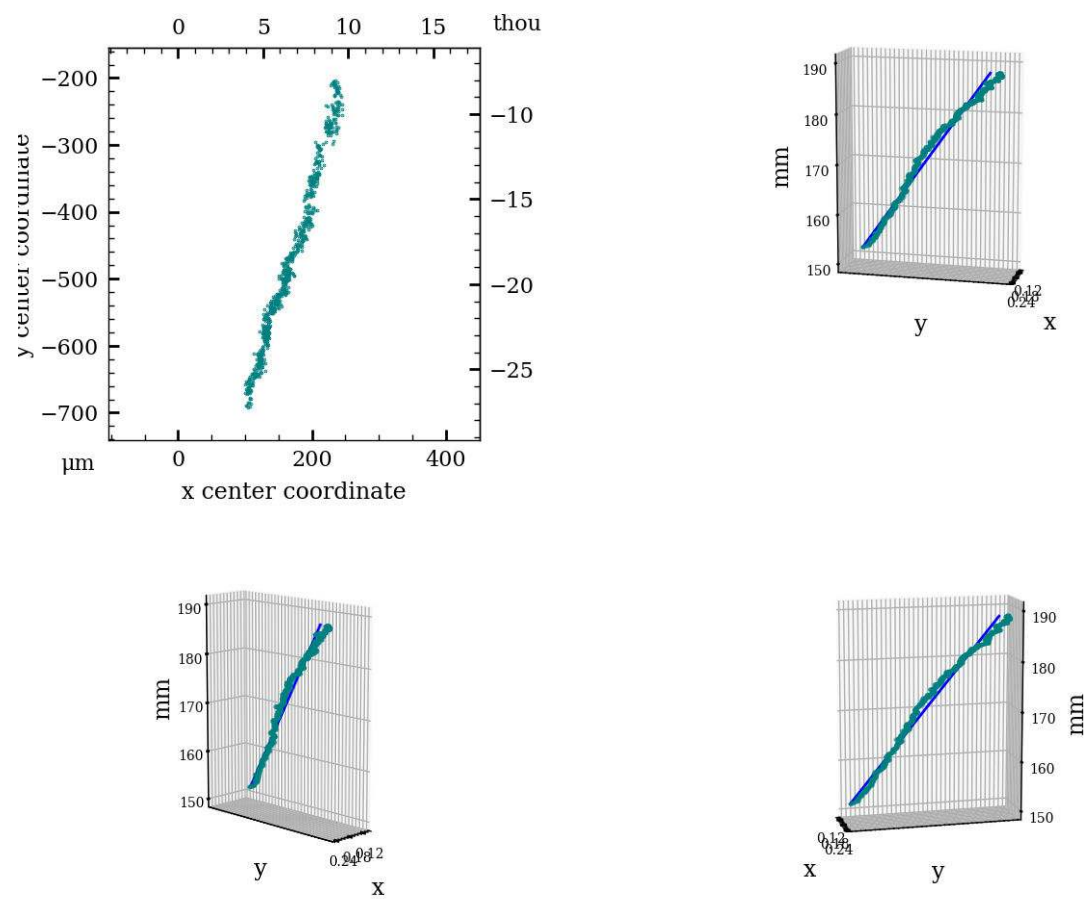


Figure 38: Coaxiality residual plots of exterior surface, MV012.

Coaxiality plots, interior surface



Coaxiality residuals from fitted axis, interior surface

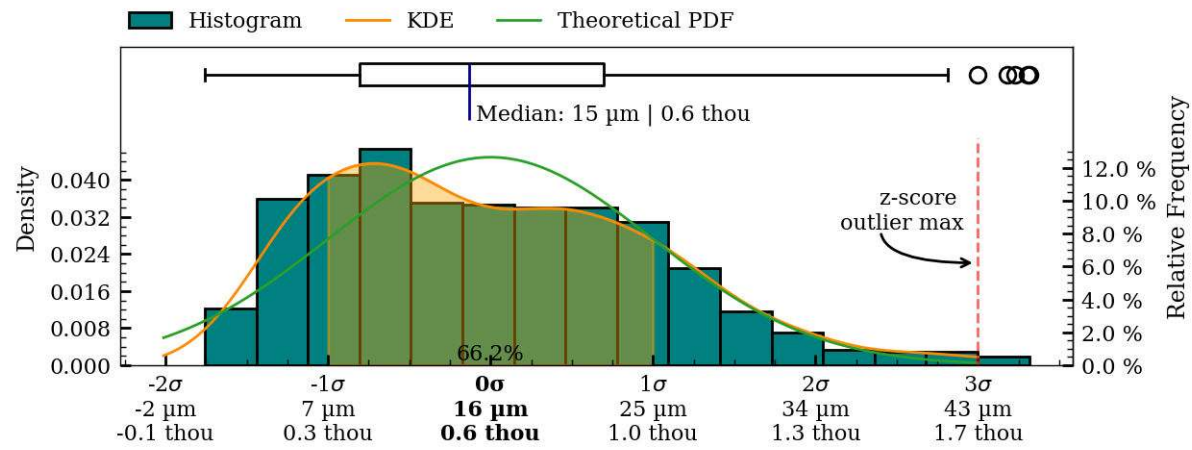
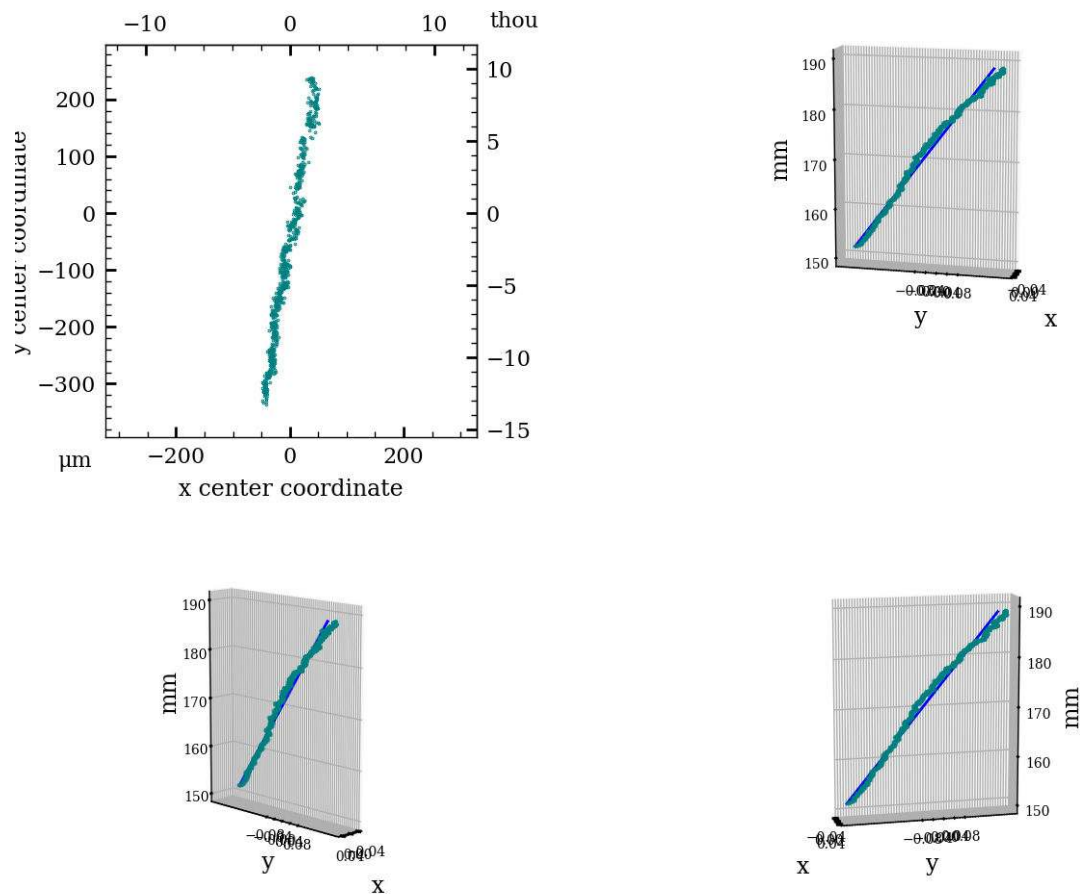


Figure 39: Coaxiality residual plots of interior surface, MV012.

Coaxiality plots, interior separately aligned surface



Coaxiality residuals from fitted axis, interior separately aligned surface

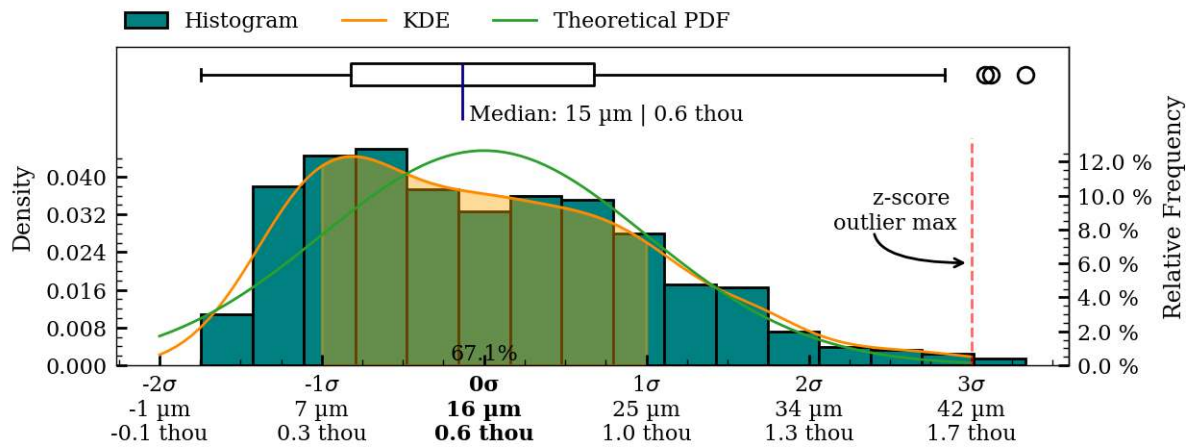


Figure 40: Coaxiality residual plots of interior_separate surface, MV012.

Surface Variability

To illustrate the overall surface deviations of the object, a surface variability heatmap has been created. This heatmap provides an accessible overview of the topography of the manufacturing precision and surface structure of the object.

The raw scan files from FreeScan Combo+ often produce geometry artifacts on the surface of the scanned object. If present these will be clearly seen, as falling outside of the heatmap range, and as such being uncolored. These geometry artifact, if present, will and increase the total range of the surface deviation statistics.

The surface variability measurements are created by fitting a number of higher-order polynomials to the two-dimensional folded profile of the scan data. This process creates an idealized mathematical representation of actual surface curvature of object, and as such provides a continuous model representation of the actual object. It is important to note that only such a non-discretized representation is sufficient to avoid introducing inconsistently varying errors in the mapping of the final surface deviation results, that the rendered heatmaps are based on.

To produce the final surface variability map, the distance from each scanned vertex to the fitted polynomial is calculated and used as the mapping function input, for applying colours to the surface of the object.

It is important to note that this variability map does not describe deviations from the original *intended* shape of the artifact (if any), as this shape (the *intended design*, so to speak) will have been lost to time. It does however provide a very informative visualization of the texture and structure of the surface and very importantly, *does* highlight potential manufacturing-relevant patterns in the surface texture (if present). Such patterns are, as an example, clearly evident on the interior surface of artifact PV001.

Exterior surface

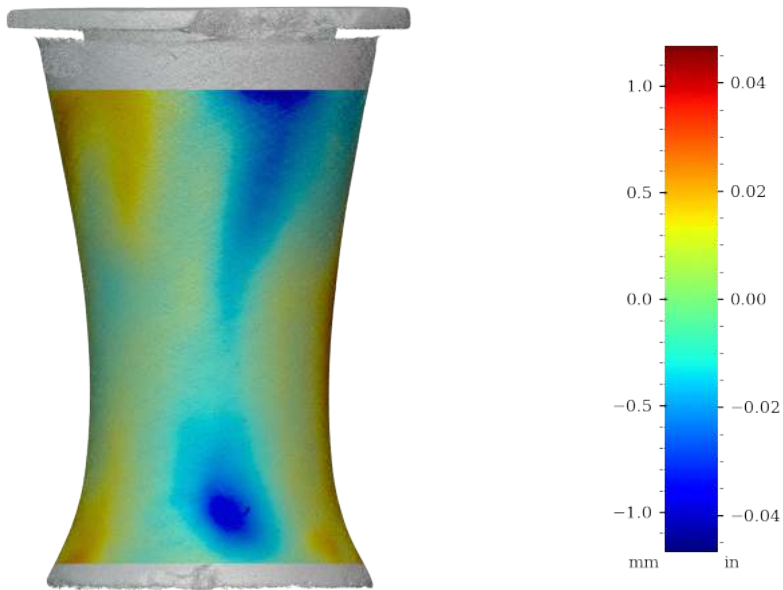


Figure 41: Surface variability heatmap of MV012, front view

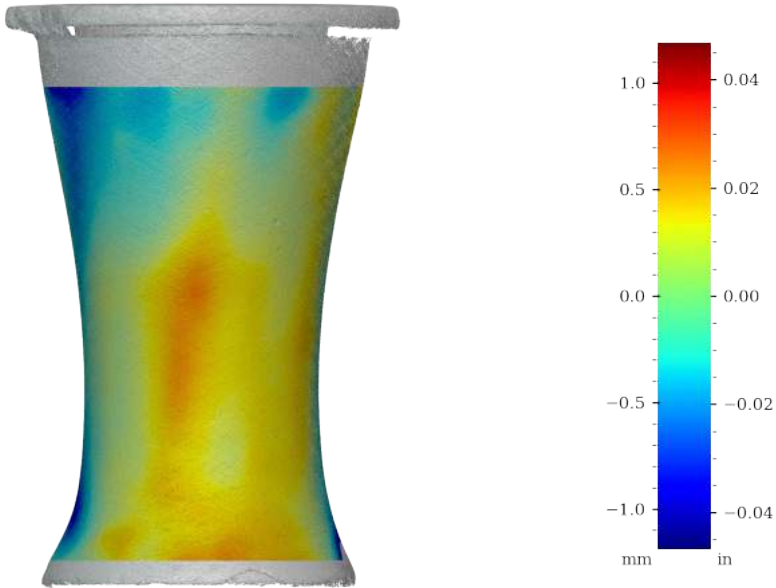


Figure 42: Surface variability heatmap of MV012, rotated 90°

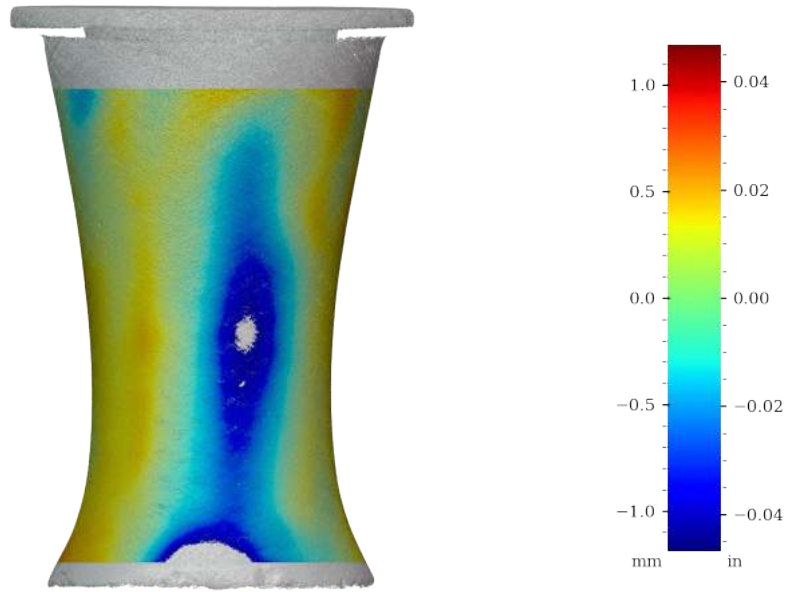


Figure 43: Surface variability heatmap of MV012, rotated 180°

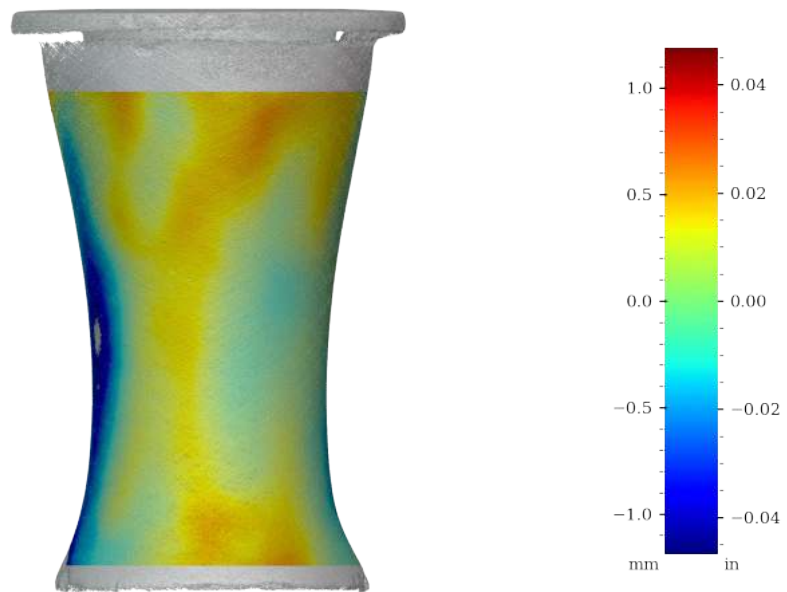


Figure 44: Surface variability heatmap of MV012, rotated 270°

Interior surface

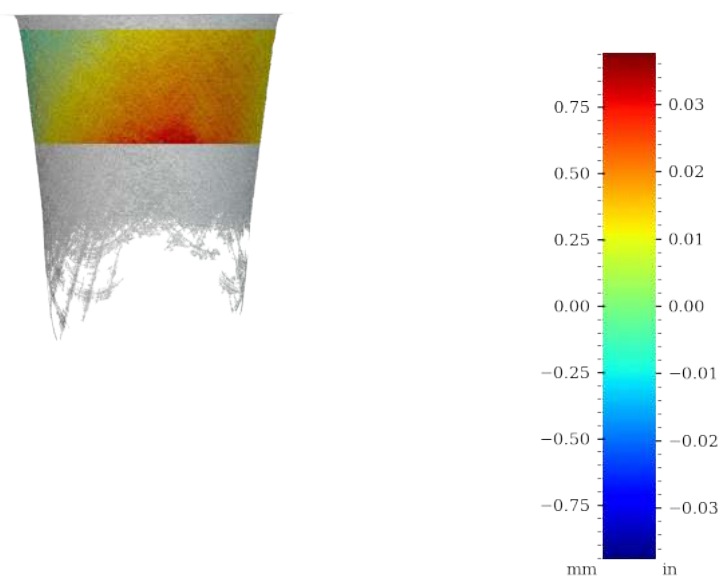


Figure 45: Surface variability heatmap of MV012, front view

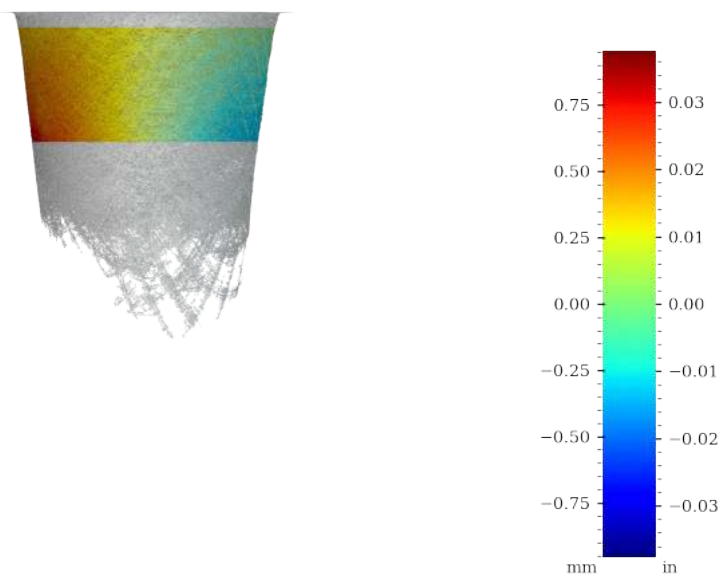


Figure 46: Surface variability heatmap of MV012, rotated 90°

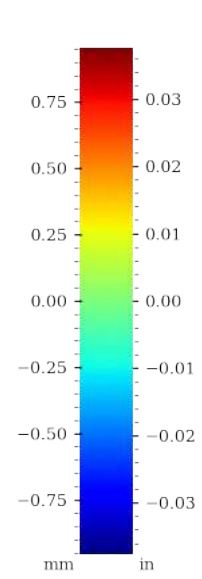


Figure 47: Surface variability heatmap of MV012, rotated 180°

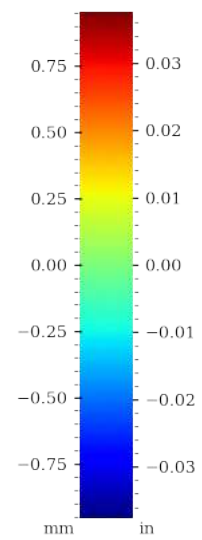


Figure 48: Surface variability heatmap of MV012, rotated 270°

Interior surface aligned separately

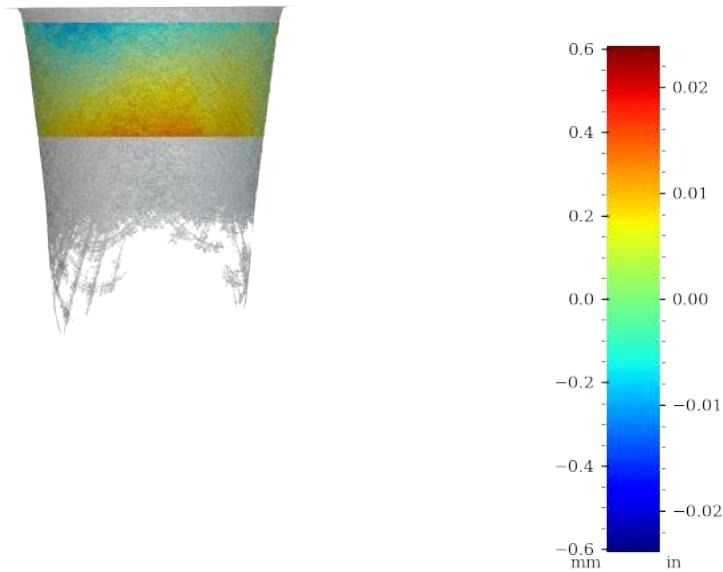


Figure 49: Surface variability heatmap of MV012, front view

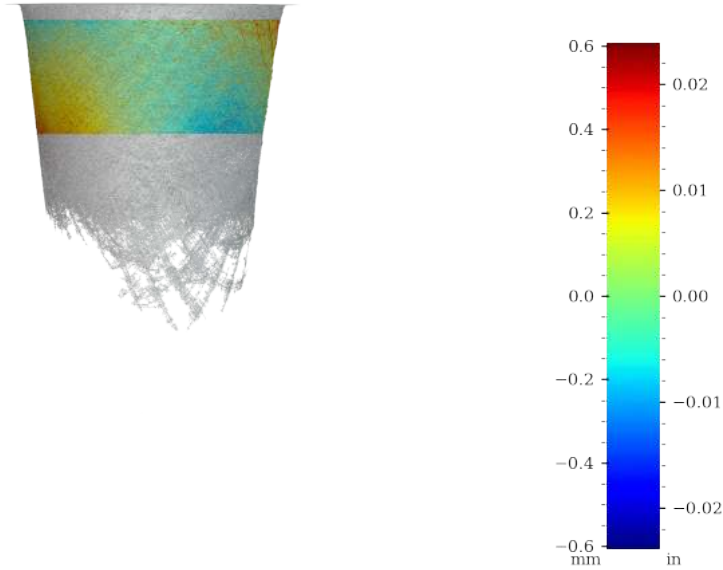


Figure 50: Surface variability heatmap of MV012, rotated 90°

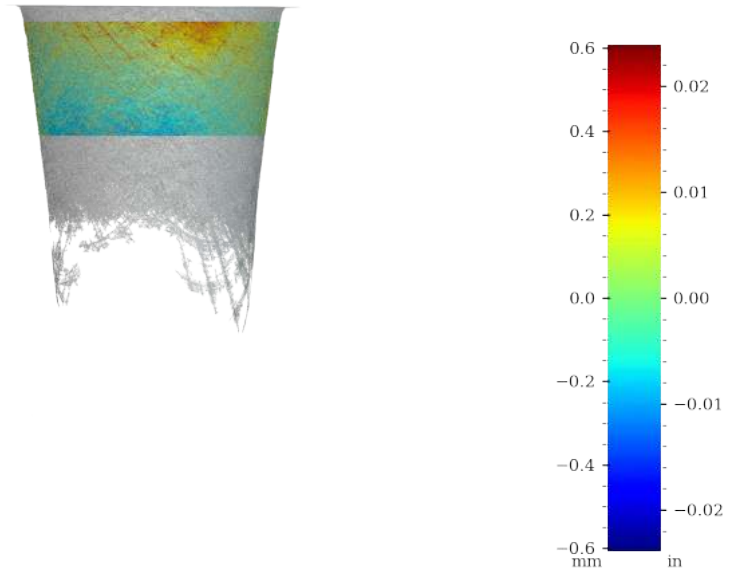


Figure 51: Surface variability heatmap of MV012, rotated 180°

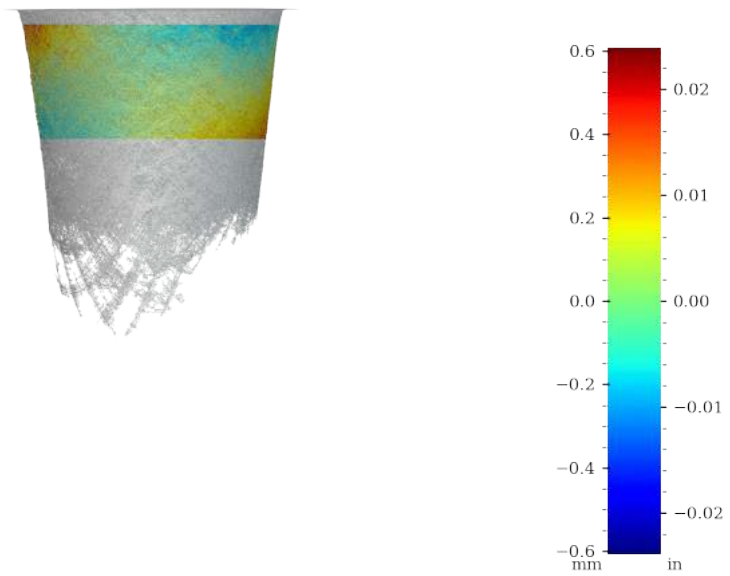


Figure 52: Surface variability heatmap of MV012, rotated 270°

Surface variability statistics

Area	MSD	RMSD	SD	Median AD	Range	Min	Max	Sample size
	mm ²	mm	mm	mm	mm	mm	mm	
Exterior	0.1682	0.410	0.245	0.154	2.810	-1.737	1.073	7010564
Interior	0.1453	0.381	0.196	0.150	2.306	-1.149	1.156	2656349
Interior separate	0.0299	0.173	0.107	0.068	1.707	-0.823	0.884	2657943
	in ²	in	in	in	in	in	in	
Exterior	0.000261	0.0161	0.0096	0.0061	0.1106	-0.0684	0.0422	7010564
Interior	0.000225	0.0150	0.0077	0.0059	0.0908	-0.0453	0.0455	2656349
Interior separate	0.000046	0.0068	0.0042	0.0027	0.0672	-0.0324	0.0348	2657943

Table 5: Surface variability statistics, MV012

Table 5 shows the statistics of the distance from the scan vertices to the best fit object model. These statistics are briefly explained below.

Histogram, KDE and Box-plot of measured surface variability - exterior surface

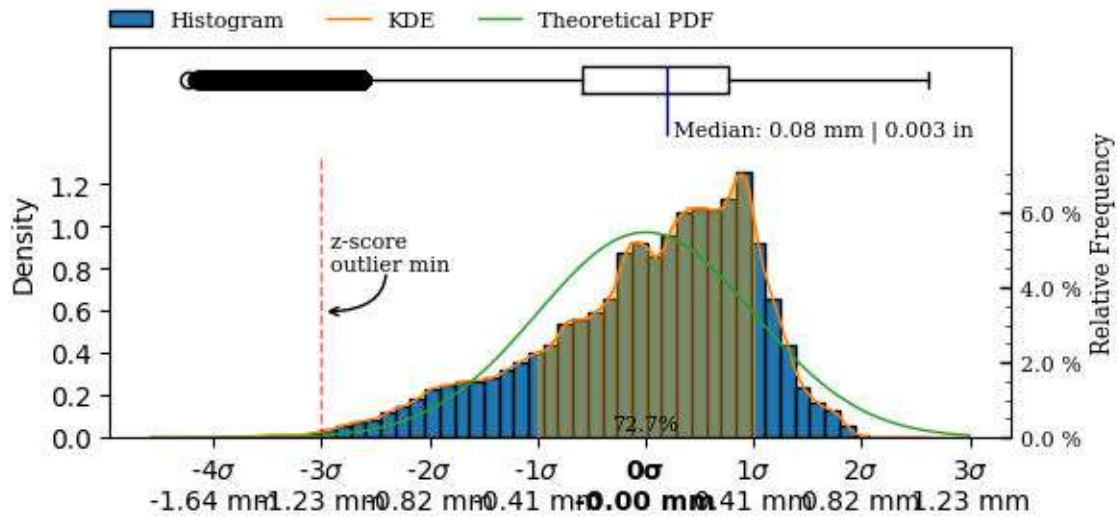


Figure 53: Exterior surface variability boxplot, kds and histogram.

Histogram, KDE and Box-plot of measured surface variability - interior surface

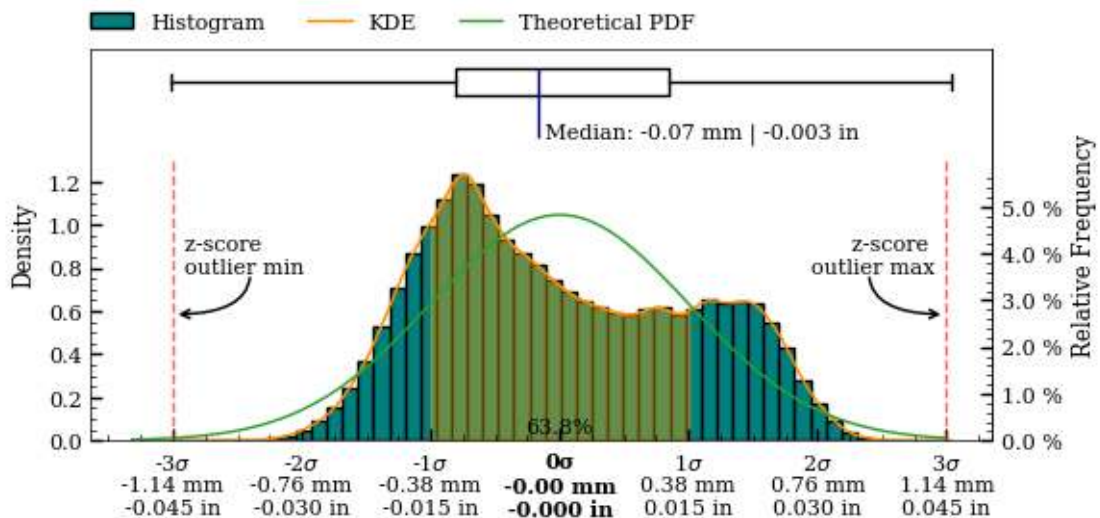


Figure 54: Interior surface variability boxplot, kds and histogram.

Histogram, KDE and Box-plot of measured surface variability - interior separately aligned surface

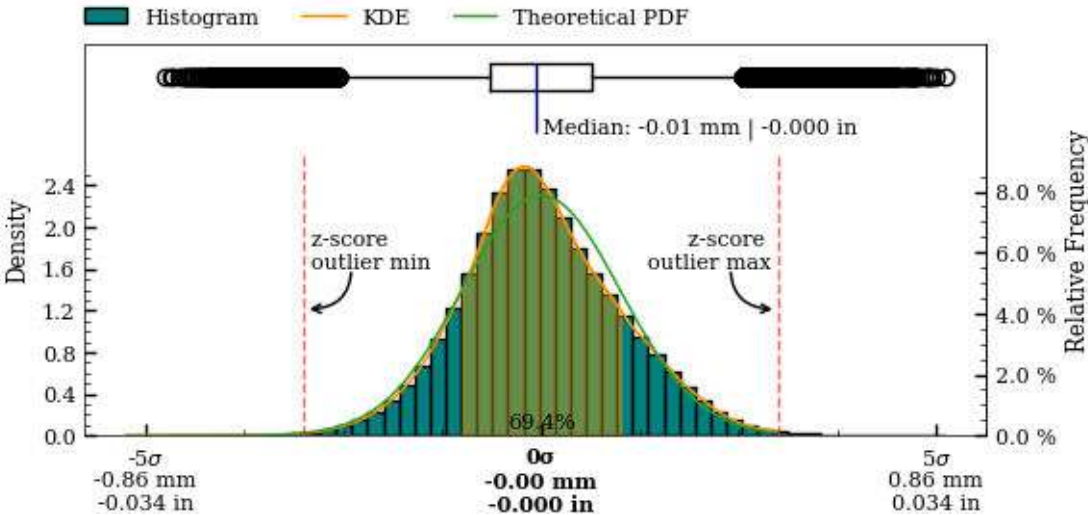


Figure 55: Interior separately aligned surface variability boxplot, kds and histogram.

Precision Score Of The Artifact

To enable valid comparison of the manufacturing precision of different artifacts, a metric that robustly quantifies the overall precision of the object is required. The considerations for such a metric will be explored in this section.

Based on these considerations, a *Precision Score* metric will be defined.

For an object to be described as having been manufactured with high precision, several qualities must be present *concurrently*, and throughout the *entire* geometry of the final object. A given object may exhibit high levels of one or more *components* of precision, but be lacking in others. For example:

- An object may present high levels of coaxiality, but lack circularity.
- An object may exhibit good circularity, but show imperfections in the surface structure.
- An object may be smoothed to perfection *without* any circularity or coaxiality.
- An object may exhibit high levels of all of the above metrics in *some* areas, but not in others.

Therefore, a precision score metric **must** account for *all* aspects of the individual, underlying precision metrics (circularity, concentricity, coaxiality and surface variability) throughout the *entire* surface area of the object.

The composite high order polynomial model, used to generate the surface variability map (described in Surface Variability, p. 40) is the best continuous mathematical representation of the object available to us (lacking any original design plans, as would normally be available in metrological analysis). This idealized model encompasses all of the above component metrics.

In the creation of the model, all scan data-points are taken into account (excluding areas with extensive damage), making it the best possible idealized representation we can achieve. When this model has been accurately created, the deviation between the model and the scanned data-points can be calculated over the non-discretized polynomials, *without* the need for an “original” CAD model (and importantly, unless such a CAD model *actually* corresponded to the original design intent, it would be an insufficient comparison basis).

Within the context of defining a valid, overall precision metric, this approach satisfies the incorporation of all of the necessary metrics:

- **Circularity:** Because the reconstructed polynomial model is revolved around the Z-plane, the idealized representation is perfectly circular, and thus incorporates the circularity component.
- **Concentricity and coaxiality:** Because the Z-axis (datum axis) is the center axis of the model, it incorporates the concentricity and coaxiality components.
- **Surface variability:** Because the model is continuous and non-discretized, it can be used accurately for all points of the scan data, and incorporates the surface variability component.

The level of precision ultimately achieved in a physical object does not share a linear relationship with its manufacturing requirements. Since continuously higher levels of final precision becomes progressively harder to achieve, an overall precision metric must take this relationship into account.

A robust statistical metric that satisfies this requirement is the *Mean Squared Deviation* (MSD or MSE). Here specifically, we can utilize the mean square of the deviations between the model ($\hat{y}$) and the data-points (y_i).

Combining all of the above considerations, we can express a well-defined *Precision Score* metric, that provides an immediately accessible way to understand the overall precision of an object, while being statistically valid. Since the Mean Squared Deviation tends towards zero as the overall precision increases, the inverse of the Mean Squared Deviation is taken to obtain a precision score metric that increases as precision increases¹³:

$$\text{Precision Score} = \frac{n}{\sum_{i=1}^n (y_i - \hat{y})^2}$$

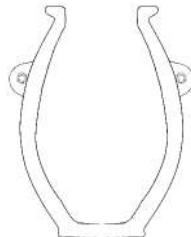
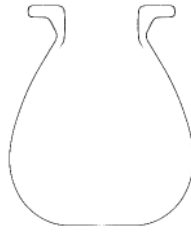
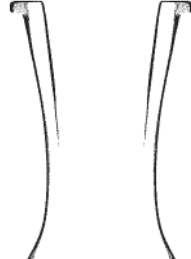
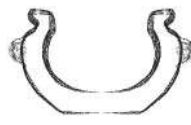
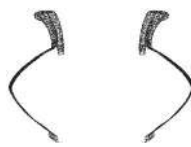
¹³The precision score unit is $\frac{1}{\text{mm}^2}$

The precision score of MV012 have been calculated separately for:

- Precision score, exterior surface: 6
- Precision score, separately aligned interior surface: 33
- Precision score, interior surface: 7
- Precision score, full surface: 8

The precision score of a Zeiss 1.00000 inch reference sphere have been calculated to 43,943 (RMSE = 0.00477 mm / 0.00010 in). The scan was obtained by Max Fomitchev-Zamilov using a Keyence VL –500 scanner with a rated accuracy of 10 microns. The precision analysis of the reference sphere scan indicates at the maximum possible precision score obtainable.

Table 6 shows the precision score of this artifact (MV012), compared to the two most precise, and the two least precise vessels currently analyzed.

Artifact		Material	Precision Score	Link to Report
		PV001 Red Granite	1980 Full: 1177 Exterior: 1980 Interior separate: 798 Interior: 722	Report Publication
		PV006 Dark grey granite	621 Full: 610 Exterior: 621 Interior separate: 479 Interior: 152	Report Publication
		MV012 Basalt	6 Full: 8 Exterior: 6 Interior separate: 33 Interior: 7	Report Publication
		RV003 Marble breccia	1.46 Full: 1.49 Exterior: 1.46 Interior separate: 1.53 Interior: 0.54	Report Publication
		MV010 Calcite (Egyptian Al-abaster)	1.17 Full: 1.32 Exterior: 1.17 Interior separate: 11 Interior: 0.17	Report Publication

Analysis Roadmap

While the current iteration of this work already provides valuable results, continued future additions and improvements will enhance their utility further. This section details planned iterative updates and improvements, to both the reports themselves, and to the underlying methodology and software they are created with.

Alignment Section

- Detailed exploration of different circle regression algorithms
- If handles are present on the vessel, exploring alignment of the vessels so the handle positions match each other
- Add optimization of the perpendicular surface deviation, with the best results of the coaxial alignment
- Align by minimizing circularity results (of rotated sample slice, to compensate for sample height distortions)

Measurements of Precision

- Section detailing how measurements perpendicular to the surface curvature are obtained
- Detailed surface area analysis, exploring the residual patterns throughout subsequent sample slices of the artifact surface
- Wall thickness deviation color map
- Robust outlier identification on circularity, to better handle analysis of damaged areas of the artifacts in addition to removal of interior crystalline structure points present in CT scans
- Layout updates to the charts and tables

Visibility of Outliers and Damaged Sections

- Identification and marking of damaged parts
- Visualization of outliers on the artifact surface

Exploration of Mathematical Primitives

- Analysis of selected curvatures and flat surfaces on the vessel in both the horizontal and vertical planes
 - Circles
 - Parabolas
 - Ellipsoids
 - Hyperbolas
 - Cones
- Implementation of robust regressions models suitable for this domain, based on RANSAC.

Metrics on Primary Features

- Measurements of features in the horizontal plane
- Measurements of features in the vertical plane
- Measurements of angles
- Measurements of volume

Exploration of Potential Design Ratios

- π , φ , e , 1, 2, 3, 4 etc.

Raw Dataset Attachments

- Including all measurement and sample coordinates as CSV-files embedded in the report
- Including an STL file of the aligned object alongside the report, for easier external replication and validation of the research results

Appendix A - Comparison Of Circularity Measurements (Z-plane vs. surface-perpendicular)

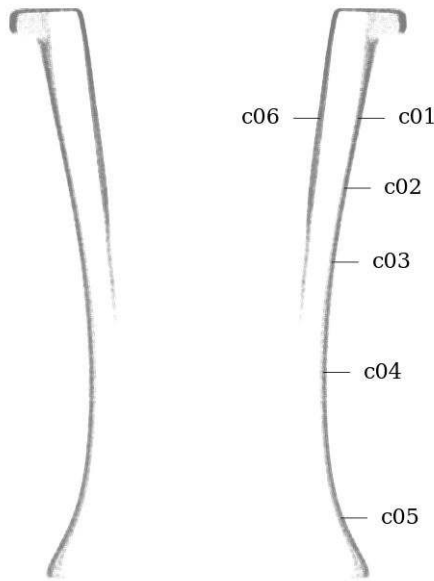


Figure 56: Circularity measurement sample locations, full mesh aligned to exterior surface

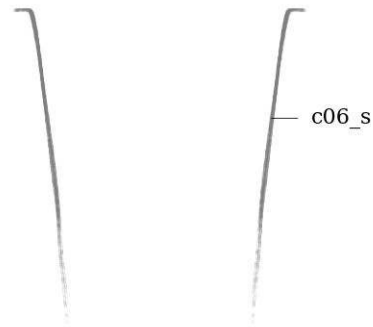


Figure 57: Circularity measurement sample location, separately aligned interior mesh

Samples perpendicular to the surface curvature

Tag	Area	Measured deviation ⁹	Residuals				Sample size	Slice		
			Range	RMSD ¹⁰	MAD ¹¹	SD		Height	Z coord.	Radius ¹²
		mm	mm	mm	mm	mm		mm	mm	mm
c01	exterior	Ø102.587±0.982	1.856	0.404	0.157	0.221	2969	0.050	157.004	51.293
c02	exterior	Ø93.084±0.762	1.324	0.333	0.123	0.182	2256	0.050	133.214	46.542
c03	exterior	Ø84.726±1.012	1.717	0.366	0.148	0.234	2056	0.050	108.050	42.363
c04	exterior	Ø78.956±1.124	1.879	0.411	0.162	0.276	1841	0.050	70.549	39.478
c05	exterior	Ø90.715±1.045	1.695	0.476	0.157	0.252	2032	0.050	21.022	45.357
c06	interior	Ø76.449±1.039	1.999	0.463	0.159	0.211	2987	0.050	157.004	38.224
c06_s	interior sep.	Ø76.508±0.550	1.092	0.208	0.081	0.107	2969	0.050	157.004	38.254

Table 7: Detailed circularity measurements at selected samples in z-plane, vessel MV012.

Samples in the Z-plane

Tag	Area	Measured deviation ⁹	Residuals				Sample size	Slice		
			Range	RMSD ¹⁰	MAD ¹¹	SD		Height	Z coord.	Radius ¹²
		mm	mm	mm	mm	mm		mm	mm	mm
c01	exterior	Ø102.744±1.061	1.761	0.402	0.166	0.235	2995	0.050	157.004	51.372
c02	exterior	Ø93.283±0.955	1.422	0.359	0.145	0.228	2336	0.050	133.214	46.642
c03	exterior	Ø84.779±1.026	1.727	0.375	0.143	0.246	2080	0.050	108.050	42.389
c04	exterior	Ø78.999±1.147	1.900	0.411	0.154	0.277	1846	0.050	70.549	39.499
c05	exterior	Ø90.974±1.222	1.781	0.528	0.151	0.311	2356	0.050	21.022	45.487
c06	interior	Ø76.240±1.094	1.963	0.479	0.182	0.243	3026	0.050	157.004	38.120
c06_s	interior sep.	Ø76.479±0.610	1.179	0.208	0.082	0.110	2989	0.050	157.004	38.239

Table 8: Detailed circularity measurements at selected samples perpendicular to vessel curvature, vessel MV012.

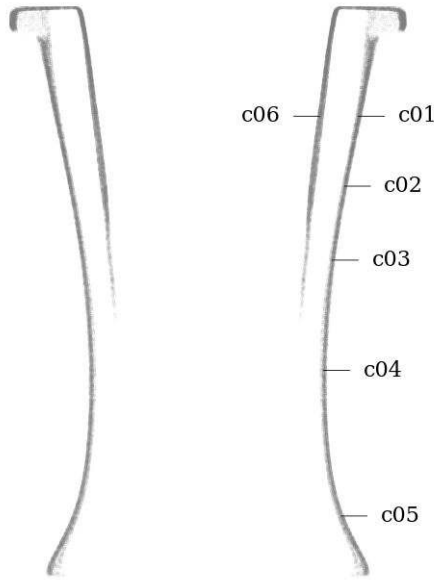


Figure 58: Circularity measurement sample locations, full mesh aligned to exterior surface

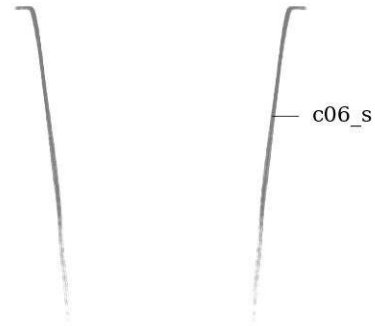


Figure 59: Circularity measurement sample location, separately aligned interior mesh

Samples perpendicular to the surface curvature

Tag	Area	Measured deviation ⁹	Residuals				Sample size	Slice		
			Range	RMSD ¹⁰	MAD ¹¹	SD		Height	Z coord.	Radius ¹²
		in	in	in	in	in		in	in	in
c01	exterior	$\text{Ø}4.0389 \pm 0.0387$	0.0731	0.0159	0.0062	0.0087	2969	0.0020	6.1813	2.0194
c02	exterior	$\text{Ø}3.6647 \pm 0.0300$	0.0521	0.0131	0.0048	0.0072	2256	0.0020	5.2447	1.8324
c03	exterior	$\text{Ø}3.3357 \pm 0.0398$	0.0676	0.0144	0.0058	0.0092	2056	0.0020	4.2539	1.6678
c04	exterior	$\text{Ø}3.1085 \pm 0.0442$	0.0740	0.0162	0.0064	0.0109	1841	0.0020	2.7775	1.5542
c05	exterior	$\text{Ø}3.5714 \pm 0.0411$	0.0667	0.0187	0.0062	0.0099	2032	0.0020	0.8276	1.7857
c06	interior	$\text{Ø}3.0098 \pm 0.0409$	0.0787	0.0182	0.0062	0.0083	2987	0.0020	6.1813	1.5049
c06_s	interior sep.	$\text{Ø}3.0121 \pm 0.0217$	0.0430	0.0082	0.0032	0.0042	2969	0.0020	6.1813	1.5061

Table 9: Detailed circularity measurements at selected samples in z-plane, vessel MV012.

Samples in the Z-plane

Tag	Area	Measured deviation ⁹	Residuals				Sample size	Slice		
			Range	RMSD ¹⁰	MAD ¹¹	SD		Height	Z coord.	Radius ¹²
		in	in	in	in	in		in	in	in
c01	exterior	$\text{Ø}4.0451 \pm 0.0418$	0.0693	0.0158	0.0065	0.0092	2995	0.0020	6.1813	2.0225
c02	exterior	$\text{Ø}3.6726 \pm 0.0376$	0.0560	0.0141	0.0057	0.0090	2336	0.0020	5.2447	1.8363
c03	exterior	$\text{Ø}3.3377 \pm 0.0404$	0.0680	0.0148	0.0056	0.0097	2080	0.0020	4.2539	1.6689
c04	exterior	$\text{Ø}3.1102 \pm 0.0452$	0.0748	0.0162	0.0060	0.0109	1846	0.0020	2.7775	1.5551
c05	exterior	$\text{Ø}3.5817 \pm 0.0481$	0.0701	0.0208	0.0060	0.0122	2356	0.0020	0.8276	1.7908
c06	interior	$\text{Ø}3.0016 \pm 0.0431$	0.0773	0.0188	0.0072	0.0096	3026	0.0020	6.1813	1.5008
c06_s	interior sep.	$\text{Ø}3.0110 \pm 0.0240$	0.0464	0.0082	0.0032	0.0043	2989	0.0020	6.1813	1.5055

Table 10: Detailed circularity measurements at selected samples perpendicular to vessel curvature, vessel MV012.

Comparison of circularity on the full vessel surface

Metric

Samples perpendicular to the surface curvature

Area	Range			Standard Deviation			RMSD			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.700	1.306	2.344	0.226	0.178	0.317	0.397	0.316	0.523	3151	0.050
Interior	1.680	1.387	2.152	0.187	0.139	0.228	0.388	0.260	0.518	760	0.050
Interior separate	1.037	0.799	1.656	0.093	0.071	0.145	0.158	0.109	0.268	760	0.050

Table 11: Detailed circularity measurements at selected samples in z-plane, vessel MV012.

Samples in the z-plane

Area	Range			Standard Deviation			RMSD			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	mm	mm	mm	mm	mm	mm	mm	mm	mm		mm
Exterior	1.733	1.333	2.924	0.268	0.203	0.466	0.410	0.324	0.713	3149	0.050
Interior	1.691	1.394	2.158	0.211	0.143	0.295	0.398	0.266	0.536	759	0.050
Interior separate	1.050	0.784	1.623	0.094	0.071	0.145	0.160	0.110	0.270	759	0.050

Table 12: Detailed circularity measurements at selected samples perpendicular to vessel curvature, vessel MV012.

Imperial

Samples perpendicular to the surface curvature

Area	Range			Standard Deviation			RMSD			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	in	in	in	in	in	in	in	in	in		in
Exterior	1.700	1.306	2.344	0.226	0.178	0.317	0.397	0.316	0.523	3151	0.050
Interior	1.680	1.387	2.152	0.187	0.139	0.228	0.388	0.260	0.518	760	0.050
Interior separate	1.037	0.799	1.656	0.093	0.071	0.145	0.158	0.109	0.268	760	0.050

Table 13: Detailed circularity measurements at selected samples in z-plane, vessel MV012.

Samples in the z-plane

Area	Range			Standard Deviation			RMSD			Slices	Slice height
	Median	Min.	Max.	Median	Min.	Max.	Median	Min.	Max.		
	in	in	in	in	in	in	in	in	in		in
Exterior	1.733	1.333	2.924	0.268	0.203	0.466	0.410	0.324	0.713	3149	0.050
Interior	1.691	1.394	2.158	0.211	0.143	0.295	0.398	0.266	0.536	759	0.050
Interior separate	1.050	0.784	1.623	0.094	0.071	0.145	0.160	0.110	0.270	759	0.050

Table 14: Detailed circularity measurements at selected samples perpendicular to vessel curvature, vessel MV012.

Circularity analysis of exterior surface - perpendicular to surface curvature

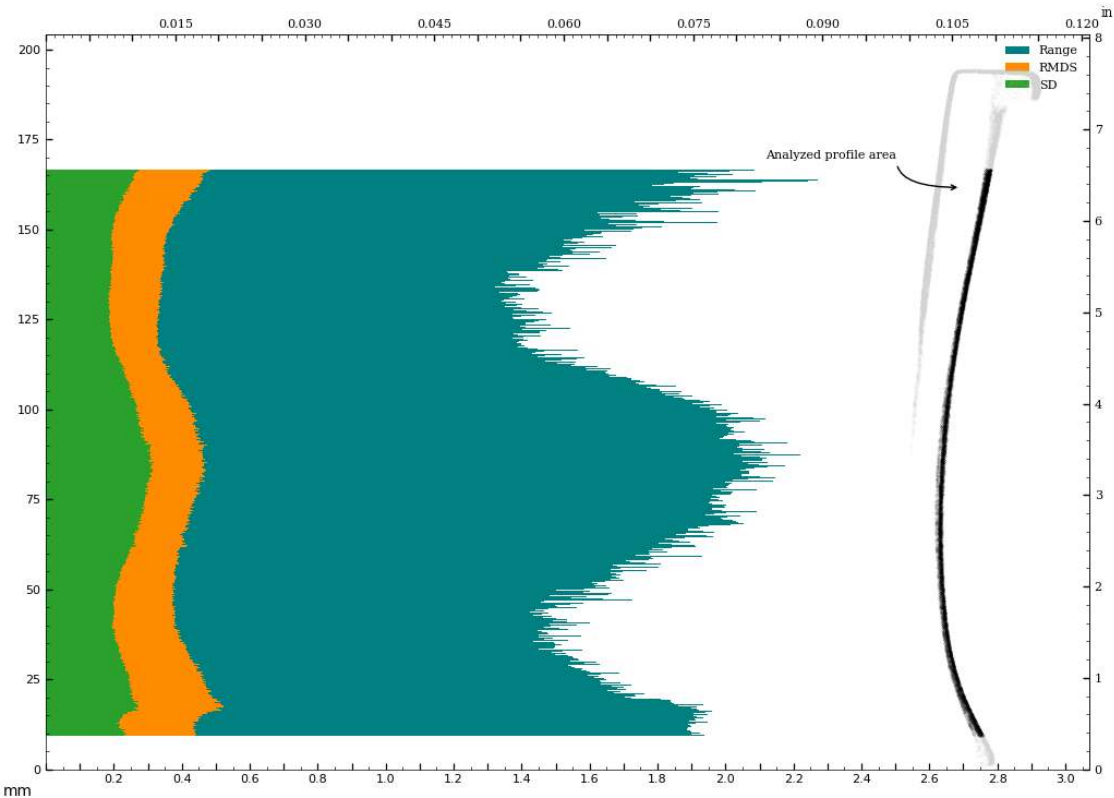


Figure 60: Circularity of exterior surface - perpendicular to surface curvature.

Circularity analysis of exterior surface - in z-plane

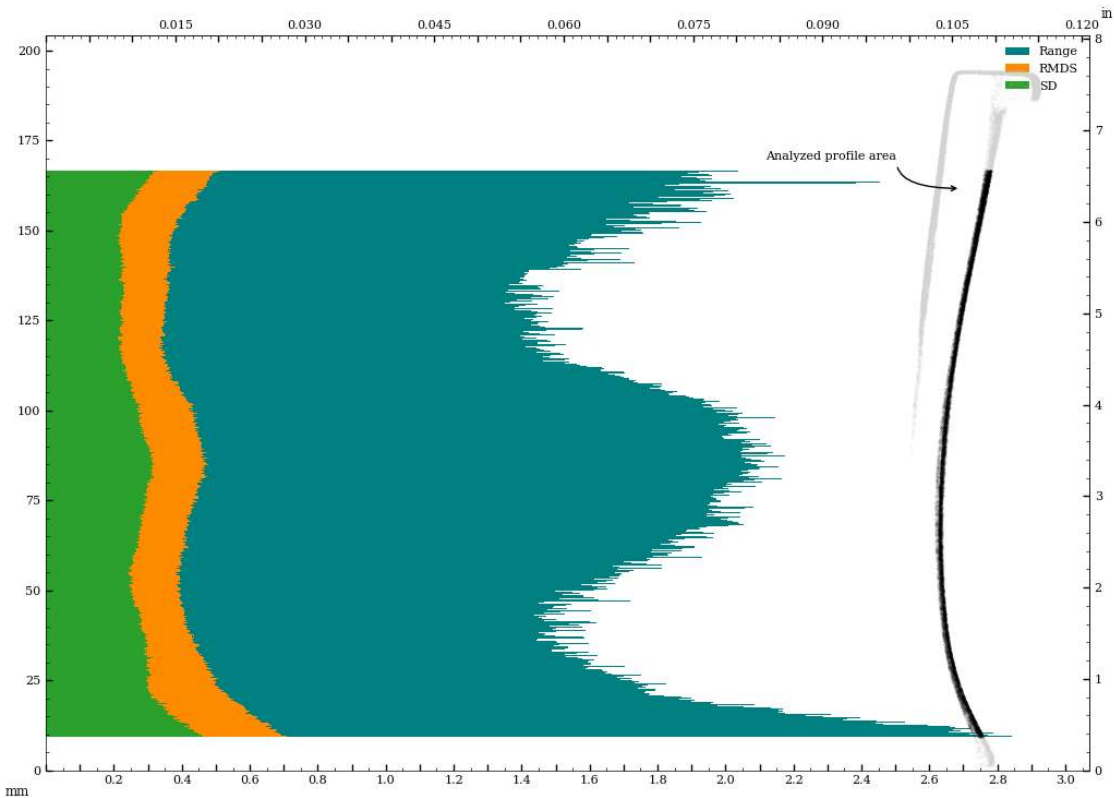


Figure 61: Circularity of exterior surface - in z-plane.

Circularity analysis of exterior surface, perpendicular to surface curvature, Standard Deviation and Root Mean Squared Deviation

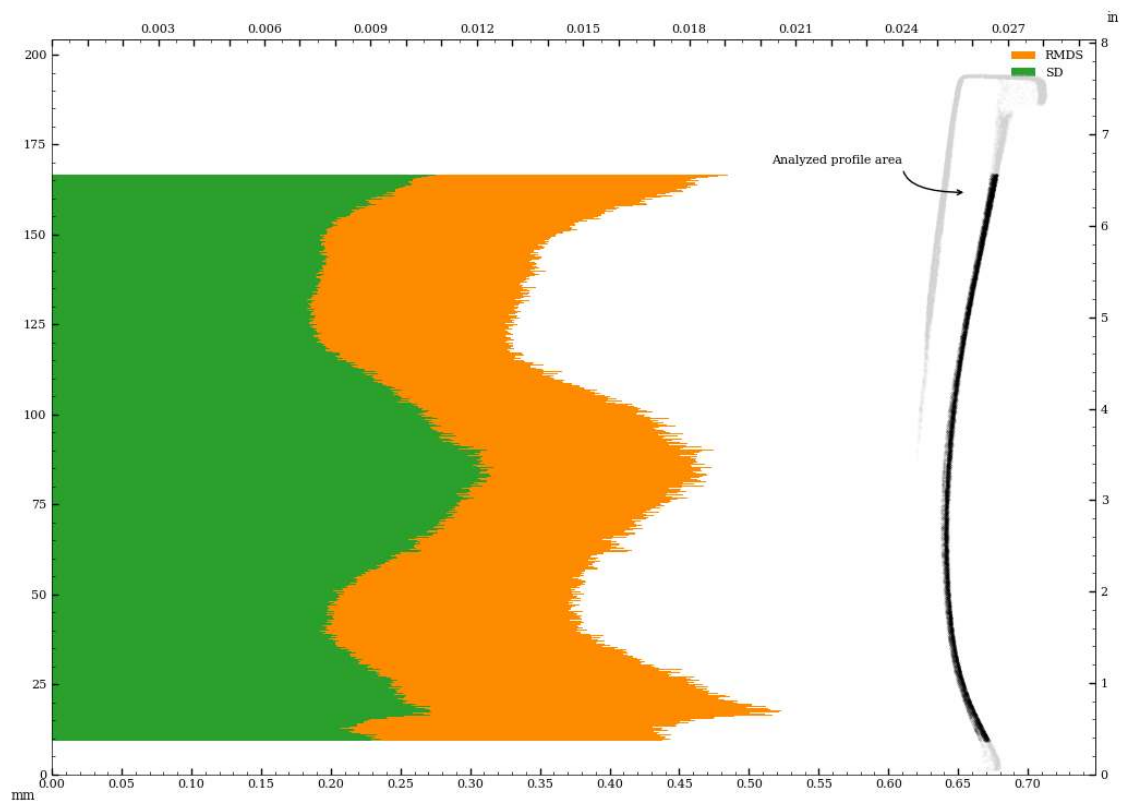


Figure 62: Vessel circularity of exterior surface, perpendicular to surface curvature, standard deviation and median absolute deviation.

Circularity analysis of exterior surface, in z-plane, Standard Deviation and Root Mean Squared Deviation

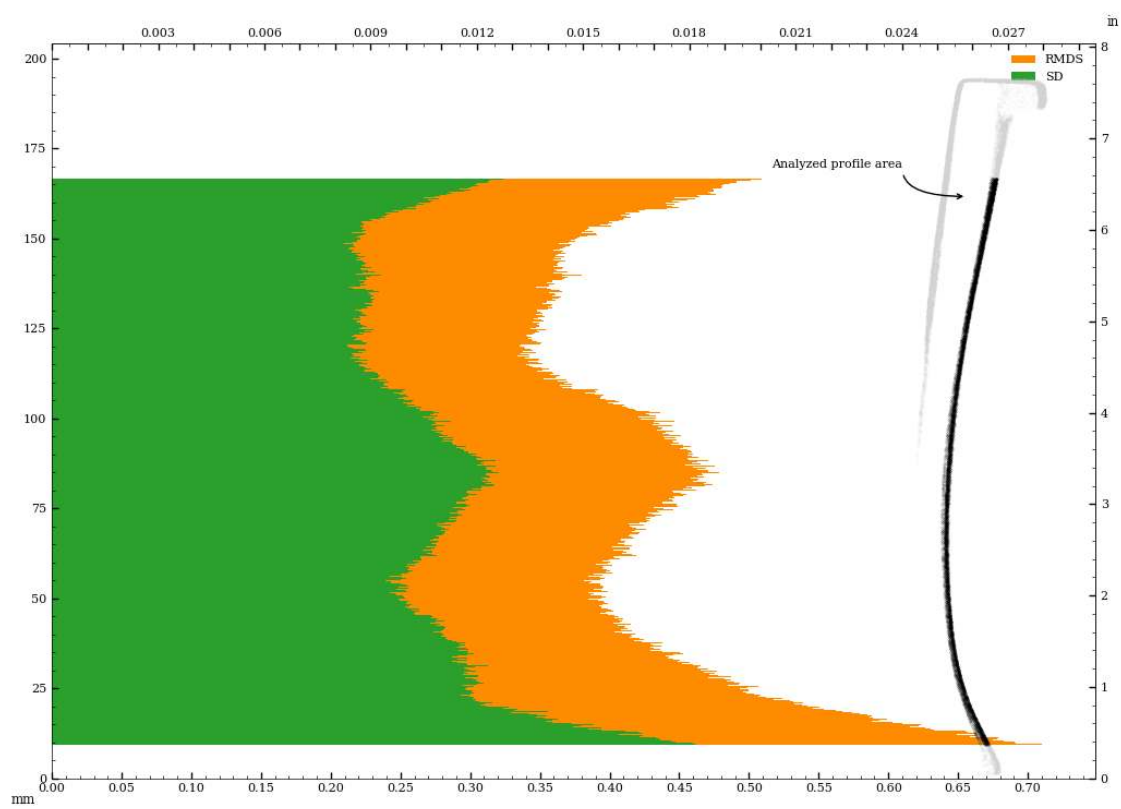


Figure 63: Vessel circularity of exterior surface, in z-plane, standard deviation and median absolute deviation.

Circularity analysis of interior surface - perpendicular to surface curvature

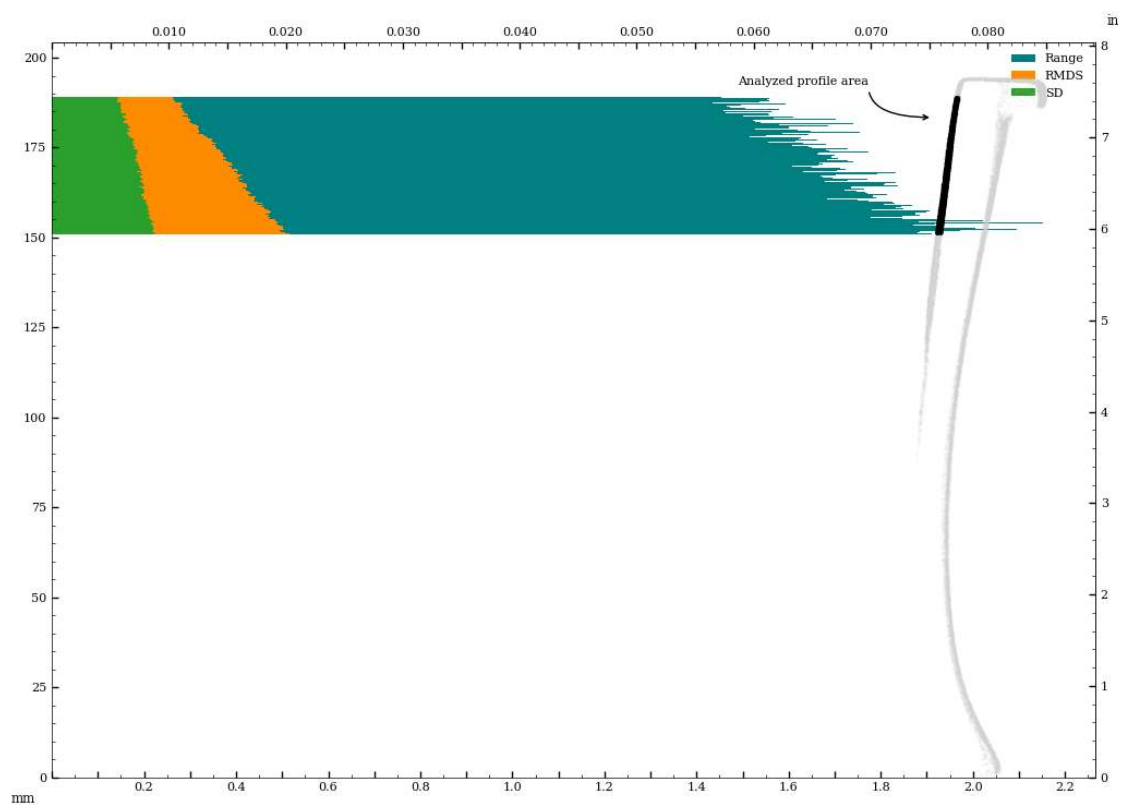


Figure 64: Circularity of interior surface - perpendicular to surface curvature.

Circularity analysis of interior surface - in z-plane

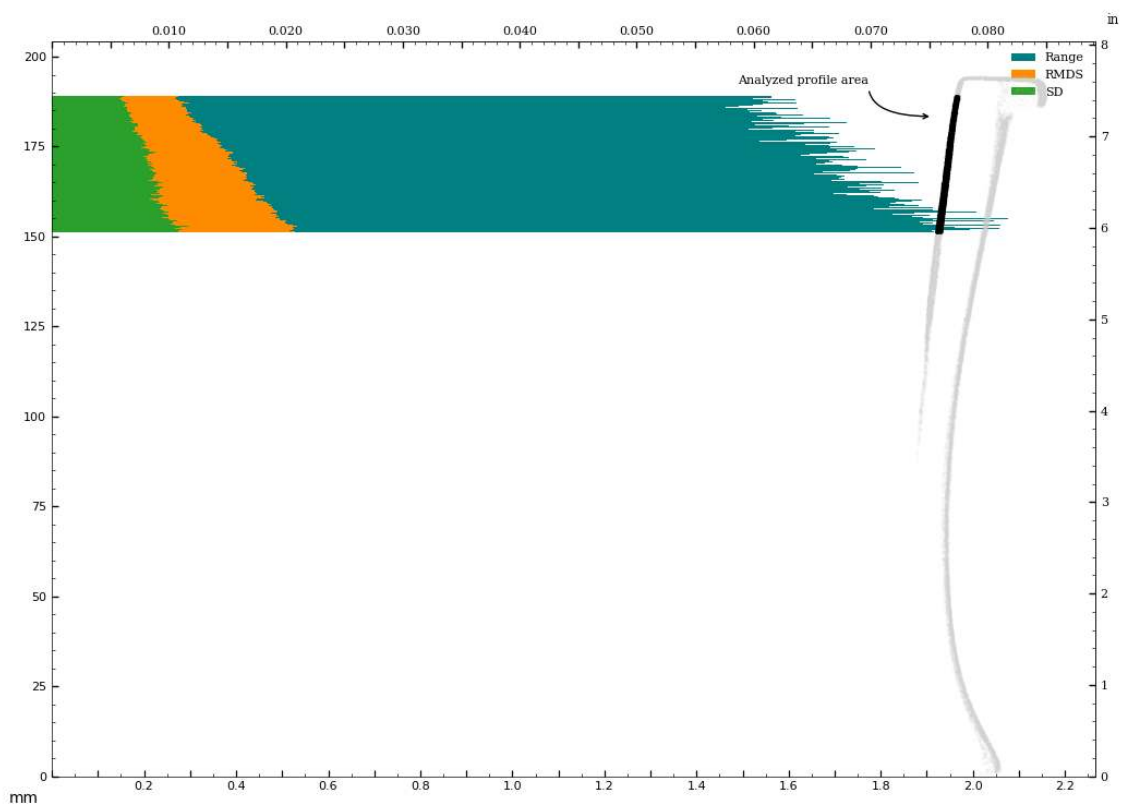


Figure 65: Circularity of interior surface - in z-plane.

Circularity analysis of interior surface, perpendicular to surface curvature, Standard Deviation and Root Mean Squared Deviation

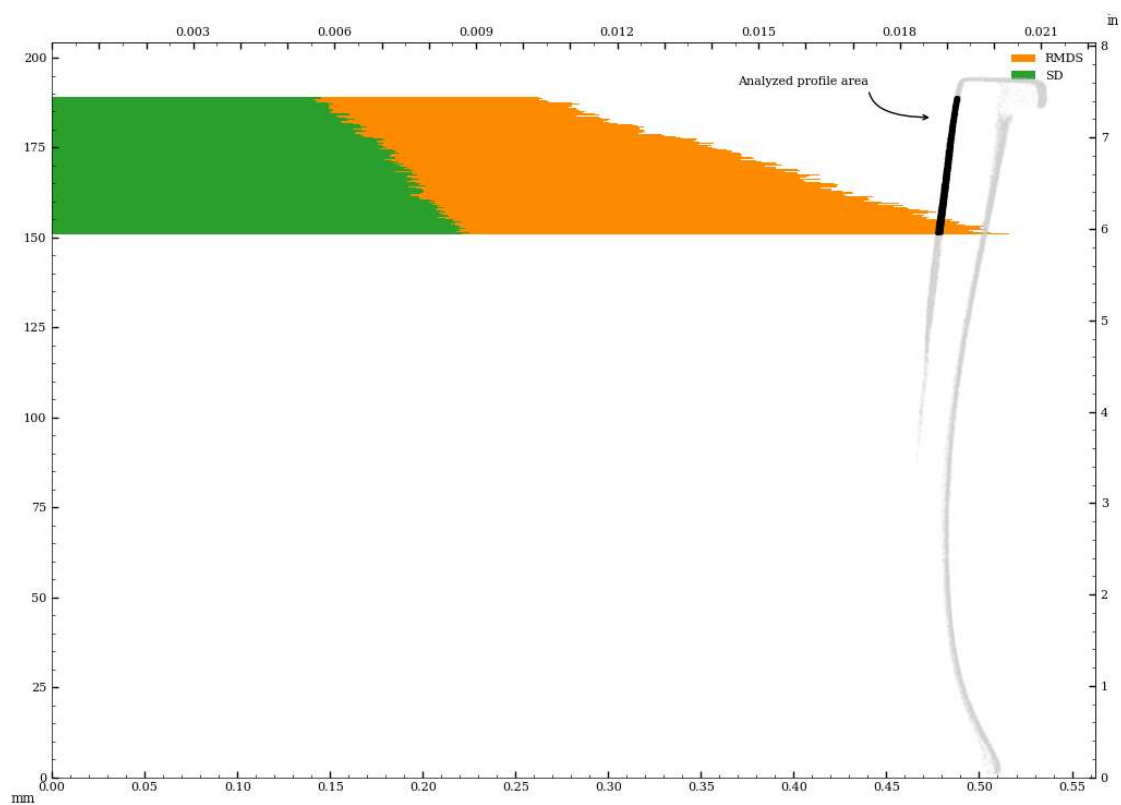


Figure 66: Vessel circularity of interior surface, perpendicular to surface curvature, standard deviation and median absolute deviation.

Circularity analysis of interior surface, in z-plane, Standard Deviation and Root Mean Squared Deviation

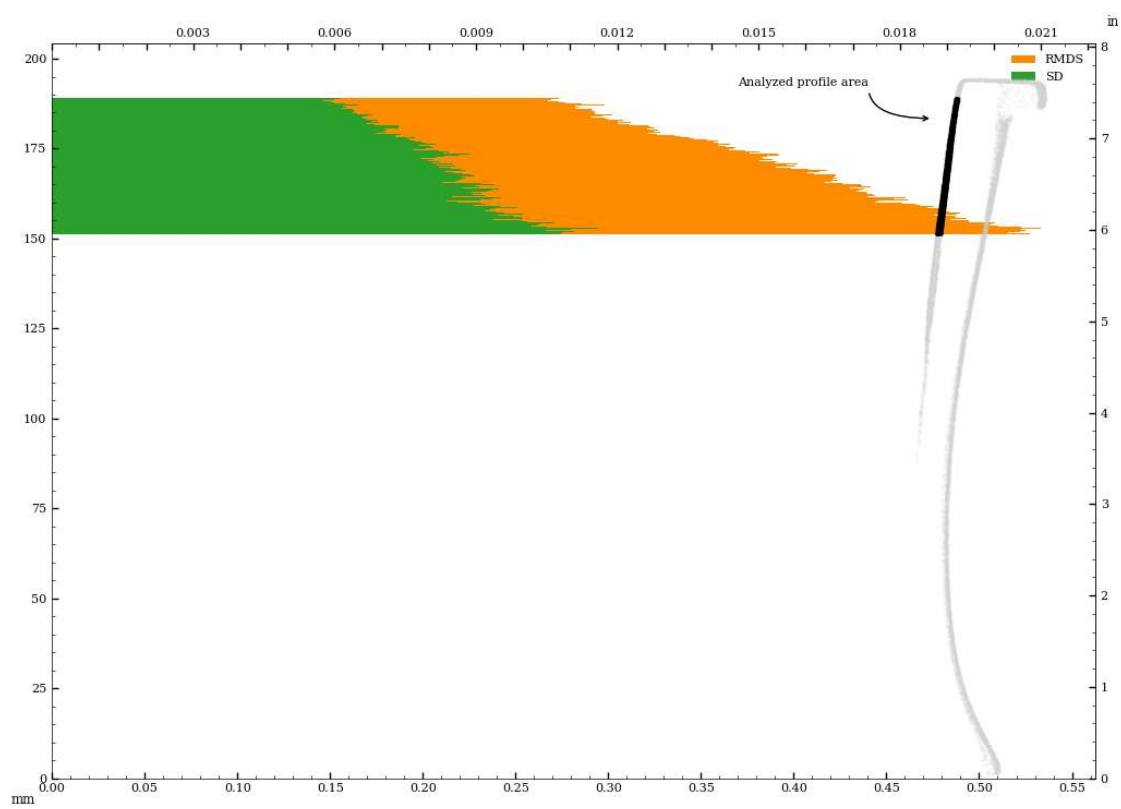


Figure 67: Vessel circularity of interior surface, in z-plane, standard deviation and median absolute deviation.

Circularity analysis of interior separately aligned surface - perpendicular to surface curvature

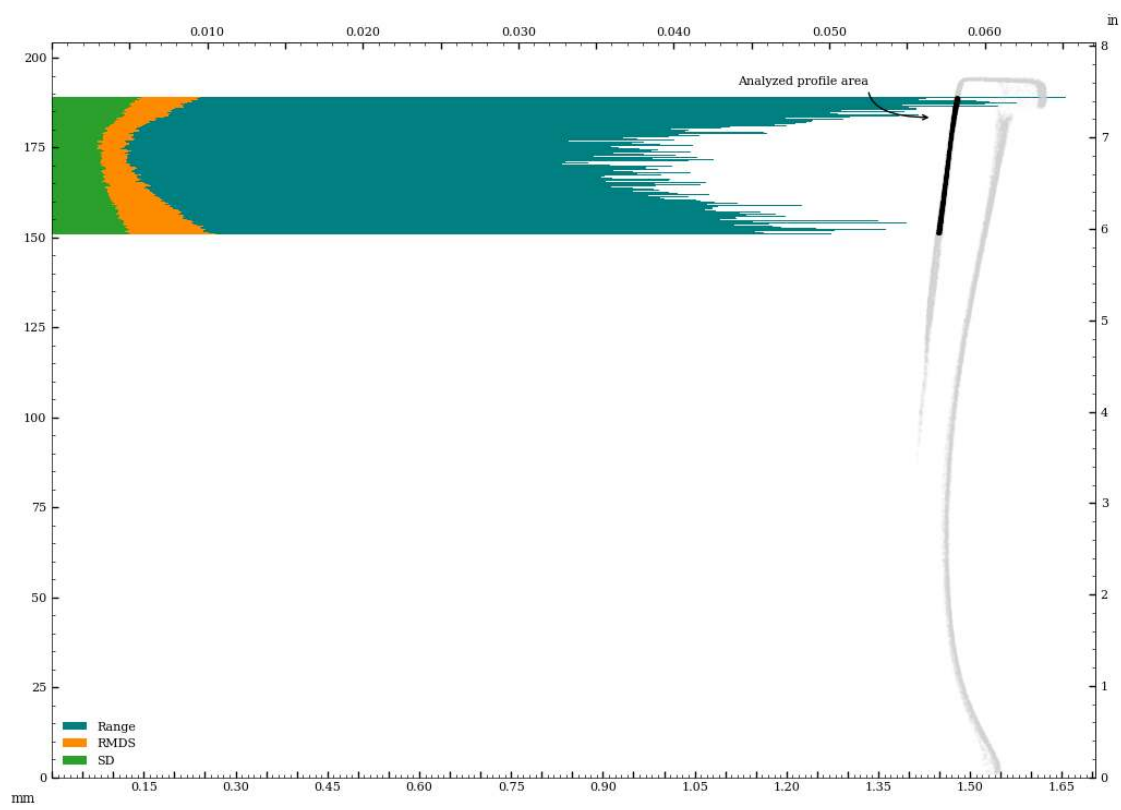


Figure 68: Circularity of interior_separate surface - perpendicular to surface curvature.

Circularity analysis of interior separately aligned surface - in z-plane

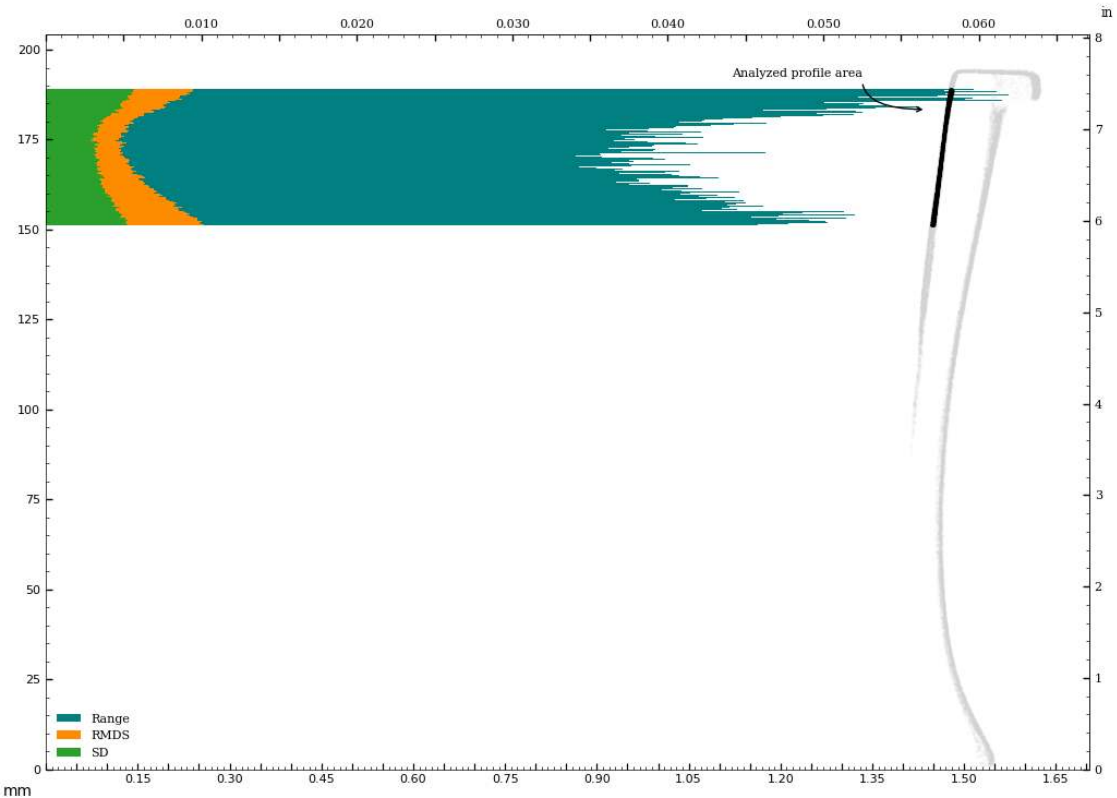


Figure 69: Circularity of interior_separate surface - in z-plane.

Circularity analysis of interior separately aligned surface, perpendicular to surface curvature, Standard Deviation and Root Mean Squared Deviation

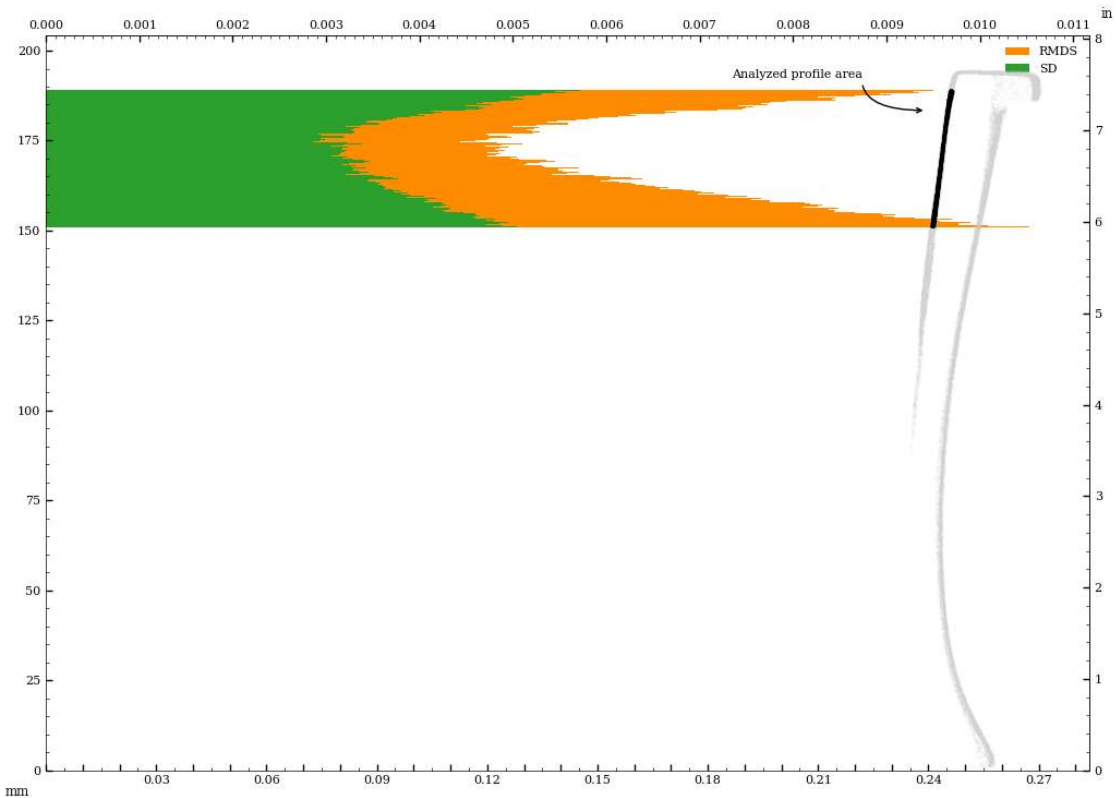


Figure 70: Vessel circularity of interior_separate surface, perpendicular to surface curvature, standard deviation and median absolute deviation.

Circularity analysis of interior separately aligned surface, in z-plane, Standard Deviation and Root Mean Squared Deviation

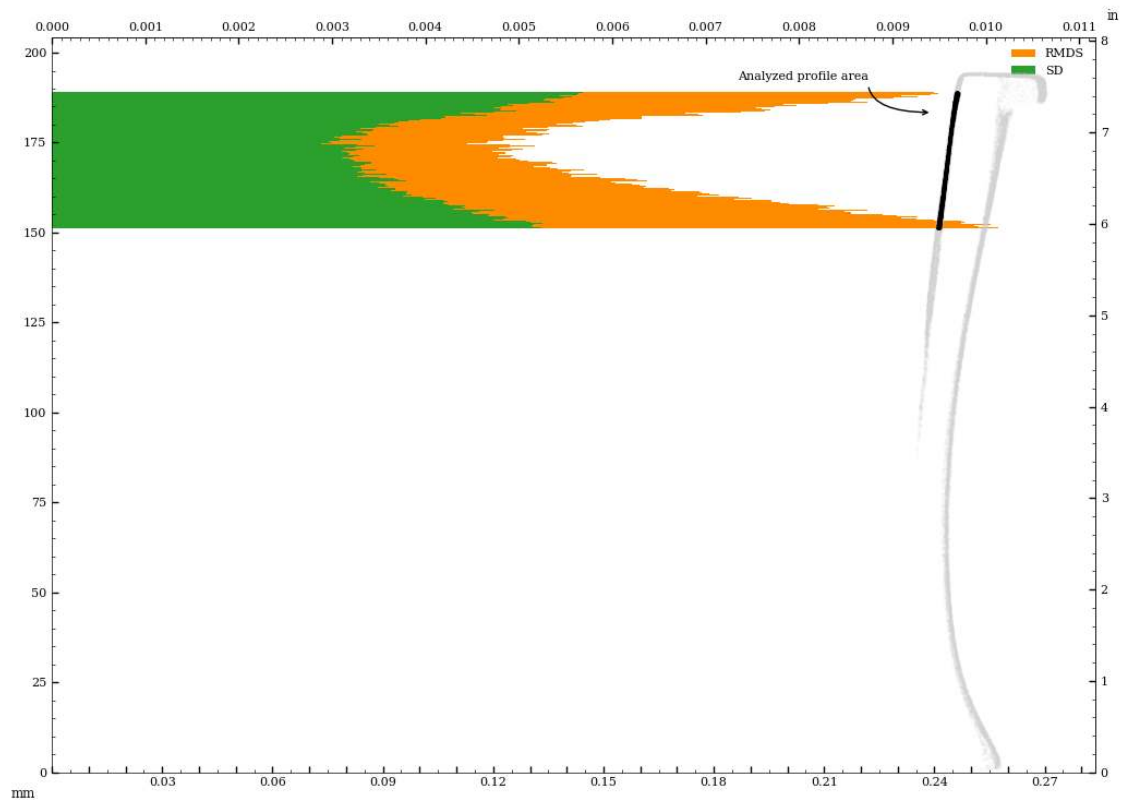


Figure 71: Vessel circularity of interior_separate surface, in z-plane, standard deviation and median absolute deviation.

Appendix B - Comparison Of Concentricity Measurements (Z-plane vs. surface-perpendicular)

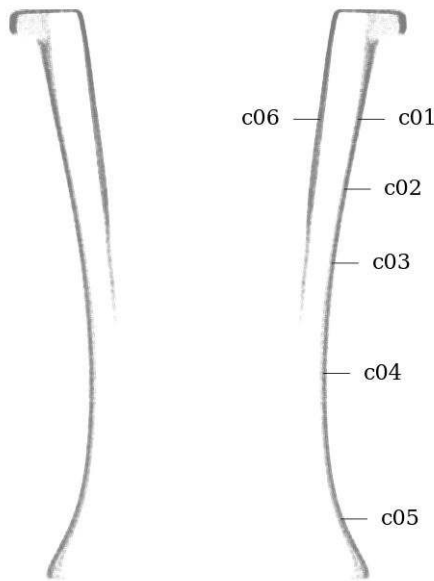


Figure 72: Circularity measurement sample locations, full mesh aligned to exterior surface

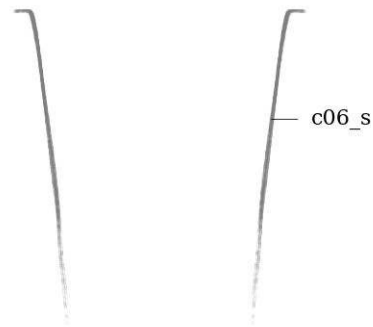


Figure 73: Circularity measurement sample location, separately aligned interior mesh

Concentricity measurements perpendicular to surface curvature

Tag	Reference	Deviation	Sample size	Circle fit residuals analysis for sample listed in Tag column						Center (x,y)
				Range full	Range inliers	RMSD full	RMDS inliers	SD full	SD inliers	
		mm		mm	mm	mm	mm	mm	mm	μm
c01	z-axis	0.443	2995	1.737	1.737	0.386	0.386	0.224	0.224	−415, 156
c02	z-axis	0.188	2336	1.320	1.319	0.341	0.340	0.179	0.178	−176, 67
c03	z-axis	0.139	2080	1.730	1.626	0.377	0.332	0.244	0.207	139, 9
c04	z-axis	0.255	1846	1.893	1.893	0.407	0.407	0.284	0.284	236, −97
c05	z-axis	0.064	2356	1.835	1.835	0.511	0.511	0.269	0.269	57, −30
c06	z-axis	0.633	3026	3.261	3.066	0.998	0.997	0.401	0.400	118, −622
c06_s	z-axis	0.250	2989	1.649	1.544	0.428	0.427	0.186	0.183	−34, −247
c01	c06	0.943								−533, 778

Concentricity measurements in z-plane

Tag	Reference	Deviation	Sample size	Circle fit residuals analysis for sample listed in Tag column						Center (x,y)
				Range full	Range inliers	RMSD full	RMDS inliers	SD full	SD inliers	
		mm		mm	mm	mm	mm	mm	mm	μm
c01	z-axis	0.443	2995	1.737	1.737	0.386	0.386	0.224	0.224	−415, 156
c02	z-axis	0.188	2336	1.320	1.319	0.341	0.340	0.179	0.178	−176, 67
c03	z-axis	0.139	2080	1.730	1.626	0.377	0.332	0.244	0.207	139, 9
c04	z-axis	0.255	1846	1.893	1.893	0.407	0.407	0.284	0.284	236, −97
c05	z-axis	0.064	2356	1.835	1.835	0.511	0.511	0.269	0.269	57, −30
c06	z-axis	0.633	3026	3.261	3.066	0.998	0.997	0.401	0.400	118, −622
c06_s	z-axis	0.250	2989	1.649	1.544	0.428	0.427	0.186	0.183	−34, −247
c01	c06	0.943								−533, 778

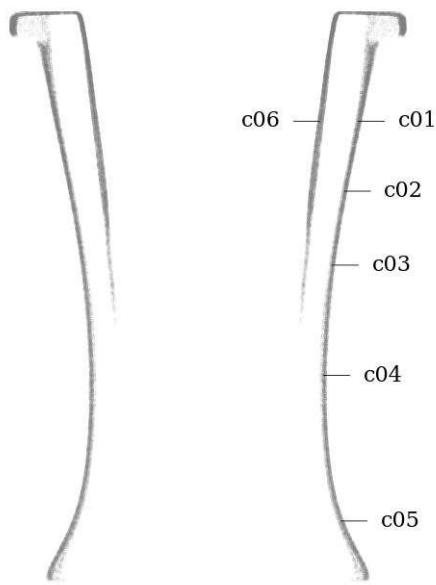


Figure 74: Circularity measurement sample locations, full mesh aligned to exterior surface

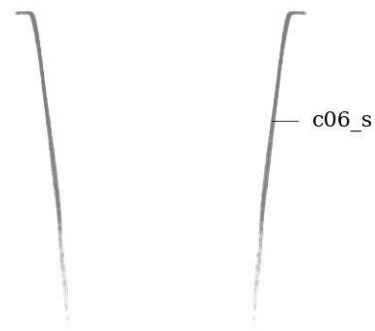


Figure 75: Circularity measurement sample location, separately aligned interior mesh

Concentricity measurements perpendicular to surface curvature

Tag	Reference	Deviation	Sample size	Circle fit residuals analysis for sample listed in Tag column						
				Range full	Range inliers	RMSD full	RMDS inliers	SD full	SD inliers	Center (x,y)
		in		in	in	in	in	in	in	thou
c01	z-axis	0.0175	2995	0.0684	0.0684	0.0152	0.0152	0.0088	0.0088	−16.3, 6.1
c02	z-axis	0.0074	2336	0.0520	0.0519	0.0134	0.0134	0.0070	0.0070	−6.9, 2.6
c03	z-axis	0.0055	2080	0.0681	0.0640	0.0148	0.0131	0.0096	0.0082	5.5, 0.4
c04	z-axis	0.0101	1846	0.0745	0.0745	0.0160	0.0160	0.0112	0.0112	9.3, −3.8
c05	z-axis	0.0025	2356	0.0722	0.0722	0.0201	0.0201	0.0106	0.0106	2.2, −1.2
c06	z-axis	0.0249	3026	0.1284	0.1207	0.0393	0.0392	0.0158	0.0158	4.6, −24.5
c06_s	z-axis	0.0098	2989	0.0649	0.0608	0.0169	0.0168	0.0073	0.0072	−1.3, −9.7
c01	c06	0.0371								−21.0, 30.6

Concentricity measurements in z-plane

Tag	Reference	Deviation	Sample size	Circle fit residuals analysis for sample listed in Tag column						
				Range full	Range inliers	RMSD full	RMDS inliers	SD full	SD inliers	Center (x,y)
		in		in	in	in	in	in	in	thou
c01	z-axis	0.0175	2995	0.0684	0.0684	0.0152	0.0152	0.0088	0.0088	−16.3, 6.1
c02	z-axis	0.0074	2336	0.0520	0.0519	0.0134	0.0134	0.0070	0.0070	−6.9, 2.6
c03	z-axis	0.0055	2080	0.0681	0.0640	0.0148	0.0131	0.0096	0.0082	5.5, 0.4
c04	z-axis	0.0101	1846	0.0745	0.0745	0.0160	0.0160	0.0112	0.0112	9.3, −3.8
c05	z-axis	0.0025	2356	0.0722	0.0722	0.0201	0.0201	0.0106	0.0106	2.2, −1.2
c06	z-axis	0.0249	3026	0.1284	0.1207	0.0393	0.0392	0.0158	0.0158	4.6, −24.5
c06_s	z-axis	0.0098	2989	0.0649	0.0608	0.0169	0.0168	0.0073	0.0072	−1.3, −9.7
c01	c06	0.0371								−21.0, 30.6